ROYAL CANADIAN NAVY
QUALIFICATION STANDARD AND PLAN (QSP)

Basic Submarine Qualification

NQUAL - AILS

MITE Course IDs:

Residential Phase - 109201

OJPR Harbour Watchkeeper - 122073

OJPR Safe Submariner - 109285

Submarine Unpressurized Escape Refresher Training (MITE 109383)

Submarine Firefighting Refresher Training (MITE 115324)

Submarine R2MR Training (MITE 122149-0006)

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RECORD OF AMENDMENTS

The Training Authority (TA) designated for this qualification is the Commander Naval Personnel and Training Group (CNPTG). Naval Training Development Centre (Atlantic) (NTDC(A)) is the designated development centre responsible for the maintenance of training. Amendments are to be requested through NTDC(A) and approved by CNPTG.

Revision	Approval Date	Authority
2.0	24/02/2020	CNPTG
2.1	21/10/2020	NTDC(A)

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FOREWORD

1. This training is based on the specification found at:

    a. A-PD-055-002/PP-001; and
    b. A-PD-055-002/PP-002.

A trainee will achieve the qualification AILS when the standards listed in all Performance Objectives have been met.

A trainee will achieve the following MITE Course IDs when standards listed in the associated Performance Objectives have been met:

    a. Residential Phase: POs 001 through 011 - MITE Course ID 109201
    b. Submarine Firefighting Refresher Training MITE Course ID 115324
    c. Submarine Unpressurized Escape Refresher Training MITE Course ID 109383
    d. BSQ OJPR Harbour Watchkeeper Part 1 - MITE Course ID 122073
    e. BSQ OJPR Safe Submariner Part 2 - MITE Course ID 109285

2. The designated TE for this training are indicated below:

TE	Phase / Module	MITE course ID	NQual
NFS(A)	All		

CHAPTER 1

TRAINING MANAGEMENT DETAILS

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AIM

1. The aim of this Qualification Standard and Plan (QSP) is to establish the requirements and training for the general duties and responsibilities required while serving in Victoria-class submarines both as a Harbour Watchkeeper and Safe Submariner – AILS. This QSP is designed to assist NTDC (A) in developing, Naval Fleet Schools in conducting and evaluating, and NPTG in validating the training.

COURSE DURATION

2. Course duration is calculated in accordance with NPTORD 4500-22, para 4.5. For this course, a period of instruction is 50 minutes in duration with 8 periods per day.

Course Duration Calculation	
Allocation of ATT	**Periods**
Sum of all EOs/POs	285
Enabling Checks (ECs)	52
Performance Checks (PCs)	OJT
Actual Training Time (ATT) Sub-total	337
Allocation of BAT, TAT, TTT and PT IAW NPTORD 4500-22, Table 5.5	
Base Administrative Time (BAT)	10
Training Administrative Time (TAT)	10
Training Travel Time (TTT)	34
Physical Fitness (PT)	24
Sub-total	78
Course Duration Calculation (ATT+BAT+TTT+TAT+PT) **400 minutes** **Partial days must be rounded up**	**52** **Trg Days**

REQUIRED COURSE STAFFING

Instructional Staff

POSITION	RANK	QUALIFICATIONS	QTY	PO / EO
Course Officer	Lt(N)	MS Eng or NCS Eng AILS, ALDE or HOD	1	001 - 011
Senior Instructor	PO1	MAR Tech, AILS, AIND, DWS	1	001 - 011
Lead Instructor	MS-PO2	MAR Tech, AILS, AIND	1	001 - 011
Instructor	S1-PO2	MAR Tech (Elect), AILS	1	002
Instructor	S1-PO2	MAR Tech (Mech), AILS	1	003 – 006 / 008
Instructor	S1-PO2	W Eng FC Tech, AILS, AKQN	1	007
Instructor	S1-PO2	W Eng Sonar Tech, AILS, AKQE	1	007
Instructor	S1-PO2	W Eng Sonar Comm, AILS, AKQK	1	007

Instructor	S1-PO2	NAV COM, AILS, AIIW	1	010

Guest Lecturers

POSITION	RANK	QTY	PO / EO
Mental Health Clinician / Educator	Civ	2	011.04
Submarine Physician's Assistant	Slt - LT(N)	1	011.04
Sea Training (Submarines) Coxswain	CPO1	1	011.04
Newly Qualified Submariner	S3 - MS	1	011.04
CANSUBFOR Chief Petty Officer	CPO1	1	011.04

CRITICAL TRAINING RESOURCES

FACILITY	NO. OF DAYS
Submarine Escape Training Facility	4
Damage Control Training Facility	4
Alongside Submarine	1
Ship Control / Machinery Control Trainers (SCT / MCT)	8
Tactical Weapons System Trainer (TWS)	1
Weapons Handling Discharge System Trainer (WHDS)	6

COURSE CAPACITY

3. Planning for the conduct of training is based on the guidelines in NPTORD 4500-40.

   a. Maximum – 20;

   b. Minimum – 10; and

   c. Optimum – 16.

PREREQUISITES

To be selected for this course the loading authority must ensure that trainees have the following prerequisites:

   a. Have a Level I Security Clearance or higher for the BSQ residential phase - MITE Crse ID 109201;

   b. Have a Level II Security Clearance or higher for Basic Submarine Qualification OJPR (Harbour Watchkeeper – Part 1) – MITE Course ID (TBD);

   c. Have a Level II Security Clearance or higher for Basic Submarine Qualification OJPR (Safe Submariner – Part 2) – MITE Course ID109285;

d. Be medically and dentally fit to the standard set out in A-MD-154-000/FP000 Medical Standards for the Canadian Forces;

e. NCMs shall complete NETP residential phase - MITE Course IDs 118531-5;

f. W Eng Techs must be qualified in:

1) Jr W Eng Tech (AKNX); or

2) Jr W Eng Tech (AKNY).

g. Be RQ – S1 qualified for the following MOSIDs:

1) Naval Combat Information Operator (ALAU);

2) Cook (ACJT);

3) Steward (ACKH); and

4) Naval Communicator (AIXC).

h. Be RQ –S3 qualified for the following MOSIDs:

1) MAR TECH 00379-04; and

2) Sonar Operator (AIXG);

i. Be PA QL6B or RQ – SLt qualified;

j. Be Med Tech QL6A qualified;

k. Be NWO Phase IV qualified for Naval Warfare Officer;

l. Be Phase VI qualified for the following MOS:

1) Marine Systems Engineering; and

2) Naval Combat Systems Engineering.

PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

PLARs shall be conducted IAW NPTORD 4500-27, DAOD 5031-1 and CFITES Vol. 12.

SPECIAL INSTRUCTIONS

Nil.

CHAPTER 2

PERFORMANCE OBJECTIVES & LESSON GUIDANCE

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PO 001 Identify Submarine Construction

1. Performance. (T0003) Identify Submarine Construction

2. Conditions

   a) Given:

      (1) Submarine;

      (2) Publications and drawings; and

      (3) Applicable references.

   b) Denied: Assistance.

   c) Environmental: Alongside submarine.

3. Standard. The submariner shall, IAW references, (T0003) identify submarine construction to include:

   a) Describing submarine layout to include:

      (1) Identifying pertinent systems reference material;

      (2) Describing submarine dimensions;

      (3) Identifying external tank locations;

      (4) Describing compartment layout;

      (5) Describing water tight integrity;

      (6) Operating doors and hatches; and

      (7) Describing fuel group layout.

   b) Identifying submarine components to include:

      (1) Identifying external and casing fittings;

      (2) Identifying shore connections;

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(3) Identifying mast and periscopes; and

c) Describing submarine buoyancy and stability.

4. References A1, A2, A3, A4, A5, A6, A7, A12, A13, B1, B2, B7, B8, B12, B13, B118, B120, B121, B122, B138, B139, B145, B146, B147, B148, B149, B150, B157

5. Task Numbers T0003

Training Limitations. Nil.

EO 001.01 – Describe Submarine Layout

001.01A – Layout and Locations

Conditions	Sequence
Classroom (S100)	Course Start of PO 001

Time	Classification	Instructor
2 x 50 mins	Unclas	AILS

Learning Objectives	Method
1) Submarine construction to include: - External tank locations - Compartment layout - Operation of all doors and hatches / Water tight integrity - Fuel group layout -System code and pipe colour coding 2) Routine casing procedures to include: - Safety precautions, hazards, and considerations - Proper authorization at sea / alongside - Purpose of a casing crawl	Tour of submarine (if platform available) Brainstorm Virtual Submarine Interactive lecture

Scenario	N/A
Activity Steps	<u>Brainstorm</u> Ask the students to brainstorm what makes a submarine different than a surface ship. Highlight these points: -How we dive -How we surface -How diesel electric submarine differs from a nuclear submarine -The value of running diesel electric submarines -How each crew member is essential(make reference to size of class compared to size of crew. -Crew members are submariners first, before their respective trades. -Less space -Water restrictions -Noise husbandry

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-Submarine allowance, per diem, and hotels
-Hygiene
-Maturity of crew
-Increased maintenance for operating under salt water, and ensuring all equipment operates as silently as possible.

Submarine Tour (Actual submarine preferred, if unavailable use Virtual Submarine or 360 camera tour)

Provide all students with an in depth tour of the the submarine to include:
-Making your way down the jetty – what you can expect to see
-Safety precautions for entering the submarine
-Going through each space explaining everything required to set the trot.
-Basic layout of the submarine
-Cover everything in the 48 hour pack
-Point out major equipment
-Bulkhead fittings and isolations
-Point out all the official space titles
-Run through a basic knowledge quiz, to include:
    -Navigating to various spaces.
    -What equipment is located in the spaces
    -How many decks are there
    -List all access points to the submarine (hatches)
    -Watertight integrity
    -Safety rounds
-If needed, walk through the submarine again
-If no platform is available on the coast, use the virtual submarine, or 360 camera tour.

Instructor will show class drawings of the compartment layout, external tank locations and the layout of the fuel groups.

Once finished virtual tour of the submarine, instructor will take the students back to 35 and 56 BKHD doors and explain their proper operation. Instructor will also explain water tight integrity and that the submarine is divided into two escape compartments.

Instructor to ask class for examples of safety considerations / hazards for working on / under the casing. Create a list on the board. Instructor to correct / add to the list and explain why these things are hazardous as well as steps to take to work safely on / under the casing.

Ensure that all students are able to access the Virtual Submarine on DLN or issued tablet.

Quiz students on navigating various spaces.

Power Point demonstation:
Instructor to ask class why and when a casing crawl might be conducted. Once answers are received from the class, instructor corrects and explains.

Instructor to ask class who they think would take part in a casing crawl. Asks individual students and makes a list on the board. Corrects, lists, and explains why certain members take part in a casing crawl.

Instructor explains the proper procedure to be granted authorization to work on under the casing while at sea / alongside.

Name major take aways from the construction lessons/activities

Resources

1)White board w/ markers
2) Computer w/ DWAN access and local projector
3) Laptops / tablets w/ access to Virtual Submarine Program
4) PP lesson EO 001.01
5) Physical 3D submarine

References

A1, A2, A3, A4, A5, A6, A7, A12, A13, B1, B2, B7, B8, B12, B13, B118, B120, B121, B122, B138, B139, B145, B146, B147, B148, B149, B150, B157

Notes to Instructor

Ensure that all students are able to access the Virtual Submarine on DLN or issued tablet.

Tell students to start becoming familar with the MCC and SCC layout.

Extra time is allotted to each lesson in PO 001 to allow for time to visit submarine if available. If submarine is not available for tour, students should spend extra time finding compartments and items on the Virtual Submarine and physical 3D model.

EO 001.02 – Identify Submarine Components

001.02 A– Submarine Components

Conditions	Sequence
Classroom (S100) Virtual Submarine Program	After EO 001.01A

Time	Classification	Instructor
2 x 50 mins	Unclas	AILS

Learning Objectives	Method
1) Locate External Hull and Casing Fittings. 2) Locate Shore Connections. 3) Locate Masts and Periscopes.	Think-Pair-Share Virtual Submarine

Scenario	N/A
Activity Steps	<u>Think-Pair-Share</u> Instructor will quickly explain what hull and casing fittings are then ask the class, on their own, to write down a list of all the External Hull and Casing Fittings. (3min) Then instructor will separate the class into small groups (2 or 3 students per group) and have them create a list of External Hull and Casing Fittings together. (3min) Once completed, the instructor will ask one group to volunteer to write their list on the board. After list is on the board, the instructor will ask the other groups if they have anything to add to the list. Once the class has listed all External Hull and Casing Fittings the instructor will correct the list, quickly explaining the purpose of each fitting. - Torpedo Embarkation Hatch - ATP Hatch - Fwd Escape Tower Hatch - Fwd and Aft SSEs - Conning Tower Upper Hatch - Fin Door - Main Access Hatch - SBS Well

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- Surface Mufflers
- Aft Escape Tower Hatch
- Main Vents (1,2,3 & 4)
- Sonar Array's (Active, Intercept and Ranging)
- 2073 10KHZ Pingers (x2)
- BQQ 10 Sail
- Fairleads
- Bollards
- Capstans
- Salvage Connections (High and Low)

Show on 3D submarine where applicable.

<u>Fittings and Shore Connections (Location and Identification)</u>

Then the instructor will use the Virtual Submarine Program (and/or pictures) to show the class what each of the fittings looks like, where they are located, and ensure they are able to identify the fitting later.

Then the instructor will use the Virtual Submarine Program (and/or pictures) to show the class what each of the Shore Connections looks like, where they are located, and ensure they are able to identify the fitting later.

- Sewage Connection
- DC Shore Connections (Fwd and Aft)
- Fwd First Filling
- Subsequent Filling
- Fire Fighting Connection
- Fresh Water Filling
- Battery Ventilation Overboard Discharge
- Chilled Water
- LP Bilge
- Lubrication Oil Filling
- AC Shore Connection
- Nitrogen Filling
- HP Air Connections
- Hydraulic Oil

Then the instructor will use the Virtual Submarine Program (and/or pictures) to show the class what each of the Masts and Periscopes looks like, where they are located, and ensure they are able to identify the fitting later.

- Mast Head Steaming Light
- Search Periscope
- Radar Mast
- Comms Mast
- Attack Periscope
- Induction Mast
- Exhaust Mast
- VEX Mast

Resources

1)White board w/markers
2) Computer w/ DWAN access and local projector
3) Laptops / tablets w/ access to virtual submarine program
4) PP lesson EO 001.02
5) Physical 3D submarine

References

A1, A2, A3, A4, A5, A6, A7, A12, A13, B1, B2, B7, B8, B12, B13, B118, B120, B121, B122, B138, B139, B145, B146, B147, B148, B149, B150, B157

Notes to Instructor

Extra time is allotted to each Lesson in PO 001 to allow for time to visit submarine if available. If submarine is not available for tour students should spend extra time finding compartments and items on the virtual submarine and physical 3D model.

EO 001.03 – Describe Buoyancy and Stability

001.03A – Buoyancy and Stability

Conditions		Sequence
Classroom (S100) Virtual Submarine Program		After EO 001.02A
Time	**Classification**	**Instructor**
2 x 50 mins	Unclas	AILS

Learning Objectives	Method
1) Describe Buoyancy to Include: - Positive Buoyancy - Negative Buoyancy - Neutral Buoyancy - Effects of Depth Changes on Buoyancy 2) Describe Stability to Include: - Longitudinal Stability - Transverse Stability - Difference between Submerged and Surfaced - Effects of Depth Changes on Stability	Interactive PowerPoint Virtual Submarine Program

Scenario	N/A
Activity Steps	Instructor to ask class to explain the difference between Positive, Negative and Neutral Buoyancy. Instructor to ask class what effects, if any, Depth Changes may have on buoyancy. Instructor to ask class to explain stability. Then explains the difference between Longitudinal and Transverse (front to back, side to side) Stability and the difference in stability when a submarine is Submerged vs on the Surface. <u>Interactive Lecture (Power Point)</u> Instructor will then give a lecture on buoyancy and stability as it relates to submarines. Instructor will utilize the interactive powerpoint while giving the lecture to allow students to see and understand how changes to buoyancy and stability are controlled by personnel within the submarine.

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SCC Tapestry/Class Handouts

Instructor will utilize the Virtual Submarine Program to show students the Submarine Control Console (SCC), and where the buoyancy and stability can be controlled from. (trim system controls, ballast system controls and main vent controls)

Emphasize the importance of integrating tapestry/SCC handout while studying EOPs. Tell them they should start studying EOPs daily.Tell them they should start studying SCC layout daily.

Resources	References
1)White board w/ markers 2) Computer w/ DWAN access and local projector	A1, A2, A3, A4, A5, A6, A7, A12, A13, B1, B2, B7, B8, B12, B13, B118, B120, B121, B122, B138, B139, B145, B146, B147, B148, B149, B150, B157

Notes to Instructor

Extra time is allotted to each Lesson in PO 001 to allow for time to visit submarine if available. If submarine is not available for tour students should spend extra time finding compartments and items on the virtual submarine and physical 3D model.

EC 001.03 – Submarine Construction

Scope	Conditions	Sequence
EO 001.01-03	Classroom S100	24 Hrs After EO 001.03A
Time	**Classification**	**Instructor**
1 x 50 mins	Unclass	AILS, AIND, AINA, AILQ

Scenario	N/A
Assessment Steps	Will be a written exam based on content from PO 001. Exam shall contain short answer questions and may contain multiple choice. Suggested breakdown of exam as follows: Layout and location – 33% Components – 33% Buoyancy and Stability – 33% INSTRUCTIONS GIVEN TO TRAINEE 1. EC 001.03 will be a written test based on content from EO 001.01-03. 2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam. 3. Do not open the booklet until you are told to do so. 4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet. 5. Work carefully. Do not spend to much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s). 6. Read each question carefully, and all responses fully, before answering the question. 7. If you have any questions during the exam, raise your hand. 8. References will/will not be provided. 9. Pass mark is 70% 10. You will have 1 x 50 min period to complete this exam. 11. Do you have any questions before you start?

Resources	References
1) Classroom 2) One test per student 3) DS solution for test	A1, A2, A3, A4, A5, A6, A7, A12, A13, B1, B2, B7, B8, B12, B13, B118, B120, B121, B122, B138, B139, B145, B146, B147, B148, B149, B150, B157

Notes to Instructor
N/A

PO 002 Operate Electrical Systems as a Harbour Watchkeeper

1. Performance. (T0019) Operate Electrical Systems as a Harbour Watchkeeper

2. Conditions

   a) Given:

      (1) Submarine;

      (2) Publications and Drawings; and

      (3) Applicable References.

   b) Denied: Assistance.

   c) Environmental: Alongside Submarine.

3. Standard. The submariner shall, IAW references, (T0019) operate submarine electrical systems as a Harbour Watchkeeper to include:

   a) Conducting rounds on the Main Battery System;

   b) Conducting rounds on the AC Electrical System; and

   c) Conducting rounds on the DC Electrical System.

4. References

A1,A2,A3,B14,B15,B16,B17,B18,B19,B20,B21,B30,B31,B32,B33,B34,B35,B39,B40, B41,B42,B43,B44,B45,B46,B48,B49,B50,B51,B52,B53,B54,B55,B117,B118,B120, B151,B152,B155,B163.

5. Task Numbers. T0019

6. Training Limitations. Limited or no access to a submarine. Student exposure may only be limited to shore based trainers and simulators.

EO 002.01 – Conduct Rounds on the Main Battery Systems as a Harbour Watchkeeper

002.01A – Main Battery

Conditions		Sequence
Classroom (S100)		Start PO 002/ After EO 001.03A
Time	**Classification**	**Instructor**
3 x 50 mins	Unclass	AINT, AJTL, AILS, AHPR, AIIX, AIND, AIMM

Learning Objectives	Method
1) Describe the Main Battery 2) Conduct a battery dip 3) Conduct a Battery Shutdown procedure for a battery fire	Interactive Lecture

Scenario	You are a member of a submarine's HPDW and must conduct routine monitoring of the Main Battery / are a member of a submarine's crew and must react to emergencies affecting the Main Battery.
Activity Steps	Instructor will introduce the Main Battery as the ultimate power source for all electrical equipment fitted onboard the submarine. Instructor will describe the construction of the battery compartments and of the batteries themselves, accesses to the compartments, the number of cells comprising each battery, the voltage range of the VPDC system and that variance's effects on ship's equipment. <u>Think pair share</u> Instructor will have the students consider based on what they have learned thus far and suggest safety concerns regarding personnel entering the Battery Compartment. Instructor will identify the legitimate concerns and raise awareness of any that were missed. Instructor will lead students to logical methods of managing exposure to these concerns to include: -Obtaining permission from the proper authority before entering the compartment, -Make the proper warning pipe, -Further precautions to be taken before entry,

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-Percautions while working in the compartment and after exit.

Instructor will further explain procedures for taking a Main Battery Dip, as the most common purpose for entering the compartment, and have students produce a temperature corrected percentage based on various dipped specific gravities. Students to be given an example of Main Battery Log to fill out during this exercise, and practice making verbal reports to the instructor.

Students to have copies of VCEOPs open to VCEOP 13.2.7 in conjunction with a whiteboard drawing of the location of Battery Ventilation valves throughout the submarine; instructor to point out that the procedure listed above will minimize egress of smoke from the Battery Compartments.

Three stars

Instructor will have students write down their 3 main takeaways from the lesson. Instructor should have some students read out their points, and will add any important points students may have missed.

Resources	References
1) White board / dry erase markers 2) 1x DWAN workstation 3) VCEOP 13 reproduced for each student 4) Main Battery logbook 5) EO 002.01A Lesson 6) Hydrometers (or photographs thereof)	A1,A2,A3,B14,B15,B16,B17,B18, B19,B20,B21,B30,B31,B32,B33, B34,B35,B39,B40,B41,B42,B43, B44,B45,B46,B48,B49,B50,B51, B52,B53,B54,B55,B117,B118,B120, B151,B152,B155,B163.

Notes to Instructor

EO 002.01 - Conduct Rounds on the Main Battery Systems as a Harbour Watchkeeper

002.01B – Main Battery Ancillary Systems

Conditions		Sequence
Classroom (S100)		After EO 002.01A
Time	**Classification**	**Instructor**
3 x 50 mins	Unclass	AINT, AJTL, AILS, AHPR, AIIX, AIND, AIMM

Learning Objectives	Method
1) Locate all Hydrogen Detection Units 2) Locate and describe Hydrogen Clearance Fans 3) Locate and describe Battery Exhaust Fans 4) Change-over Battery Ventilation Systems 5) Operate Battery Agitation and explain when required 6) Operate Battery Cooling and explain when required	Interactive Lecture

Scenario	You are a member of a submarine's HPDW and must conduct routine monitoring of the Main Battery.
Activity Steps	Instructor will ask the students to expand on the discussions from the previous lecture concerning safety issues relating to the Main Battery, to consider if there were issues that themselves could be managed, rather than simply managing the exposure to them (i.e. ventilating explosive gases instead of simply minimizing the presence of personnel in the compartment). Instructor will introduce the Hydrogen Detection Units (HDUs) and associated Hydrogen Detection Control Units (HDCUs), describe their functions, and show where their readings can be seen. Instructor will plot their locations on a diagram of the Battery Ventilation system. Instructor will discuss Hydrogen Eliminators and plot their locations on a diagram of the internal layout of the submarine. Instructor will explain the design of maintaining constant airflow in the Battery Compartments and draw the students' attention to the physical differences between the Battery Exhaust Fans and Hydrogen Clearance Fans, in explanation of their functional differences.

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Students should open VCSOP 131 and locate on the tapestry diagram of the MCC the controls required to change-over Battery Ventilation, and explain back to the instructor when/why this would be necessary. Instructor will show video and have students change-over battery ventilation on the virtual submarine. (MCT can be used if available).

Instructor will explain the function of Battery Agitation in the prevention of electrolyte stratification, and its routine usage. Instructor will demonstrate the physical phenomenon that occurs in the agitation system when the submarine experiences reduced atmospheric pressures, and how this is ameliorated by running the agititaion blower during such evolutions.

Instructor is to explain the basic function of cooling systems generally, and the role of the Battery Cooling system specifically. Draw the students attention to their cellular phones and how they may get warm to the touch while charging. Instructor to detail battery parameters and when to run Battery Cooling.

Review

Resources	References
1)White board / dry erase markers 2) 1x DWAN workstation 3) VCSOP 131 reproduced for each student 4) EO 002.01B Lesson	A1,A2,A3,B14,B15,B16,B17,B18, B19,B20,B21,B30,B31,B32,B33, B34,B35,B39,B40,B41,B42,B43, B44,B45,B46,B48,B49,B50,B51, B52,B53,B54,B55,B117,B118,B120, B151,B152,B155,B163.

Notes to Instructor

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EO 002.02 – Conduct Rounds on the Submarine DC Electrical System as a Harbour Watchkeeper

002.02A – Variable Potential Direct Current System

Conditions		Sequence
Classroom (S100)		After EO 002.01B
Time	**Classification**	**Instructor**
2 x 50 mins	Unclass	AINT, AJTL, AILS, AHPR, AIIX, AIND, AIMM

Learning Objectives	Method
1) Describe the function of the Main VPDC System 2) Identify the Fwd/Aft Battery Switchboards 3) Describe how to switch on Secondary Lighting 4) Describe the Control and Indication functions for both the Main and Auxiliary Power Breakers	Interactive Lecture

Scenario	You are a member of submarine's HPDW and must react to losses of electrical power, changes in the state of the electrical systems, and must be capable of monitoring associated indications.
Activity Steps	Instructor will detail the voltage range of the system (call-back to Main Battery lecture). Identify that alongside the submarine is fed from shore supply, point out location of Shore Supply connections on exterior diagram of sub or 3D model. Ask the students, based on what they have learned of submarine construction, to estimate the location of the DC Shore Supply hull penetrations inside the boat; correct as necessary. Instructor will draw the VPDC system on the white board (alternatively show a slide of the simplified system diagram); pair explanations of components with photographs thereof (Battery Switchboards and DCLCs). Instructor will cover equipment fed from DCLCs and show pictures of the equipment. Instructor have students on their own write out equipment found on each load center (5min). After students are complete ask for some answers and display correct lists.

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Using a diagram or photograph of a Battery Switchboard, identify and explain the function of: the Isolating (laying-off) Links, Shore Supply Links, Mid-Point Links, Auxiliary Power Fuse Board (with specific mention of the normal and alternate supplies for the 140kW MGs and DCLCs), Main and Auxiliary Power Breakers, and the Risers. Identify the controls for Secondary Lighting, restrictions on its use, and how/where to identify if conditions are suitable for its use.

Returning to a diagram of the VPDC system, have the students identify the location of the Main and Auxiliary Power Breakers, and identify one such breaker from the photographs of all major VPDC equipment. Draw attention to the size of the unit and point out its distance from watchkeeping positions; ask the students if there would be a logical preference regarding method of operation. Describe the available positions of operation and the implications on system function were local control to be used. Describe additional locations for indications of the breaker status.

(On an internal layout diagram, mark the locations and names of each equipment fed from the DCLCs). Have each student match a machine to the DCLC that normally supplies it, based on their understanding of the design logic. Point out the locations of all “green reset” buttons.

<u>Review (3 Stars)</u>

Instructor asks students to write down the 3 main points they took away from this lesson (3 min). Have some students read out their points.

If not in student points instructor will reiterate the following:
-System redundancy (fwd/aft swbds)
-Battery storage
-Battery is always DC electricity
-Most important equipment has a DC supply, as Our primary electricy source is DC
-Without DC (battery) there is no other electricity

Resources	References

1) EO 002.02A Lesson 2) 3D Submarine Model 3)White board/dry erase markers 4) Photographs of major VPDC components	A1,A2,A3,B14,B15,B16,B17,B18, B19,B20,B21,B30,B31,B32,B33, B34,B35,B39,B40,B41,B42,B43, B44,B45,B46,B48,B49,B50,B51, B52,B53,B54,B55,B117,B118,B120, B151,B152,B155,B163.

Notes to Instructor

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EO 002.02 – Conduct Rounds on the Submarine DC Electrical System as a Harbour Watchkeeper

002.02B – 24 Volt DC System

Conditions		Sequence
Classroom (S100)		After EO 002.02A
Time	**Classification**	**Instructor**
2 x 50 mins	Unclass	AINT, AJTL, AILS, AHPR, AIIX, AIND, AIMM

Learning Objectives	Method
1) Describe the function of the 24VDC System 2) Locate the 24V panels and LP Battery 3) Conduct an Earth Check on the 24VDC System	Interactive Lecture

Scenario	You are a member of submarine's HPDW and must conduct rounds of the 24VDC System IAW Trot Sheet.
Activity Steps	Instructor will describe what 24VDC is used for, and how it is generated onboard; identify that not all services using 24V power are vital to safety, i.e. not needed during emergencies. List all Grade 1 services and point out how their loss would complicate other emergency situations. Introduce the concept of the battery back-up (could be likened to a UPS). Describe the capacity/endurance of the LP Battery and factors that could affect it (VICES usage). On an internal diagram of the submarine (incident board) mark the locations of all major 24VDC System components and describe their function (TRUs, LP Battery, Grade 1 Supply Panel, Grade 2 Distibution Panel). Explain that the system, like other electrical systems, breaks down to individual fuse panels. Utilizing either a photo or diagram of the Grade 1 Supply Panel, describe how to conduct an earth check on the 24V system, and how to interpret the results. Get the students to logic out what should be done in the case of a defect being discovered. Review (Three Star)

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Instructor will asks students to write down their 3 main takeaways from this lesson (3 min). Instructor will have some of the students read out their points.

Resources	References
1) EO 002.02B Lesson 2)White board/dry erase markers 3) Incident board 4) Photos of major 24VDC equipment 5) Copy of Trot Sheet	A1,A2,A3,B14,B15,B16,B17,B18, B19,B20,B21,B30,B31,B32,B33, B34,B35,B39,B40,B41,B42,B43, B44,B45,B46,B48,B49,B50,B51, B52,B53,B54,B55,B117,B118,B120, B151,B152,B155,B163.

Notes to Instructor

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EO 002.03 – Conduct Rounds on the Submarine AC Electrical System as a Harbour Watchkeeper

002.03A – 440 Volt AC System

Conditions		Sequence
Classroom (S100)		After EO 002.02B
Time	**Classification**	**Instructor**
3 x 50 mins	Unclass	AINT, AJTL, AILS, AHPR, AIIX, AIND, AIMM

Learning Objectives	Method
1) Locate and describe the 440VAC System components 2) Describe Emergency Interconnecting Cables 3) Locate major equipment and starters fed by the 440VAC System 4) Locate the Ventilation Control Panel and describe what items are operated from the panel	Interactive Lecture

Scenario	You are a member of a submarine's HPDW and must be capable of resetting all major equipment onboard; maintaining the habitability of the submarine; and assisting as a member of crew with emergency procedures.
Activity Steps	The instructor is to explain the 440VAC diagram to the class, include functions of 140kW Motor-Generator Sets (including starters), AC Shore Supply Hull Connection and Hull Penetrator, AC Shore Supply Link Box, Fwd/Aft AC Switchboards, and AC Load Centres. Instructor to introduce Emergency Interconnectors as functionally similar to surface ship casualty power, excepting that individual equipments are not fed directly, rather the entire LC is energized from its opposite unit. This necessitates making considerations regarding electrical loading and priorities for restoration of power. Instructor will detail parameters and procedures for use of the cables. Utilizing an internal diagram of the submarine (incident board), the instructor will mark the locations of major 440VAC equipment (including starter panels) throughout the boat including: Ship's Stores Refrigeration System compressors, Trim Pumps, 24VDC Transformer Rectifier Units, Battery Agitation Blower, Main Hydraulic Pump, External Hydraulic Pump, Steering & Hydroplanes AC

Hydraulic Pump, Galaxy Static Frequency Converters, Mk 36 Static Frequency Converters, Air Treatment Units, Ship's Exhaust Fan, MR/ER Axial Flow Fan, LP Bilge Pump, Chilled Water System Compressors, Ship's Fresh Water Cooling Pumps, Ship's Sea Water Cooling Pump, Battery Cooling Pump. Also identify the locations of the Galley 440VAC Isolation Switch, and the Ventilation Control Panel.

Using a projected photograph of the Ventilation Control Panel, identify it as the remote control for both ATUs, the Ship's Exhaust Fan, and the Ventilation Heaters. Describe how to reset the panel, including how/why the heaters must be reset after changing speed of ATU2, and the purpose of that interlock.

Review

Resources

1) EO 002.03 Lesson
2) White board/dry erase markers
3) 440VAC System diagram handout/slide
4) Incident board
5) Photographs of major 440VAC equipment.

References

A1,A2,A3,B14,B15,B16,B17,B18, B19,B20,B21,B30,B31,B32,B33, B34,B35,B39,B40,B41,B42,B43, B44,B45,B46,B48,B49,B50,B51, B52,B53,B54,B55,B117,B118,B120, B151,B152,B155,B163.

Notes to Instructor

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EC 002.03 - Operate Submarine Electrical Systems as a Harbour Watchkeeper

Scope	Conditions	Sequence
EO 002.01-03	Classroom	24 Hrs After EO 002.03A
Time	**Classification**	**Instructor**
1 x 50 min	Unclass	AILS, AIND, AINA, AILQ

Scenario	N/A
Assessment Steps	Will be a written exam based on content from PO 004. Exam shall contain short answer questions and may contain multiple choice. INSTRUCTIONS GIVEN TO TRAINEE 1. EC 003.03 will be a written test based on content from EO 003.01, 003.02 and 003.03. 2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam. 3. Do not open the booklet until you are told to do so. 4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet. 5. Work carefully. Do not spend to much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s). 6. Read each question carefully, and all responses fully, before answering the question. 7. If you have any questions during the exam, raise your hand. 8. References will/will not be provided. 9. Pass mark is 70% 10. You will have 1 x 50 min period to complete this exam. 11. Do you have any questions before you start?

Resources	References
1) Classroom 2) One test per student 3) DS Solution for test	A1,A2,A3,B14,B15,B16,B17,B18, B19,B20,B21,B30,B31,B32,B33, B34,B35,B39,B40,B41,B42,B43, B44,B45,B46,B48,B49,B50,B51,

B52,B53,B54,B55,B117,B118,B120,
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Notes to Instructor
N/A

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PO 003 Operate Air Systems as a Harbour Watchkeeper

1. Performance. (T0005) Operate Air Systems as a Harbour Watchkeeper

2. Conditions

   a) Given:

      (1) Submarine;

      (2) Publications and Drawings; and

      (3) Applicable References.

   b) Denied: Assistance.

   c) Environmental: Alongside Submarine.

3. Standard. The submariner shall, IAW references, (T0005) operate air systems as a Harbour Watchkeeper, to include:

   a) Operate the HP Air System:

      (1) Conducting Rounds;

      (2) Charging the HP Air System;

      (3) Emergency Blow; and

      (4) Normal Blow.

   b) Operate the LP Air System:

      (1) Auxiliary Vent & Blow (AV&B);

      (2) Inflate shaft seal;

      (3) EBS;

      (4) BIBS;

      (5) VEX System; and

(6) Induction System.

4. References

A1,A2,A3,B56,B57,B58,B59,B60,B61,B62,B63,B64,B65,B66,B67,B68,B69,B70,B71, B72,B117,B118,B120,B130,B132,B160,B164,B171.

5. Task Numbers. T0005

6. Training Limitations. Limited or no access to a submarine. Student exposure may be limited to shore based trainers and simulators.

EO 003.01 – Operate HP Air System as a Harbour Watchkeeper

003.01A – Basics of HP Air System

Conditions		Sequence
Classroom (S100)		Start PO 003/After EO 002.03A
Time	**Classification**	**Instructor**
4 x 50 mins	Unclass	AILS, AIND, AINA, AILQ

Learning Objectives	Method
1) Function of the Air Systems 2) Components of the HP Air Systems to include; - HP Air Bottle Groups -GIVs -Ring Main Integrity Valves -Bulkhead Isolating Valves -HP Air Compressors -Normal Blows -Emergency Blows	Think-Pair-Share Problem Solving Matching Interactive lecture 3 Stars

Scenario	NA
Activity Steps	Instructor will explain that the function of the HP Air System is to compress, store, process and distribute clean dry air at 280 bar (2 min). <u>Think-Pair-Share</u> Instructor will ask students, on their own and without talking, to list what they think the main components of the HP Air system would be on a sheet of paper. Instructor will let the students know that it does not matter if they get any correct this is just to get them thinking about the system (2min). Instructor will ask students to pair with another student to discuss and revise their list (3 min). Instructor will ask if any group thinks they have a correct list and have them list it on a white board. Instructor will correct the list on board. - HP Air Bottle Groups

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-GIV's
-Ring Main Integrity Valves
-Bulkhead Isolating Valves
-HP Air Compressors
-Normal Blows
-Emergency Blows

Problem Solving (Build System)

Instructor will add the corresponding technical symbols next to the components and ask students individually to combine the components into an operation system diagram to the best of their ability. Instructor can allow some class discussion but each student should attempt their own diagram (10 min).

Instructor will monitor students and correct as required.

Once students have had 10 mins to attempt their own the instructor will draw a simplified system diagram on the board and explain the function, HP Air flow and answer questions.

After presenting the simplified system, the instructor will ask students if they remember from the start of class what is supplied by HP Air system and associated HP Air cross-connects. As students give examples the instructor will add them to the diagram on the board until the complete submarine diagram is presented.

Instructor can then present the official diagram for students to see.

Matching

Instructor displays numbered photos of HP Air system components with a corresponding list of number component names. Students are to match the components with the photo on a sheet of paper (5 min).

Instructor will monitor progress.

Instructor will review answers.

Virtual Tour of System:
Instructor will show students location of components (using Virtual Submarine, 3D Sub model, or internal submarine diagram)

Review (3 Star)

Instructor will have students write down their 3 main points from the lesson (3 min). Instructor will have some students read out their points.

If not covered in the students points, the instructor will review the following:
-Cross Connections
-Location of HPA Compressors
-Bottle Groups

Resources	References
1)3D Submarine (if available) 2) Paper and pen/pencil 3)One computer with DWAN access 4) Whiteboard 5) EO 004.01 lesson	A1,A2,A3,B56,B57,B58,B59,B60, B61,B62,B63,B64,B65,B66,B67, B68,B69,B70,B71,B72,B117,B118, B120,B130,B132,B160,B164,B171.

Notes to Instructor

Material will be assessed in EC 004.03.

EO 003.01 - Operate HP Air System as a Harbour Watchkeeper

003.01B – Operation of HP Air system

Conditions		Sequence
Classroom		After EO 003.01A
Time	**Classification**	**Instructor**
2 x 50 mins	Unclass	AILS, AIND, AINA, AILQ

Learning Objectives	Method
1) Operate the HP Air system to include; -HP Air charge -Normal and Emergency Blows	Unstructured Brainstorm Interactive Lecture Think-Pair-Share

Scenario	You are on watch as a harbour watchkeeper. You are tasked with conducting an HP Air Charge with use of both HP Air compressors/shore charge IAW VCSOPs. On completion you will need to restore system to SOP line-up.
Activity Steps	Instructor reads scenario <u>Brainstorm (Unstructured)</u> Instructor asks class to come up with safety considerations prior to operating this system. Instructor writes all suggestions on board (5 min). Once students are done instructor can eliminate suggestion not relevant to this system and highlights/discusses important safety considerations such as: -Before starting any equipment, there are pre-start checks that must be conducted. For HPAC (High Pressure Air Compressor) ensure all pre-start checks are complete, to include: -Fourth stage column drain is open -Power supplies available -Starter selected to remote -Oil level -Sea water cooling as well as freshwater cooling must be running prior to flash up of HPAC's

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Additional considerations for an Air Charge:
-Associated trips with HPAC's
-Monitor system pressure while air charge is ongoing

Interactive Lecture

Instructor will explain HP Air system operation in relation to Normal and Emergency Blows. (2 mins)

Instructor will cover the basic principles of Normal and Emergency Blow operation to include:
-Control
-Indication
-Limitations
-Safety

Think-Pair-Share

Instructor will ask students on their own to list **components** of the HP Air system and associated **cross-connects** (2 min). Then with a partner revise their lists (3 min).

Instructor will review list and reiterate the increase in knowledge from before the class.

Resources	References
1) Whiteboard 2) Paper and pen/pencil 3) One computer with DWAN access 4) 3D Submarine (if available) 5) EO 004.01 Lesson	A1,A2,A3,B56,B57,B58,B59,B60, B61,B62,B63,B64,B65,B66,B67, B68,B69,B70,B71,B72,B117,B118, B120,B130,B132,B160,B164,B171.

Notes to Instructor

Material will be assessed in EC 004.03. Operation will be practiced and assessed on OJPR.

EO 003.02 – Operate the LP Air System as a Harbour Watchkeeper

003.02A – Basics of the AV&B system

Conditions		Sequence
Classroom (S100)		After EO 003.01B
Time	**Classification**	**Instructor**
2 x 50 mins	Unclass	AILS, AIND, AINA, AILQ

Learning Objectives	Method
1) Function 2) Components	Think-Pair-Share Problem Solving Matching

Scenario	NA
Activity Steps	Instructor will explain what AV&B is used for on the submarine, to include: -System Function -System Components -System Operation -Control and Indication <u>Think-Pair-Share</u> Instructor will ask students, on their own and without talking, to list what they think the main components of the AV&B system would be on a sheet of paper. Instructor will let the students know that it does not matter if they get any correct this is just to get them thinking about the system (2min). Instructor will ask students to pair with another student to discuss and revise their list (3 min). Instructor will ask if any group thinks they have a correct list and have them list it on a white board. Instructor will correct the list on board <u>Problem Solving (Build System)</u> Instructor will add the corresponding technical symbols next to the components and asks students individually to combine the components into an operation

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system diagram to the best of their ability. Instructor can allow some class discussion but each student should attempt their own diagram (10 min).

Instructor will monitor students and correct as required.

Once students have had 10 mins to attempt their own, the instructor will draw a simplified system diagram on the board and explain the function, air flow, and then answer questions.

After presenting the simplified system, the instructor will ask students if they remember from the start of class what is supplied by the AV&B system. As students give examples the instructor will add them to the diagram on the board until the complete submarine diagram is presented.

Instructor can then present the official/engineering diagram for students to see.

Matching

Instructor displays numbered photos of the AV&B system components with a corresponding list of numbered component names. Students are to match the components with the photo on a sheet of paper (5 min).

Instructor will monitor progress.

Instructor will review answers.

Virtual tour of HPA system:
Instructor will show students location of components (using Virtual Submarine, 3D Sub model, or internal submarine diagram)

Provide students time to navigate Virtual Submarine and put on an Air Charge.

Resources	References
1)Slide Set cards 2) Projector 3)DWAN Access	A1,A2,A3,B56,B57,B58,B59,B60, B61,B62,B63,B64,B65,B66,B67, B68,B69,B70,B71,B72,B117,B118,

4)Tablets with Virtual Sub B120,B130,B132,B160,B164,B171.

Notes to Instructor

Theory will be assessed in EC 003.03. Practical demonstration and assessment during OJPR.

EO 003.02 - Operate the LP Air System as a Harbour Watchkeeper

003.02B- Operation of the AV&B System

Conditions		Sequence
Classroom (S100)		After EO 003.02A
Time	**Classification**	**Instructor**
2 x 50 mins	Unclass	AILS, AIND, AINA, AILQ

Learning Objectives	Method
1)Operate the AV&B System to include: -Operation of Vent & Blow valves -Putting a blow on M Port -Monitor AV&B pressure from surveillance and locally	Unstructured Brainstorm Interactive Lecture 3 Stars

Scenario	You are on watch as a harbour watchkeeper. Line up trim for firefighting from M port/ Stbd Aft Trim IAW VCSOPs. On completion restore system to SOP line-up.
Activity Steps	<u>Brainstorm (Unstructured)</u>: Instructor asks class to come up with safety considerations prior to operating this system. Instructor writes all suggestions on board (5 min). Once students are done instructor can eliminate suggestions not relevant to this system and highlight/discusses important safety considerations such as: - Combined Vent and Blow cocks are to be left in the VENT position unless the appropriate tank is being blown. -Ensure vent is shut before putting blow on tank -Appropriate warning pipes are made -Appropriate PPE is used while operating AV&B and associated systems <u>Interactive Lecture:</u> Instructor will explain monitoring the AV&B system as a harbour watchkeeper. (2 mins)

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Instructor will cover the basic principles of monitoring the AV&B system to include:
-Control
-Indication
-Limitations
-Safety

<u>3 Stars:</u>
-Line up for Firefighting
-Warning Pipes
-Precautions

Resources	References
1)Whiteboard	1)
2) Paper and pen/pencil	2)
3)One computer with DWAN access	
4) 3D Submarine (if available)	
5) EO 004.01 Lesson	

Notes to Instructor

Material will be assessed in EC 004.03. Operation will be practiced and assessed on OJPR

EO 003.02 - Operate the LP Air System as a Harbour Watchkeeper

003.02C – Basics of the LP Blow System

Conditions		Sequence
Classroom S100		After EO 003.02B
Time	**Classification**	**Instructor**
2 x 50 mins	Unclass	AILS, AIND, AINA, AILQ

Learning Objectives	Method
1) Function 2) Components	Think-Pair-Share Problem Solving Matching

Scenario	N/A
Activity Steps	Instructor will list the components of the LP Blow System. Instructor will explain that the LP Blow system is used for the following: - To enable the submarine to achieve full buoyancy - To Purge / Ventilate various compartments - To test watertight integrity by carrying out a Vacuum test - Surfacing the submarine on the LP blower Think-Pair-Share Instructor will ask students, on their own and without talking, to list what they think the main components of the AV&B system would be on a sheet of paper. Instructor will let the students know that it does not matter if they get any correct this is just to get them thinking about the system (2min). Instructor will ask students to pair with another student to discuss and revise their list (3 min). Instructor will ask if any group thinks they have a correct list and have them list it on a white board. Instructor will correct the list on board

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Problem Solving (Build System)

Instructor will add the corresponding technical symbols next to the components, and ask students individually to combine the components into an operation system diagram to the best of their ability. Instructor can allow some class discussion but each student should attempt their own diagram (10 min).

Instructor will monitor students and correct as required.

Once students have had 10 mins to attempt their own, the instructor will draw a simplified system diagram on the board and explain the function, air flow, and then answer questions.

After presenting the simplified system, the instructor will ask students if they remember from the start of class what is supplied by the AV&B system. As students give examples the instructor will add them to the diagram on the board until the complete submarine diagram is presented.

Instructor can then present the official/engineering diagram for students to see.

Matching

Instructor displays numbered photos of the AV&B system components with a corresponding list of numbered component names. Students are to match the components with the photo on a sheet of paper (5 min).

Instructor will monitor progress.

Instructor will review answers.

Instructor will show students location of components (using virtual submarine, 3D Sub model, or internal submarine diagram)

Resources	References
1)Slide Set cards 2) Projector 3)DWAN Access 4)Tablets with Virtual Sub	A1,A2,A3,B56,B57,B58,B59,B60, B61,B62,B63,B64,B65,B66,B67, B68,B69,B70,B71,B72,B117,B118, B120,B130,B132,B160,B164,B171.

Notes to Instructor

EO 003.02 - Operate the LP Air System as a Harbour Watchkeeper

003.02D – Operation of the LP Blow System

Conditions	Sequence
Classroom S100	After EO 003.02C

Time	Classification	Instructor
1 x 50 mins	Unclass	AILS, AIND, AINA, AILQ

Learning Objectives	Method
1) Operation of the LP Blow System to include: -Operation and Pre-Starts of the LP Blower -Operation of the VEX System -Operation of the Induction System	Interactive Lecture

Scenario	You are on watch as a harbour watchkeeper. Achieve maximum buoyancy (Blow A Round) IAW VCSOPs. Systems must be returned to SOP lineup on completion.
Activity Steps	Interactive Lecture With aid of the Virtual Submarine Program, instructor will demonstrate location of equipment related to blowing a round. Instructor will also demonstrate proper procedure for blowing a round, draining down the VEX system, and draining down the Induction system. Proper voice procedures will also be demonstrated during the interactive lecture. <u>3 Stars:</u> -LP Blower pre-starts -Drain down and open LP Master Blows - etc. etc.

Resources	References

1)Slide Set cards	A1,A2,A3,B56,B57,B58,B59,B60,
2) Projector	B61,B62,B63,B64,B65,B66,B67,
3)DWAN Access	B68,B69,B70,B71,B72,B117,B118,
4)Tablets with Virtual Sub	B120,B130,B132,B160,B164,B171.

Notes to Instructor

NTDC(A) Approval – 21/10/2020
NPTG HQ Approval 24/02/2020
Version No: 2.1
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EC 003.02 - Operate Submarine Air Systems as a Harbour Watchkeeper

Scope	Conditions	Sequence
EO 003.01-02	Classroom	24 Hrs After EO 003.02D
Time	**Classification**	**Instructor**
1 x 50 mins	Unclass	AILS, AIND, AINA, AILQ

Scenario	N/A
Assessment Steps	Will be a written exam based on content from PO 003. Exam shall contain short answer questions and may contain multiple choice. Suggested breakdown of exam as follows: HP Air System - 50% LP Air System - 50% INSTRUCTIONS GIVEN TO TRAINEE 1. EC 003.02 will be a written test based on content from EO 003.01 and 003.02. 2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam. 3. Do not open the booklet until you are told to do so. 4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet. 5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s). 6. Read each question carefully, and all responses fully, before answering the question. 7. If you have any questions during the exam, raise your hand. 8. References will/will not be provided. 9. Pass mark is 70% 10. You will have 1 x 50 mins period to complete this exam. 11. Do you have any questions before you start?

Resources	References

1) Classroom 2) One test per student 3) DS Solution for test	A1,A2,A3,B56,B57,B58,B59,B60, B61,B62,B63,B64,B65,B66,B67, B68,B69,B70,B71,B72,B117,B118, B120,B130,B132,B160,B164,B171.

Notes to Instructor
N/A

PO 004 Operate Submarine Cooling Systems as a Harbour Watchkeeper

1. Performance. (T0020) Operate Submarine Cooling Systems as a Harbour Watchkeeper

2. Conditions

   d) Given:

      (1) Submarine;

      (2) Publications and Drawings; and

      (3) Applicable References.

   e) Denied: Assistance.

   f) Environmental: Alongside Submarine.

3. Standard. The submariner shall, IAW references, (T0020) operate submarine cooling systems as a Harbour Watchkeeper to include:

   a) Sea Water System;

   b) Fresh Water Systems; and

   c) Chilled Water System.

4. References

A1,A2,A3,B47,B74,B75,B76,B77,B78,B79,B106,B108,B109,B110,B111,B112,B113, B114,B115,B116,B117,B118,B120,B160,B161,B164,B170.

5. Task Numbers. T0020.

6. Training Limitations. Limited or no access to a submarine. Student exposure may be limited to shore based trainers and simulators

EO 004.01 – Operate Sea Water Cooling System as a Harbour Watchkeeper

004.01A – Basics of Sea Water Cooling

Conditions		**Sequence**
Classroom (S100)		Start PO 004/ After EO 003.02D
Time	**Classification**	**Instructor**
2 x 50 mins	Unclass	AILS, AIND, AINA, AILQ

Learning Objectives	**Method**
1) Function of the Sea Water Cooling System 2) Components of the Sea Water Cooling System to include: - Battery Cooling Heat Exchanger SW 229 -FW/SW Heat Exchanger SW 218/219 -SAN System SW 402 to include Relief Valve.	Think-Pair-Share Problem Solving Matching 3 Stars

Scenario	NA
Activity Steps	<u>Think-Pair-Share</u> Instructor will ask students, on their own and without talking, to list what they think the main components of a sea water cooling system would be on a sheet of paper. Instructor will let the students know that it does not matter if they get any correct this is just to get them thinking about the system (2min). Instructor will ask students to pair with another student to discuss and revise their list (3 min). Instructor will ask if any group thinks they have a correct list and have them list it on a white board. Instructor will correct the list on board and briefly explain. -Sea Water Inlet/ Hull and Backup Valves -Sea Water Outlet/Hull and Backup Valves -Isolation Valves -Pumps -Heat Exchangers -Strainers

Interactive Lecture (Power point)

Lecture will include why the submarine has sea water cooling and what it cools.

Problem Solving (Build System)

Instructor will add the corresponding technical symbols next to the components and asks students individually to combine the components into an operation system diagram to the best of their ability. Instructor can allow some class discussion but each student should attempt their own diagram (10 min).

Instructor will monitor students and correct as required.

Once students have had 10mins to attempt their own the instructor will draw a simplified system diagram on the board and explain the function, why components are where they are, show water flow, and answer questions.

After presenting the simplified system, the instructor will ask students if they remember from the start of class what is cooled by the sea water. As students give examples the instructor will add them to the diagram on the board until the complete submarine diagram is presented.

Instructor can then present the official diagram for students to see.

Matching

Instructor displays numbered photos of sea water cooling components with a corresponding list of number component names. Students are to match the components with the photo on a sheet of paper (5 min).

Instructor will monitor progress.

Instructor will review answers.

Instructor will show students location of components (using Virtual Submarine, 3D Sub model, or internal submarine diagram)

Stars:

Instructor will ask students to list on a paper the three main takeaways they got from this lesson. (5 min)

Points the instructor may want to reiterate.

-Mimic Panel Operation (MCC)
-Equipment cooled by Sea Water Cooling
-Components

Resources	References
1)3D Submarine (if available) 2) Paper and pen/pencil 3)One computer with DWAN access 4) Whiteboard 5) EO 004.01 lesson	A1,A2,A3,B47,B74,B75,B76,B77, B78,B79,B106,B108,B109,B110, B111,B112,B113,B114,B115,B116, B117,B118,B120,B160,B161,B164, B170.

Notes to Instructor
Material will be assessed in EC 004.03.

EO 004.01 - Operate Sea Water Cooling System as a Harbour Watchkeeper

004.01B – Operation of Sea Water Cooling

Conditions		**Sequence**
Classroom		After EO 004.01A
Time	**Classification**	**Instructor**
1 x 50 mins	Unclass	AILS, AIND, AINA, AILQ

Learning Objectives	**Method**
1) Operate the Seawater System to include: -Conducting Rounds -Control -Indication -Abnormal Line-Up -Limitations -Safety	Unstructured Brainstorm Interactive Lecture Think-Pair-Share

Scenario	You are on watch as a harbour watchkeeper and divers are going to be operating in the vicinity of the submarine. Part of the diver down procedure is to tag out the sea water cooling inlet and outlet hull and backup valves. On completion you will need to restore system to SOP line-up.
Activity Steps	Instructor reads scenario <u>Interactive Lecture</u> Instructor will explain sea water cooling system operation in relation to tagging out and restoring system as part of dive operations as a harbour watchkeeper using VCSOPs. Instructor will cover the basic principles of sea water cooling system operation to include: -Control -Indication -Abnormal Line-Up -Limitations

-Safety

Think-pair-share

Instructor will ask students on their own to list **components** of the Sea Water Cooling system and **what is cooled** by sea water (2 min). Then with a partner revise their lists (3 min).

Instructor will review list and reiterate the increase in knowledge from before the class.

Resources	References
1)Whiteboard 2) Paper and pen/pencil 3)One computer with DWAN access 4)Projector 5) EO 004.01 Lesson	A1,A2,A3,B47,B74,B75,B76,B77, B78,B79,B106,B108,B109,B110, B111,B112,B113,B114,B115,B116, B117,B118,B120,B160,B161,B164, B170.

Notes to Instructor
Material will be assessed in EC 004.03. Operation will be practiced and assessed on OJPR.

EO 004.02 – Operate Fresh Water Cooling System as a Harbour Watchkeeper

004.02A – Basics of Fresh Water Cooling

Conditions		**Sequence**
Classroom (S100)		After EO 004.01B
Time	**Classification**	**Instructor**
2 x 50 mins	Unclass	AILS, AIND, AINA, AILQ

Learning Objectives	**Method**
1)Function and Services Supplied 2) Components - Pumps -Tanks -Heat Exchanger -Shore Connection	Think-Pair-Share Problem Solving Matching

Scenario	NA
Activity Steps	Instructor will explain that fresh water is used for cooling on the submarine. What it is used to cool, to include: a. Fwd 140 KW MG Air Cooler AW 603 b. Chilled Water Plant Condensers AW 409 / 410 c. Aft 140 KW MG Air Cooler AW 215 d. Port / Stbd HP Air Compressors AW 207 / 208 <u>Think-Pair-Share</u> Instructor will ask students, on their own and without talking, to list what they think the main components of a fresh water cooling system would be on a sheet of paper. Instructor will let the students know that it does not matter if they get any correct, this is just to get them thinking about the system (2min). Instructor will ask students to pair with another student to discuss and revise their list (3 min).

Instructor will ask if any group thinks they have a correct list and have them list it on a white board.
Instructor will correct the list on board

Problem Solving (Build System)

Instructor will add the corresponding technical symbols next to the components and asks students individually to combine the components into an operation system diagram to the best of their ability. Instructor can allow some class discussion but each student should attempt their own diagram (10 min).

Instructor will monitor students and correct as required.

Once students have had 10 mins to attempt their own, the instructor will draw a simplified system diagram on the board and explain the function, water flow, and answer questions.

After presenting the simplified system, the instructor will ask students if they remember from the start of class what is cooled by the fresh water. As students give examples the instructor will add them to the diagram on the board until the complete submarine diagram is presented.

Instructor can then present the official/engineering diagram for students to see.

Matching

Instructor displays numbered photos of the fresh water cooling components with a corresponding list of numbered component names. Students are to match the components with the photo on a sheet of paper (5 min).

Instructor will monitor progress.

Instructor will review answers.

Instructor will show students location of components (using Virtual Submarine, 3D Sub model, or internal submarine diagram)

Resources	References
1) EO 004.01 Lesson 2) Projector 3)DWAN Access 4)Tablets with Virtual Sub 5) 3D Submarine Model if available	A1,A2,A3,B47,B74,B75,B76,B77, B78,B79,B106,B108,B109,B110, B111,B112,B113,B114,B115,B116, B117,B118,B120,B160,B161,B164, B170.

Notes to Instructor
Theory will be assessed in EC 004.03. Practical demonstration and assessment during OJPR.

EO 004.02 - Operate Fresh Water Cooling System as a Harbour Watchkeeper

004.02B- Operation of Fresh Water Cooling System

Conditions		**Sequence**
Classroom (S100)		After EO 004.02A
Time	**Classification**	**Instructor**
1 x 50 mins	Unclass	AILS, AIND, AINA, AILQ

Learning Objectives	**Method**
1)Operate Fresh Water Cooling System to include: -Conducting rounds -Control -Indication -Abnormal Line-Ups -Prerequisites -Limitations -Safety	Unstructured Brainstorm Interactive Lecture Think-Pair-Share

Scenario	You are on watch as a harbour watchkeeper in Halifax harbour and the HPA bottles are low and require an air charge.
Activity Steps	Instructor reads scenario <u>Interactive Lecture</u> Instructor will explain fresh water cooling system operation in relation to conduting an air charge as a harbour watchkeeper using VCSOPs. Instructor will cover the basic principles of fresh water cooling system operation to include: -Control (show controls on MCC tapestry/diagram) -Indication -Abnormal Line-Up -Limitations -Safety -Topping up Expansion Tank

Think-pair-share

Instructor will ask students on their own to list components of the Fresh Water Cooling system and what is cooled by fresh water (2 min). Then with a partner revise their lists (3 min).

Instructor will review list and reiterate the increase in knowledge from before the class.

Resources	References
1)Whiteboard 2) Paper and pen/pencil 3)One computer with DWAN access 4)Projector 5) EO 004.01 Lesson	A1,A2,A3,B47,B74,B75,B76,B77, B78,B79,B106,B108,B109,B110, B111,B112,B113,B114,B115,B116, B117,B118,B120,B160,B161,B164, B170.

Notes to Instructor
Theory will be assessed in EC 004.03. Practical demonstration and assessment during OJPR.

EO 004.03 – Operate Chilled Water Cooling System as a Harbour Watchkeeper

004.03A – Basics of Chilled Water Cooling

Conditions		**Sequence**
Classroom		After EO 004.02B
Time	**Classification**	**Instructor**
2 x 50 mins	Unclass	AILS, AIND, AINA, AILQ

Learning Objectives	**Method**
1) Function of the Chilled Water Cooling System 2) Components of the Chilled Water Cooling System to include; - Pumps -Tank -Shore Connection -Services Supplied	Think-Pair-Share Problem Solving Matching

Scenario	N/A
Activity Steps	Instructor will explain what chilled water is and what is supplied by the chilled water to include: a. Ships Refrigeration Plant b. ATU's c. Main Hydraulic Plant Rep Tank Cooler d. External Hydraulic Plant Rep Tank Cooler e. Steering & Hydroplanes Hydraulic Plant Rep Tank Cooler f. Fwd Trim Pump <u>Think-Pair-Share</u> Instructor will ask students, on their own and without talking, to list what they think the main components of a chilled water cooling system would be on a sheet of paper. Instructor will let the students know that it does not matter if they get any correct. This is just to get them thinking about the system (2min). Instructor will ask students to pair up with another student to discuss and revise their list (3 min).

Instructor will ask if any group thinks they have a correct list and have them list it on a white board.
Instructor will correct the list on board

Problem Solving (Build System)

Instructor will add the corresponding technical symbols next to the components and asks students individually to combine the components into an operation system diagram to the best of their ability. Instructor can allow some class discussion but each student should attempt their own diagram (10 min).

Instructor will monitor students and correct as required.

Once students have had 10 mins to attempt their own, the instructor will draw a simplified system diagram on the board, then explain function, water flow and answer questions.

After presenting the simplified system, the instructor will ask students if they remember from the start of class what is cooled by the chilled water. As students give examples, the instructor will add them to the diagram on the board until the complete submarine diagram is presented.

Instructor can then present the official/engineering diagram for students to see.

Matching

Instructor displays numbered photos of the chilled water cooling components with a corresponding list of numbered component names. Students are to match the components with the photo on a sheet of paper (5 min).

Instructor will monitor progress.

Instructor will review answers.

Instructor will show students location of components (using Virtual Submarine, 3D Sub model, or internal submarine diagram)

Resources	References
1) EO 004.03 Lesson 2) Projector 3)DWAN Access 4)Tablets with Virtual Sub 5) 3D Submarine Model if available	A1,A2,A3,B47,B74,B75,B76,B77, B78,B79,B106,B108,B109,B110, B111,B112,B113,B114,B115,B116, B117,B118,B120,B160,B161,B164, B170.

Notes to Instructor
Theory will be assessed in EC 004.03. Practical demonstration and assessment during OJPR.

EO 004.03 - Operate Chilled Water Cooling System as a Harbour Watchkeeper

004.03B – Operate Chilled Water Cooling System

Conditions		**Sequence**
Classroom		After EO 004.03A
Time	**Classification**	**Instructor**
1 x 50 mins	Unclass	AILS, AIND, AINA, AILQ

Learning Objectives	**Method**
1) Conduct rounds on AC Plants 2) Top up chilled water expansion tank to include reading the sight glass	Unstructured Brainstorm Interactive Lecture Think-Pair-Share

Scenario	On duty as a harbour watchkeeper alongside Halifax and shore chillers are unavailable. Submarine is required to run onboard chillers and watchkeeper must do rounds on the system.
Activity Steps	Instructor reads scenario <u>Interactive Lecture</u> Instructor will explain chilled water cooling system rounds as a harbour watchkeeper using VCSOPs including topping up the chilled water expansion tank. Instructor will cover the basic principles of chilled water cooling system operation to include: -Control -Indication -Abnormal Line-Up -Limitations -Safety -When to top up Expansion Tank -How to read the Sight Glass <u>Think-pair-share</u>

Instructor will ask students on their own to list components of the Chilled Water Cooling system and what is cooled by chilled water (2 min). Then with a partner revise their lists (3 min).

Instructor will review list and reiterate the increase in knowledge from before the class.

Resources	References
1) EO 004.03 Lesson 2) Projector 3) DWAN Access 4) Tablets with Virtual Sub	A1,A2,A3,B47,B74,B75,B76,B77, B78,B79,B106,B108,B109,B110, B111,B112,B113,B114,B115,B116, B117,B118,B120,B160,B161,B164, B170.

Notes to Instructor
Theory will be assessed in EC 004.03. Practical demonstration and assessment during OJPR.

EC 004.03 - Operate Submarine Cooling Systems as a Harbour Watchkeeper

Scope	Conditions	Sequence
EO 004.01-03	Classroom	24 Hrs After EO 004.03A
Time	**Classification**	**Instructor**
1 x 50 mins	Unclass	AILS, AIND, AINA, AILQ

Scenario	N/A
Assessment Steps	Will be a written exam based on content from PO 004. Exam shall contain short answer questions and may contain multiple choice. Suggested breakdown of exam as follows: Fresh Water Cooling System – 33% Salt Water Cooling System – 33% Chilled Water Cooling System – 33% INSTRUCTIONS GIVEN TO TRAINEE 1. EC 004.03 will be a written test based on content from EO 004.01, 004.02 and 004.03. 2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam. 3. Do not open the booklet until you are told to do so. 4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet. 5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s). 6. Read each question carefully, and all responses fully, before answering the question. 7. If you have any questions during the exam, raise your hand. 8. References will/will not be provided. 9. Pass mark is 70% 10. You will have 1 x 50 min period to complete this exam. 11. Do you have any questions before you start?

Resources	References
1) Classroom 2) One test per student 3) DS solution for test	1) 2)

Notes to Instructor
N/A

Limitations	Practical component will be assessed on OJPR
Impact of Failure	IAW NPTORD 4500-54

PO 005 Operate Trim, Bilge and Ballast Systems as a Harbour Watchkeeper

1. Performance. (T0007) Operate Trim, Bilge and Ballast Systems as a Harbour Watchkeeper

2. Conditions

   a) Given:

      (1) Submarine;

      (2) Publications and Drawings; and

      (3) Applicable References.

   b) Denied: Assistance.

   c) Environmental: Alongside Submarine.

3. Standard. The submariner shall, IAW references, (T0007) operate Trim, Bilge and Ballast Systems as a Harbour Watchkeeper to include:

   a) Trim System;

   b) Bilge System; and

   c) Ballast System.

4. References

A1,A2,A3,B89,B90,B91,B92,B93,B94,B95,B96,B97,B98,B99,B100,B102,B103,B104, B105,B117,B118,B120,B137,B160,B172.

5. Task Numbers. T0007

6. Training Limitations. Limited or no access to a submarine. Student exposure may be limited to shore based trainers and simulators.

EO 005.01 – Operate Trim System as a Harbour Watchkeeper

005.01A – Basics of Trim System

Conditions		**Sequence**
Classroom (S100)/Trainer		Start PO 005/After EO 004.03B
Time	**Classification**	**Instructor**
4 x 50 mins	Unclass	AILS, AIND, AINA, AILQ

Learning Objectives	**Method**
1)Function of the Trim System -Trim -Firefighting Line Up -Garbage Ejector Wash Down -WSC Spray 2) Components of the Trim System to include: -Pumps -System Valves -Cross Connections -Tanks 3) Location of the Trim System components: -Pumps -Valves -Cross Connections -Hose Connections -Tanks -Contents Gauges -Dip Rods 4) Operation of the Trim System -Prerequisites -Limitations -Local -Remote	Problem Solving Matching SCT Demonstration Power Point Presentation

-Indication
-Abnormal Line Ups
-Safety

Scenario	NA
Activity Steps	Instructor will take class to SCT and give a demonstration of how the trim system is operated from the SCC, to include: -Tracing a path for the water to move in order to know which valves need to be open/shut, and what pump to use. -Moving water from one tank to another, demonstrating how the gauges/indications work, size of tanks, explain what they would see on surveillance (if it worked). -Discuss the valves that are left open in SOP line up, and why. -How to put a blow on trim tanks for Fire Fighting. -Show what would happen with Loss of 24V, tie it into the Electrical. -Discuss consequences of pumping a tank dry, or over filling a tank. -Emphasize importance of ensuring Trim system is in SOP prior to putting a blow on for Fire Fighting. -Discuss other capabilities of Trim system other than trimming. <u>Problem Solving (Build System)</u> Instructor will add the corresponding technical symbols next to the components and ask students individually to combine the components into an operation system diagram to the best of their ability. Instructor can allow some class discussion but each student should attempt their own diagram (10 min). Instructor will monitor students and correct as required. Once students have had 10 mins to attempt their own, the instructor will draw a simplified system diagram on the board, then explain the function, water flow and answer questions. After presenting the simplified system, the instructor will ask students if they remember from the start of class what the function of the Trim System is. As students give examples the instructor will add them to the diagram on the board until the complete submarine diagram is presented.

Instructor can then present the official diagram for students to see.

<u>Matching</u>

Instructor displays numbered photos of Trim System components with a corresponding list of number component names. Students are to match the components with the photo on a sheet of paper (5 min).

Instructor will monitor progress.

Instructor will review answers.

Instructor will show students location of components (using Virtual Submarine, 3D Sub Model, or Internal Submarine Diagram)

Resources	References
1)3D Submarine (if available) 2) Paper and pen/pencil 3)One computer with DWAN access 4) Whiteboard 5) EO 004.01 lesson	A1,A2,A3,B47,B74,B75,B76,B77, B78,B79,B106,B108,B109,B110, B111,B112,B113,B114,B115,B116, B117,B118,B120,B160,B161,B164, B170.

Notes to Instructor

Material will be assessed in EC 005.01.

EO 005.02 - Operate Bilge and Ballast as a Harbour Watchkeeper

005.02A – Operate Bilge System

Conditions		Sequence
Classroom		After EO 005.01A
Time	**Classification**	**Instructor**
6 x 50 mins	Unclass	AILS, AIND, AINA, AILQ

Learning Objectives	Method
1) Function of the HP Bilge System - Rapid de-ballasting - Series/Parallel with Ballast Pump 2)Components of the HP Bilge System - Pump - System valves - Cross connections - Tanks - Services supplied 3)Location of the HP Bilge System components - Pump - System valves - Cross connection - Rapid suction - Tanks - Services supplied 4)Operation of the HP Bilge System - Control - Indication - Abnormal line-ups - Prerequisites - Limitations - Safety 5)Function of the LP Bilge System	Problem Solving Matching SCT Demonstration Power Point Presentation

6) Components of the LP Bilge System
- Pump
- System valves
- Tanks
- Cross connections
7)Location of the LP Bilge System components
- Pump
- System valves
- Tanks
- Cross connections
8)Operation of the LP Bilge System
- Control
- Indication
- Abnormal line-ups
- Prerequisites
- Limitations
- Safety

Scenario	NA
Activity Steps	Instructor will take class to SCT and give a demonstration of how the Bilge System is operated from the SCC, to include: -Tracing a path for the water to move in order to know which valves need to be open/shut, and what pump to use. -Discussing Hull and Back up valves, and where they are operated. -Rapid de-ballasting -Operating pumps in series/parallel -Demonstrate how gauges/flow meters/indications work -Demonstrate how to adjust the speed of the Bilge pump <u>Problem Solving (Build System)</u> Instructor will add the corresponding technical symbols next to the components and asks students individually to combine the components into an operation system diagram to the best of their ability. Instructor can allow some class discussion but each student should attempt their own diagram (10 min).

Instructor will monitor students and correct as required.

Once students have had 10 mins to attempt their own, the instructor will draw a simplified system diagram on the board and explain the function and water flow and answer questions.

After presenting the simplified system, the instructor will ask students if they remember from the start of class what the function of the Trim System is. As students give examples, the instructor will add them to the diagram on the board until the complete submarine diagram is presented.

Instructor can then present the official diagram for students to see.

<u>Matching</u>

Instructor displays numbered photos of Bilge System components with a corresponding list of numbered component names. Students are to match the components with the photo on a sheet of paper (5 min).

Instructor will monitor progress.

Instructor will review answers.

Instructor will show students location of components (using Virtual Submarine, 3D Sub Model, or Internal Submarine Diagram)

Resources	References
1)Whiteboard 2) Paper and pen/pencil 3)One computer with DWAN access 4) EO 004.01 Lesson	A1,A2,A3,B47,B74,B75,B76,B77, B78,B79,B106,B108,B109,B110, B111,B112,B113,B114,B115,B116, B117,B118,B120,B160,B161,B164, B170.

Notes to Instructor
Material will be assessed in EC 005.02. Operation will be practiced and assessed on OJPR.

EO 005.02 – Operate Bilge and Ballast as a Harbour Watchkeeper

005.02B – Operate Ballast System

Conditions		Sequence
Classroom (S100)		After EO 005.02A
Time	**Classification**	**Instructor**
5 x 50 mins	Unclass	

Learning Objectives	Method
1)Function of the Ballast System - Rapid de-ballasting - Series/Parallel with HP bilge pump 2)Components of the Ballast System - Pump - System valves - Cross connections - Tanks - Services supplied 3)Location of the Ballast System components - Pumps - System valves - Cross connections - Rapid suction - Tanks - Content gauges - Dip rods - Services supplied 4)Operation of the Ballast System - Control - Indication	Problem Solving Matching SCT Demonstration Power Point Presentation

- Abnormal line-ups
- Prerequisites
- Limitations
- Safety

Scenario	NA
Activity Steps	Instructor will take class to SCT and give a demonstration of how the Ballast system is operated from the SCC, to include: -Tracing a path for the water to move in order to know which valves need to be open/shut, and what pump to use. -Discussing Hull and Back up valves, and where they are operated. -Rapid de-ballasting -Operating pumps in series/parallel -Demonstrate how gauges/flow meters/indications work -Demonstrate how to adjust the speed of the Ballast pump <u>Problem Solving (Build System)</u> Instructor will add the corresponding technical symbols next to the components and asks students individually to combine the components into an operation system diagram to the best of their ability. Instructor can allow some class discussion but each student should attempt their own diagram (10 min). Instructor will monitor students and correct as required. Once students have had 10 mins to attempt their own the instructor will draw a simplified system diagram on the board and explain the function and water flow and answer questions. After presenting the simplified system, the instructor will ask students if they remember from the start of class what the function of the Trim System is. As students give examples the instructor will add them to the diagram on the board until the complete submarine diagram is presented. Instructor can then present the official diagram for students to see.

Matching

Instructor displays numbered photos of Ballast System components with a corresponding list of number component names. Students are to match the components with the photo on a sheet of paper (5 min).

Instructor will monitor progress.

Instructor will review answers.

Instructor will show students location of components (using Virtual Submarine, 3D Sub Model, or Internal Submarine Diagram)

Resources	References
1)Slide set cards 2) Projector 3)DWAN access 4)Tablets with Virtual Sub	A1,A2,A3,B47,B74,B75,B76,B77, B78,B79,B106,B108,B109,B110, B111,B112,B113,B114,B115,B116, B117,B118,B120,B160,B161,B164, B170.

Notes to Instructor

Theory will be assessed in EC 005.03. Practical demonstration and assessment during OJPR.

EC 005.02 – Operate Trim, Bilge and Ballast Systems as a Harbour WatchKeeper

Scope	Conditions	Sequence
EO 005.01-02	Classroom	24 Hrs After EO 005.02B
Time	**Classification**	**Instructor**
1 x 50 min	Unclass	AILS, AIND, AINA, AILQ

Scenario	N/A
Assessment Steps	Will be a written exam based on content from EO 001.02. Exam shall contain short answer questions and may contain multiple choice. Suggested breakdown of exam as follows: INSTRUCTIONS GIVEN TO TRAINEE 1. EC 005.02 will be a written test based on content from EO 005.01-02. 2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam. 3. Do not open the booklet until you are told to do so. 4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet. 5. Work carefully. Do not spend to much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s). 6. Read each question carefully, and all responses fully, before answering the question. 7. If you have any questions during the exam, raise your hand. 8. References will/will not be provided. 9. Pass mark is 70% 10. You will have 1 x 50 mins period to complete this exam. 11. Do you have any questions before you start?

Resources	References

1) Classroom 2) One test per student 3) DS solution for test	A1,A2,A3,B47,B74,B75,B76,B77, B78,B79,B106,B108,B109,B110, B111,B112,B113,B114,B115,B116, B117,B118,B120,B160,B161,B164, B170.

Notes to Instructor
N/A

PO 006 Operate the Hydraulic Systems as a Harbour Watchkeeper

1. Performance. (T0006) Operate the Hydraulic Systems as a Harbour Watchkeeper

2. Conditions

g) Given:

(1) Submarine;

(2) Publications and Drawings; and

(3) Applicable References.

h) Denied: Assistance.

i) Environmental: Alongside Submarine.

3. Standard. The submariner shall, IAW references, (T0006, T0008) operate the Hydraulic Systems as a Harbour Watchkeeper to include:

a) Main and External Hydraulic Systems; and

b) Steering and Hydroplanes System.

4. References

A1,A2,A3,B80,B81,B82,B83,B84,B85,B86,B87,B88,B117,B118,B120,B122,B123,B124, B125,B126,B127,B160,B161.

5. Task Numbers. T0006, T0008.

6. Training Limitations. Limited or no access to a submarine. Student exposure may be limited to shore based trainers and simulators.

EO 006.01 – Operate the Hydraulic System as a Harbour Watchkeeper

006.01A – Basics of Hydraulic Plants

Conditions		Sequence
Classroom (S100)		Start PO 006/ After EO 005.02B
Time	**Classification**	**Instructor**
2 x 50 mins	Unclass	AILS, AIND, AINA, AILQ

Learning Objectives	Method
1) Function of the Hydraulic Power Plants	Think-Pair-Share
2) Components of the Hydraulic Power Plants	Problem Solving
3) Operating Modes of the Hydraulic Power Plants	Matching
4) Control and Indication of the Hydraulic Power Plants	PowerPoint

Scenario	NA
Activity Steps	Instructor will explain that hydraulic fluid is used for operating mechanical equipment on the submarine. List some items that operates with hydraulic fluid (2min). <u>Think-Pair-Share</u> Instructor will ask students, on their own and without talking, to list what they think the main components of a Hydraulic Power Plant would be on a sheet of paper. Instructor will let the students know that it does not matter if they get any correct this is just to get them thinking about the system (2min). Instructor will ask students to pair with another student to discuss and revise their list (3 min). Instructor will ask if any group thinks they have a correct list and have them list it on a white board. Instructor will correct the list on board. -Hydraulic pump -Replenishing tank -Accumulator -Unloader valve -Fire isolation valves -Relief valves

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-Bypass valves
-Plant supply and return filters
-Plant non-return valves

Problem Solving (Build System)

Instructor will add the corresponding technical symbols next to the components, and ask students individually to combine the components into an operation system diagram to the best of their ability. Instructor can allow some class discussion but each student should attempt their own diagram (10 min).

Instructor will monitor students and correct as required.

Once students have had 10 minutes to attempt their own, the instructor will draw a simplified system diagram on the board, explain the function, hydraulic fluid flow and answer questions.

After presenting the simplified system, the instructor will ask students if they remember from the start of class what is operated by hydraulics.

Instructor can then present the official diagram for students to see.

Interactive lecture (PowerPoint)

Instructor will use the PowerPoint to cover material in lecture.

Resources	References
1)3D Submarine (if available) 2) Paper and pen/pencil 3) One computer with DWAN access 4) Whiteboard 5) Virtual Submarine 6) EO 006.01 lesson	A1,A2,A3,B47,B74,B75,B76,B77, B78,B79,B106,B108,B109,B110, B111,B112,B113,B114,B115,B116, B117,B118,B120,B160,B161,B164, B170.

Notes to Instructor

Material will be assessed in EC 006.04. Operation will be practiced and assessed on OJPR

EO 006.01 - Operate the Hydraulic System as a Harbour Watchkeeper

006.01B – Operation of Hydraulic Plants

Conditions		Sequence
Classroom (S100)		After EO 006.01A
Time	**Classification**	**Instructor**
2 x 50 mins	Unclass	AILS, AIND, AINA, AILQ

Learning Objectives	Method
1) Operate the Hydraulic Plants	Unstructured Brainstorm Interactive Lecture Think-Pair-Share PowerPoint

Scenario	You are on watch as a harbour watchkeeper in Halifax, and FMF Cape Scott are going to be operating masts and periscopes. You will need to operate the main and external hydraulic plants within the operating parameters IAW TAB.
Activity Steps	Instructor reads scenario <u>Brainstorm (Unstructured)</u> Instructor asks class to come up with safety considerations prior to operating this system. Instructor writes all suggestions on board (5 min). Once students are done instructor can eliminate suggestion not relevant to this system and highlight/discuses important safety considerations such as: -checking bridge and fin clear of personnel. -checking proper line up of hydraulic systems IAW VCSOPs. -Operating pressure -Obtaining DWS permission -Man aloft <u>Interactive Lecture</u> Instructor will explain hydraulic plant operation in relation to operating as a harbour watchkeeper using VCSOPs.

Instructor will cover the basic principles of hydraulic plant operation to include:
-Control
-Indication
-Abnormal line-up
-Limitations
-Safety
-Rounds (What would a milky hydraulic fluid indicate)
-Operating Parameters

Think-pair-share

Instructor will ask students on their own to list **components** of hydraulic plant and how they work (2 min). Then with a partner revise their lists (3 min).

Instructor will review list and reiterate the increase in knowledge from before the class.

Resources	References
1) Whiteboard 2) Paper and pen/pencil 3) One computer with DWAN access 4) Virtual Submarine 5) EO 006.02 Lesson	A1,A2,A3,B47,B74,B75,B76,B77, B78,B79,B106,B108,B109,B110, B111,B112,B113,B114,B115,B116, B117,B118,B120,B160,B161,B164, B170.

Notes to Instructor

Material will be assessed in EC 006.02. Operation will be practiced and assessed on OJPR.

EO 006.01 – Operate Hydraulic system as a Harbour Watchkeeper

006.01C – Basics of Main Hydraulics system

Conditions		Sequence
Classroom (S100)		After EO 006.01B
Time	**Classification**	**Instructor**
1 x 50 mins	Unclass	AILS, AIND, AINA, AILQ

Learning Objectives	Method
1)Function of the Main Hydraulic system	Think-Pair-Share
2) Components of the Main Hydraulic system	Matching
3) Operating Modes of the Main Hydraulic system	PowerPoint
4) Control and Indication of the Main Hydraulic system	

Scenario	NA
Activity Steps	Instructor will explain what the Main Hydraulic system is used for on the submarine, to include; -System Function -System Components - System Operation -Control and Indication <u>Think-Pair-Share</u> Instructor will ask students, on their own and without talking, to list what they think the main components of the main hydraulics system would be on a sheet of paper. Instructor will let the students know that it does not matter if they get any correct, this is just to get them thinking about the system (2min). Instructor will ask students to pair with another student to discuss and revise their list (3 min). Instructor will ask if any group thinks they have a correct list and have them list it on a white board. Instructor will correct the list on board. <u>Matching</u>

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Instructor displays numbered photos of the main hydraulic system components with a corresponding list of numbered component names. Students are to match the components with the photo on a sheet of paper (5 min).

-Main Vents
-GIV
-Ballast Hull valve
-Hydraulic operated trim valve
-Induction Hull Valve

Instructor will monitor progress.

Instructor will review answers.

Instructor will show students location of components (using Virtual Submarine, 3D Sub model, or internal submarine diagram)

<u>Interactive Lecture (PowerPoint)</u>

Instructor will use PowerPoint to cover material in lecture

Resources	References
1) Whiteboard 2) Paper and pen/pencil 3) One computer with DWAN access 4) Virtual Submarine 5) EO 006.02 Lesson	A1,A2,A3,B47,B74,B75,B76,B77, B78,B79,B106,B108,B109,B110, B111,B112,B113,B114,B115,B116, B117,B118,B120,B160,B161,B164, B170.

Notes to Instructor

Material will be assessed in EC 006.04. Operation will be practiced and assessed on OJPR.

EO 006.01 - Operate Hydraulic System as a Harbour Watchkeeper

006.01D- Operation of Main Hydraulic System

Conditions		Sequence
Classroom (S100)		After EO 006.01C
Time	**Classification**	**Instructor**
2 x 50 mins	Unclass	AILS, AIND, AINA, AILQ

Learning Objectives	Method
1)Operate the Main Hydraulic system	Brainstorm Interactive Lecture Think-Pair-Share PowerPoint

Scenario	As a harbour watchkeeper you are required to pump the bilge.
Activity Steps	Instructor will read the scenario and explain that the main hydraulics are used to operate LP bilge system valves, and show where they are on the SCC (trainer or tapestry). <u>Brainstorm (Unstructured)</u> Instructor asks class to come up with equipment that this system operates. Instructor writes all suggestions on board (5 min). Once students are done instructor can eliminate suggestions not relevant to this system and highlight/discusses important safety considerations such as: -GIV -Hull valves -Search and Attack Periscope -Bulkhead doors (35 and 56) <u>Interactive Lecture</u> Instructor will explain the main hydraulic system operation in relation to operating as a harbour watchkeeper using VCSOPs.

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Instructor will cover the basic principles of the main hydraulic system operation to include:
-Control
-Indication
-Abnormal line-up
-Limitations
-Safety

Think-pair-share

Instructor will ask students on their own to list equipment operated by the main hydraulic system and how they work (2 min). Then with a partner revise their lists (3 min).

Instructor will review list, and reiterate to increase in knowledge from before the class.

Resources	References
1) Whiteboard 2) Paper and pen/pencil 3) One computer with DWAN access 4) Virtual Submarine 5) EO 006.01 Lesson	A1,A2,A3,B47,B74,B75,B76,B77, B78,B79,B106,B108,B109,B110, B111,B112,B113,B114,B115,B116, B117,B118,B120,B160,B161,B164, B170.

Notes to Instructor

Material will be assessed in EC 006.02. Operation will be practiced and assessed on OJPR

EO 006.01 - Operate Hydraulic System as a Harbour Watchkeeper

006.01E – Basics of External Hydraulic System

Conditions		Sequence
Classroom (S100)		After EO 006.01D
Time	**Classification**	**Instructor**
1 x 50 mins	Unclas	AILS, AIND, AINA, AILQ

Learning Objectives	Method
1)Function of the External Hydraulic system 2) Components of the External Hydraulic system 3) Operating Modes of the External Hydraulic system 4) Control and Indication of the External Hydraulic system	Think-Pair-Share PowerPoint

Scenario	
Activity Steps	Instructor will explain what the External Hydraulic system is used for on the submarine, to include; -System Function -System Components - System Operation -Control and Indication <u>Think-Pair-Share</u> Instructor will ask students, on their own and without talking, to list what they think the main components of the External Hydraulics system would be on a sheet of paper. Instructor will let the students know that it does not matter if they get any correct, this is just to get them thinking about the system (2min). Instructor will ask students to pair with another student to discuss and revise their list (3 min). Instructor will ask if any group thinks they have a correct list and have them list it on a white board. Instructor will correct the list on board <u>Interactive Lecture</u>

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Instructor will use PowerPoint to cover material in lecture

Resources	References
1) Whiteboard 2) Paper and pen/pencil 3) One computer with DWAN access 4) Virtual Submarine 5) EO 006.01 Lesson	A1,A2,A3,B47,B74,B75,B76,B77, B78,B79,B106,B108,B109,B110, B111,B112,B113,B114,B115,B116, B117,B118,B120,B160,B161,B164, B170.

Notes to Instructor

Material will be assessed in EC 006.02. Operation will be practiced and assessed on OJPR

EO 006.01 - Operate Hydraulic System as a Harbour Watchkeeper

006.01F – Operation of External Hydraulics System

Conditions		Sequence
Classroom (S100)		After EO 006.01E
Time	**Classification**	**Instructor**
2 x 50 mins	Unclas	AILS, AIND, AINA, AILQ

Learning Objectives	Method
1)Operate the External Hydraulic system	Brainstorm Interactive Lecture Think-Pair-Share PowerPoint

Scenario	As a harbour watchkeeper, FMF is doing work and requires the comms mast to be raised.
Activity Steps	Istructor reads the scenario. Talks about safety considerations. Discusses procedure for raising the comms mast. <u>Brainstorm (Unstructured)</u> Instructor asks class to come up with other equipment that this system operates. Instructor writes all suggestions on board (5 min). Once students are done instructor can eliminate suggestions not relevant to this system and highlight/discusses important safety considerations such as: -Masts -Weapons Systems -Towed Array -Capstan -Anchor Windlass -Extend/Retract Foreplanes <u>Interactive Lecture</u>

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Instructor will explain the main hydraulic system operation in relation to operating as a harbour watchkeeper using VCSOPs.

Instructor will cover the basic principles of the External Hydraulic system operation to include:

-Control
-Indication
-Abnormal line-up
-Limitations
-Safety

Think-pair-share

Instructor will ask students on their own to list equipment operated by the External Hydraulic system and how they work (2 min). Then with a partner revise their lists (3 min).

Instructor will review list and reiterate to increase in knowledge from before the class.

Resources	References
1) Whiteboard 2) Paper and pen/pencil 3) One computer with DWAN access 4) Virtual Submarine 5) EO 006.01 Lesson	A1,A2,A3,B47,B74,B75,B76,B77, B78,B79,B106,B108,B109,B110, B111,B112,B113,B114,B115,B116, B117,B118,B120,B160,B161,B164, B170.

Notes to Instructor
Material will be assessed in EC 006.02. Operation will be practiced and assessed on OJPR

EO 006.02 – Operate Steering and Hydroplanes Hydraulic System

006.02A – Basics of Steering and Hydroplane system

Conditions		Sequence
		After EO 006.01F
Time	**Classification**	**Instructor**
3 x 50 mins	Unclas	AILS, AIND, AINA, AILQ

Learning Objectives	Method
1)Function of the Hydroplanes and Steering Hydraulic system 2) Components of the Hydroplanes and Steering system 3) Operating Modes of the Hydroplanes and Steering Hydraulic system 4) Control and Indication of the Hydroplanes and Steering Hydraulic system	Think-Pair-Share Matching PowerPoint

Scenario	N/A
Activity Steps	Instructor will explain what the Hydroplanes and Steering Hydraulic system are used for on the submarine, to include; -System function -System Components - System Operation -Control and Indication <u>Think-Pair-Share</u> Instructor will ask students, on their own and without talking, to list what they think the main components of the Hydroplanes and Steering system would be on a sheet of paper. Instructor will let the students know that it doesn't matter if they get any correct this is just to get them thinking about the system (2min).

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Instructor will ask students to pair with another student to discuss and revise their list (3 min).
Instructor will ask if any group thinks they have a correct list and have them list it on a white board.
Instructor will correct the list on board

Ineractive Lecture

Instructor will use PowerPoint to cover material in lecture

Matching

Instructor displays numbered photos of the Hydroplanes and Steering system components with a corresponding list of numbered component names. Students are to match the components with the photo on a sheet of paper (5 min).

Instructor will monitor progress.

Instructor will review answers.

Instructor will show students location of components (using Virtual Submarine, 3D Sub model, or internal submarine diagram)

Resources	References
1) Whiteboard 2) Paper and pen/pencil 3) One computer with DWAN access 4) Virtual Submarine 5) EO 006.02 Lesson	A1,A2,A3,B47,B74,B75,B76,B77, B78,B79,B106,B108,B109,B110, B111,B112,B113,B114,B115,B116, B117,B118,B120,B160,B161,B164, B170.

Notes to Instructor

Material will be assessed in EC 006.02. Operation will be practiced and assessed on OJPR

EO 006.02 - Operate Steering and Hydroplanes Hydraulic System

006.02B – Operation of Hydroplanes and Steering system

Conditions		Sequence
Classroom (S100)		After EO 006.02A
Time	**Classification**	**Instructor**
2 x 50 mins	Unclass	AILS, AIND, AINA, AILQ

Learning Objectives	Method
1)Operate the Hydroplane and Steering Hydraulic system	Brainstorm Interactive Lecture Think-Pair-Share PowerPoint

Scenario	N/A
Activity Steps	Instructor states what equipment is operated by the system: -Aft Hydroplanes -Rudder <u>Interactive Lecture</u> Instructor will explain the Hydroplanes and Steering system operation in relation to operating as a harbour watchkeeper using VCSOPs. Instructor will cover the basic principles of the main hydraulic system operation to include: -Control -Indication -Abnormal line-up -Limitations -Safety

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Interactive Lecture

Instructor will use PowerPoint to cover all material in lecture

Think-pair-share

Instructor will ask students on their own to list **components,** of the Hydroplanes and Steering system, and how they work (2 min). Then with a partner revise their lists (3 min).

Instructor will review list and reiterate to increase in knowledge from before the class.

Resources	References
1) Whiteboard 2) Paper and pen/pencil 3) One computer with DWAN access 4) Virtual Submarine 5) EO 006.02 Lesson	A1,A2,A3,B47,B74,B75,B76,B77, B78,B79,B106,B108,B109,B110, B111,B112,B113,B114,B115,B116, B117,B118,B120,B160,B161,B164, B170.

Notes to Instructor

Material will be assessed in EC 006.02. Operation will be practiced and assessed on OJPR

EC 006.02 – Operate Hydraulic Systems as a Harbour Watchkeeper

Scope	Conditions	Sequence
EO 006.01-02	Classroom	24 Hrs After EO 006.02B
Time	**Classification**	**Instructor**
1 x 50 mins	Unclass	AILS, AIND, AINA, AILQ

Scenario	N/A
Assessment Steps	Will be a written exam based on content from EO 006.01-02. Exam shall contain short answer questions and may contain multiple choice. Suggested breakdown of exam as follows: Main Hydraulic Plant – 10% Main Hydraulic System – 10% External Hydraulic Plan – 10% External Hydraulic System – 10% Steering and Hydroplanes Plant – 10% Steering and Hydroplanes System – 10% Planes and Steering – 20% OMC – 20% INSTRUCTIONS GIVEN TO TRAINEE 1. EC 006.02 will be a written test based on content from EO 006.01-02. 2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam. 3. Do not open the booklet until you are told to do so. 4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet. 5. Work carefully. Do not spend to much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s). 6. Read each question carefully, and all responses fully, before answering the question. 7. If you have any questions during the exam, raise your hand.

	8. References will/will not be provided. 9. Pass mark is 70% 10. You will have a 1 x 50 mins period to complete this exam. 11. Do you have any questions before you start?

Resources	References
1) Classroom 2) One test per student 3) DS Solution for test	A1,A2,A3,B47,B74,B75,B76,B77, B78,B79,B106,B108,B109,B110, B111,B112,B113,B114,B115,B116, B117,B118,B120,B160,B161,B164, B170.

Notes to Instructor
N/A

PO 007 Operate Combat Systems as a Harbour Watchkeeper

1. Performance. (T0021) Operate Combat Systems as a Harbour Watchkeeper

2. Conditions

   a) Given:

      (1) Submarine;

      (2) Publications and Drawings; and

      (3) Applicable References.

   b) Denied: Assistance.

   c) Environmental: Alongside Submarine.

3. Standard. The submariner shall, IAW references, (T0021) conduct rounds on Combat Systems as a Harbour Watchkeeper to include:

   a) Operating the Communications Equipment;

   b) (T0011, T0015) Conducting rounds on Weapon Systems and Small Arms; and

   c) Tactical Weapon Systems.

4. References

A1,A2,A3,B5,B33,B34,B118,B133,B134,B135,B136,B140,B141,B150,B153,B158,B159, B160,B162,B165,B166,B167,B168.

5. Task Numbers. T0011, T0015, T0021

6. Training Limitations. Limited or no access to a submarine. Student exposure may be limited to shore based trainers and simulators.

EO 007.01 – Operate the communication equipment

007.01A –Internal Communications

Conditions		Sequence
TWIS and Classroom		Start PO 007/ After EO 006.02B
Time	**Classification**	**Instructor**
2 x 50 mins	Unclass	

Learning Objectives	Method
1) Safety Precautions 2) Function and Components of Internal Communications	Self-Study Interactive Lecture Tour

Scenario	The Submarine is home port along side the jetty. You are asssisting FMF trail staff with harbour acceptance trials of internal communications.
Activity Steps	1. The student will be given access to the TWIS trainer and handouts of the functions of and layout of Internal Communications equipment. 2. The instructor will provide all information to the students on functions of internal communications, layout diagrams and a tour the TWIS trainer explaining the function of each piece of equipment. 3. The student will be expected to explain functions and placement for each type of internal communications to include: a.Vices b.VCS 446; c.DC Net; d.Sound powered; and e.PRC's 4. The students will be quizzed on the function and layout of internal comms in PC 007 5. The instruction will do a review with the students at the end of the internal comms unit. 3 Stars -Purpose of Internal Comms -Use of internal Comms

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	-Idenfify DC Nets

Resources	References
1) TWIS 2) 3D submarine model 3) Hand outs	A1,A2,A3,B5,B33,B34,B118,B133, B134,B135,B136,B140,B141,B150, B153,B158,B159,B160,B162,B165, B166,B167,B168.

Notes to Instructor
Nil

EO 007.01 – Operate the communication equipment

007.01B – Operating Internal Communications Equipment

Conditions		Sequence
TWIS and Classroom		After EO 007.01A
Time	**Classification**	**Instructor**
2 x 50 mins	Unclass	AILO

Learning Objectives	Method
1) Safety Precautions 2) Operating different types of internal communication	Self-Study Interactive Lecture Tour Demo

Scenario	The Submarine is home port along side the jetty. You are asssisting FMF trail staff with harbour acceptance trials of internal communications.
Activity Steps	1. The student will be given access to the TWIS trainer and handouts on operating Internal Communications equipment. 2. The instructor will provide all information to the students on operating internal communications, and a tour of the TWIS to include a demo on operating internal comms equipment. 3. The student will be expected to operate each type of internal communications to include: a.Vices b.VCS 446; c.DC Net; d.Sound powered; and e.PRC's 4. The student will be given time to practice operating each piece of equipment in preparation for a practical exam. 5. The instructor will test each student on operating each piece of internal comms equipment.

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Resources	References
1) TWIS 2) 3D submarine model 3) Hand out	A1,A2,A3,B5,B33,B34,B118,B133, B134,B135,B136,B140,B141,B150, B153,B158,B159,B160,B162,B165, B166,B167,B168.

Notes to Instructor

Nil

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EO 007.01 – Operate the communication equipment

007.01C – External Communications

Conditions		Sequence
TWIS and Classroom		After EO 007.01B
Time	**Classification**	**Instructor**
1 x 50 mins	Unclass	AILO

Learning Objectives	Method
1) Safety Precautions 2) Function and Components of External Communications	Self-Study Interactive Lecture Tour Demo

Scenario	The Submarine is home port along side the jetty. You are asssisting FMF trial staff with harbour acceptance trials of external communications.
Activity Steps	1. The student will be given access to the TWIS trainer and handouts of the functions of and layout of External Communications equipment. 2. The instructor will provide all information to the students on functions of external communications, layout diagrams, and a tour with the TWIS trainer explaining the function of each piece of equipment. 3. The student will be expected to explain functions and placement for each type of external communications to include: a.VHF b.PRC's; c.BWA; d.VLF/LF fin loops; e.SATCOM; f.COMM mast; and g.QHE/SS2 antenna 4. The students will be quizzed on the function and layout of external Comms in PC 007 5. The instruction will do a review with the students at the end of the internal comms unit. 3 Stars

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	-Purpose of External Comms -Use of External Comms -Identify DC Nets

Resources	References
1) TWIS 2) 3D submarine model 3) Hand out	A1,A2,A3,B5,B33,B34,B118,B133, B134,B135,B136,B140,B141,B150, B153,B158,B159,B160,B162,B165, B166,B167,B168.

Notes to Instructor
Nil

EO 007.01 – Operate the communication equipment

007.01D – Operating External Communications Equipment

Conditions		Sequence
TWIS and Classroom		After EO 007.01C
Time	**Classification**	**Instructor**
1 x 50 mins	Unclass	

Learning Objectives	Method
1) Safety Precautions 2) Operating different types of internal communication	Self-Study Interactive Lecture Tour Demo

Scenario	The Submarine is home port alongside the jetty. You are assisting FMF trial staff with harbour acceptance trials of external communications
Activity Steps	1. The student will be given access to the TWIS trainer and handouts on operating External Communications equipment. 2. The instructor will provide all information to the students on operating External communications, and a tour of the TWIS to include a demo on operating external Comms Equipment. 3. The student will be expected to operate each type of external communications to include: a.VHF b.PRCs; c.BWA; d.VLF/LF fin loops; e.SATCOM; f.COMM mast; and g.QHE/SS2 antenna 4. The student will be given time to practice operating each piece of equipment in preparation for a practical exam. 5. The instructor will test each student on operating each piece of internal Comms equipment.

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Resources	References
1) TWIS 2) 3D submarine model 3) Handout	A1,A2,A3,B5,B33,B34,B118,B133, B134,B135,B136,B140,B141,B150, B153,B158,B159,B160,B162,B165, B166,B167,B168.

Notes to Instructor
Nil

EO 007.02 – Conduct Rounds On the Weapon Systems

007.02A – Conduct Rounds on the Weapons Systems and Small Arms

Conditions		Sequence
S-100 Trainer		After EO 007.01D
Time	**Classification**	**Instructor**
3 x 50 mins	Unclass	AILS, AIND, AINA, AILQ

Learning Objectives	Method
1) Conduct Weapons Rounds of the WSC to include: a. Torpedo Tube Status: o Outboard Operating Pressure o Rear Door Tally o Rear Door Chained and Locked b. Secondary Display Decoder Indications c. Hydraulic Control Packs Line up d. WSC Bilge e. Tube Operator Console f. Submerged Signal Ejector Breach Shut g. MK 48 Torpedo Stowage h. Otto Fuel/Hazmat Spill (VCEOP) i. Ammunition/Pyrotechnic Lockers Secured j. Hatch Procedures k. Air Pressures to include o AV&B o HPA o EBS	Hand outs Guided Discussion Self-Study

Scenario	As a harbour watchkeeper part of your duties are conducting rounds on the boat. You are to conduct rounds in the WSC.
Activity Steps	Instructor will hand out the Rounds Sheet and give the students a chance to read. Instructor will take the students through the rounds sheet explaining what they are checking.

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Instructor will explain the procedure for opening the ATP and Weapons Embarkation Hatch.

Instructor will take small groups to conduct rounds. Other students will self-study.

Resources	References
1. WH&DS Trainer 2. Hand Outs	A1,A2,A3,B5,B33,B34,B118,B133, B134,B135,B136,B140,B141,B150, B153,B158,B159,B160,B162,B165, B166,B167,B168.

Notes to Instructor

Give handout to students day before class. Practical assessment will be conducted during OJPR.

EO 007.02 – Conduct Rounds On the Weapon Systems

007.02B – Locate and Describe the Major Components of the Weapons Handling and Discharge System

Conditions		Sequence
S-100 Trainer		After EO 007.02A
Time	**Classification**	**Instructor**
3 x 50 mins	Unclass	AILS, AIND, AINA, AILQ

Learning Objectives	Method
Locate and describe the major components of the Weapons Handling Discharge System to include the following: a. Torpedo Tube layout b. Tanks to include: • WRTT • TOT • WTT c. ATP to include: • Drains • Inflatable Seals • Inflatable Seal Reducing Station • ATP 821 • Firing Air Reducing Stations • TOT Vents • ATP Bilge	Guided discussion Handouts Self Study

Scenario	
Activity Steps	Instructor will hand out info Sheets and give the students a chance to read.

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Instructor will give the students a guided tour of the Weapons Trainer pointing out the locations of all the pieces and describing all the pieces of the Weapons Handling and Discharge System.

Resources	References
1. WH&DS Trainer 2. Hand Outs	A1,A2,A3,B5,B33,B34,B118,B133, B134,B135,B136,B140,B141,B150, B153,B158,B159,B160,B162,B165, B166,B167,B168.

Notes to Instructor

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EO 007.02 – Conduct Rounds on Weapons Systems

007.02C – Describe the SSE

Conditions		Sequence
S-100 Trainer and Classroom		After EO 007.02B
Time	**Classification**	**Instructor**
10 x 50 mins	Unclass	

Learning Objectives	Method
1)Safety Precautions 2)Function of the SSE 2)Components of the SSE 3)Operation of the SSE 4)Pyrotechnics 5)Operation of Pyrotechnics .	Self-Study Interactive Lecture

Scenario	You are at sea on the submarine on patrol and are assisting the TI firing the SSE.
Activity Steps	1) The student will be given access to the SSE trainer in S-100, a handout on the SSE, and videos/pictures on the operational modes of the SSE. 2) The student will conduct self-study of the handout material, and watch the SSE videos in preparation for the interactive lecture at the SSE trainer in S-100. (Students will self-study while waiting for tours/runs in the trainer). 3) The instructor will provide the student will all the required information on the SSE, and a tour of the Trainer to include: identifying the components of the SSE, and a demo on the operation of the SSE. 4) a.The student will be expected to describe the function of the SSE. b.Identify and locate the components of the SSE to include: - Interlocks - Air System - Hydraulic System - Local Operation Panel and - Sequencing Manifold

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	c. Identify components and state purpose of Pyrotechnics/Countermeasures to include: - Safety - Function - Types - Characteristics - Stowage (location of Pyro/Ammo locker) - Accidental Activation d. Complete a quiz consisting of a diagram of the SSE and fill in the blanks identifying the components of the SSE. 5) The instructor will review the quiz with the students.

Resources	**References**
1)SSE Trainer in S-100 2)Videos 3)Pictures	A1,A2,A3,B5,B33,B34,B118,B133, B134,B135,B136,B140,B141,B150, B153,B158,B159,B160,B162,B165, B166,B167,B168.

Notes to Instructor
A block diagram of the interlocks will help students understand how and why these interlocks are in place. Practical will be assessed during EC 011.01

EO 007.02 – Operate Combat Systems as a Harbour Watchkeeper

007.02D – Locate and Describe the Small Arms.

Conditions		Sequence
Class Room (S100)		After EO 007.02C
Time	**Classification**	**Instructor**
2 x 50 mins	Unclass	AILS

Learning Objectives	Method
Locate and describe Small Arms types to include: • C8 • Sig Sauer • Signal Flare Pistols • Line Throwing Devices • Stowage/Security Location Small Arms Log Book Locate and describe pyrotechnics and their operation • Function • Types • Construction • Flooding Arrangments • Pyro Activation IAW VCEOP	Handouts Guided Discussion

Scenario	NA
Activity Steps	Instructor will hand out info sheets and give the students a chance to read. Instructor will discuss with the class where all the weapons are located, small arm rounds and the small arms log. Other students will self study and quiz each other. (Flash Cards)

Resources	References
1) Handout 2) Inert Rounds 3) Flash Cards	A1,A2,A3,B5,B33,B34,B118,B133, B134,B135,B136,B140,B141,B150, B153,B158,B159,B160,B162,B165, B166,B167,B168.

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Notes to Instructor

Practical will be assessed during EC 011.01

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EC 007.02– Conduct Rounds on the Weapon Systems as a Harbour Watchkeeper

Scope	Conditions	Sequence
EO 007.02	Classroom S-100	24 Hrs After EO 007.02D
Time	**Classification**	**Instructor**
1 x 50 mins	Unclas	AILS

Scenario	You are a harbour watchkeeper and are required to do rounds on the SSE.
Assessment Steps	INSTRUCTIONS GIVEN TO TRAINEE 1. EC 007.02 will be a written test based on content from EO 007.02. 2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam. 3. Do not open the booklet until you are told to do so. 4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet. 5. Work carefully. Do not spend to much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s). 6. Read each question carefully, and all responses fully, before answering the question. 7. If you have any questions during the exam, raise your hand. 8. References will/will not be provided. 9. Pass mark is 70% 10. You will have 1 x 50 mins period to complete this exam. 11. Do you have any questions before you start?

Resources	References
1) Test 2) DS Solution	A1,A2,A3,B5,B33,B34,B118,B133, B134,B135,B136,B140,B141,B150, B153,B158,B159,B160,B162,B165, B166,B167,B168.

Notes to Instructor
Practical component of this EO will be assessed during EC 011.01

Limitations	Practical component will be assessed on OJPR
Impact of Failure	IAW NPTORD 4500-54

EO 007.03 – Conduct Rounds On the Tactical Weapons Systems

007.03A - Locate and Describe the SONAR Systems.

Conditions		**Sequence**
Classroom (S-100) Submarine (If Available) TWS or Fred Sherwood Trainer		After EO 007.02D
Time	**Classification**	**Instructor**
2 x 50 mins	Unclass	W-Eng Sonar Tech

Learning Objectives	**Method**
1) What is SONAR? 2) What affects the propagation of sound? 3) What are hydrophones and how are they different from transducers? 4) Locate and describe the "Wet Bits" of the SONAR Suite. - BQQ-10 Sail, Bow, AI&R - 2008 UWT (Underwater Telephone) - 2073 EUWT (Emergency Underwater Telephone) - Echo Sounders - 2071 Augmenter - 2004 Bathythermograph 5) Locate and Describe the "Dry Bits" of the SONAR Suite. - BQQ-10 Sail, Bow, AI&R - 2008 UWT (Underwater Telephone) - 2073 EUWT (Emergency Underwater Telephone) - Echo Sounders - 2071 Augmenter - 2004 Bathythermograph	Power Point White Board Submarine Tour / Virtual Sub

Scenario	N/A
Activity	Instructor will go over basics of SONAR:

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Steps	
	SONAR is the acronym for **SO**und **N**avigation **A**nd **R**anging. It is a system for the detection of objects under water by emitting sound pulses and detecting or measuring their return after being reflected. It can also be used for navigation and communication with other vessels. There are two categories of SONAR, Active and Passive. Passive sonar is essentially listening for sound made by vessels. Active sonar emits pulses of sounds and listens for echoes. Think of SONAR as underwater **RADAR** (**RA**dio **D**etection **A**nd **R**anging). Think-Pair-Share Instructor will ask students on their own to write down some of the factors they think can affect how sound moves through water. (2 min) Instructor will ask students to discuss with the student next to them, to help with improving their lists. (3 min) Instructor will then ask if someone would like to read their list. Instructor will confirm or correct students list and briefly explain each factor. (10min) Note: Instructor to ensure factors below are covered. ***Factors*** *that can affect how sound moves through water:* • *Environment (Temperature, Salinity, and Pressure)* • *Frequency (High Frequency vs Low Frequency), and* • *Noise (Natural & Artificial).* Instructor will describe transducers. A transducer is a device that converts energy from one form into another. A hydrophone is a transducer that converts underwater sound waves into electrical signals, similar to a microphone. In most active SONAR systems the same transducers serve as both projectors and hydrophones, but there are good reasons in some cases to use separate hydrophones for reception. In addition passive search and surveillance SONAR, as well as passive ranging SONAR, use only hydrophones. Instructor Ask the class the following questions:

	-Ask the students how they think one submarine communicates with another? -Ask the students how a submarine can detect a surface vessel? -Ask the students which SONAR systems might have Transducers or Hydrophones and why? Instructor will go through Power Point Slides (HMCS WINDSOR SONAR FAMILIARIZATION.pptx) of equipment and where they are located around the submarine. Physical 3D sub and Virtual Sub should be utilized where applicable. Instructor ask the students to write down their 3 main takeaways from this lesson. Have a couple students read out their points. (10 min) If the Submarine is available, then schedule a visit to the sub so the students can point out and name the visible SONARs and equipment.

Resources	References
1) 3D Submarine (if available) 2) Whiteboard w/ Markers 3) Computer w/ DWAN access and Local Projector 4) Submarine (if available) 5) Virtual Submarine	A1,A2,A3,B5,B33,B34,B118,B133, B134,B135,B136,B140,B141,B150, B153,B158,B159,B160,B162,B165, B166,B167,B168.

Notes to Instructor
Nil

EC 007.03– Operate Combat Systems as a Harbour Watchkeeper

Scope	**Conditions**	**Sequence**
EO 007.03 and 007.01	Classroom S-100	24 Hrs After EO 007.03A
Time	**Classification**	**Instructor**
1 x 50 mins	Unclas	AILS

Scenario	You are a harbour watchkeeper and are required to do rounds on the SSE.
Assessment Steps	INSTRUCTIONS GIVEN TO TRAINEE 1. EC 007.03 will be a written test based on content from EO 007.01 and EO 007.03. 2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam. 3. Do not open the booklet until you are told to do so. 4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet. 5. Work carefully. Do not spend to much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s). 6. Read each question carefully, and all responses fully, before answering the question. 7. If you have any questions during the exam, raise your hand. 8. References will/will not be provided. 9. Pass mark is 70% 10. You will have 1 x 50 mins period to complete this exam. 11. Do you have any questions before you start?

Resources	**References**
1) Test 2) DS Solution	A1,A2,A3,B5,B33,B34,B118,B133, B134,B135,B136,B140,B141,B150, B153,B158,B159,B160,B162,B165, B166,B167,B168.

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Notes to Instructor
Nil

Limitations	Practical component will be assessed on OJPR
Impact of Failure	IAW NPTORD 4500-54

PO 008 Operate Submarine Domestic Systems as a Harbour Watchkeeper

1. Performance. (T0017) Operate Submarine Domestic Systems as a Harbour Watchkeeper

2. Conditions

   a) Given:

      (1) Submarine;

      (2) Publications and Drawings; and

      (3) Applicable References.

   b) Denied: Assistance.

   c) Environmental: Alongside Submarine.

3. Standard. The submariner shall, IAW references, (T0017) operate Submarine Domestic Systems as a Harbour Watchkeeper, to include:

   a) Operating Submarine Domestic Systems; and

   b) (T0018) Conducting Rounds in the Garbage Ejector.

4. References. A1, A2, A3, B4, B90, B91, B107, B113, B137, B160.

5. Task Numbers. T0017, T0018.

6. Training Limitations. Limited or no access to a submarine. Student exposure may be limited to shore based trainers and simulators.

EO 008.01 – Operate Submarine Domestic Systems

008.01A – Basics of Domestic Fresh Water System

Conditions		Sequence
Classroom (S100)		Start PO 008/ After EO 007.03A
Time	**Classification**	**Instructor**
2 x 50 mins	Unclas	AILS, AIND, AINA, AILQ

Learning Objectives	Method
1)Function of Domestic Fresh Water 2)Components of Domestic Fresh Water - Tanks - System Valves - ROD - Water Heaters 3)Location of Domestic Fresh Water Components - Tanks - System Valves - ROD Plant - Water Heaters 4)Operation of Domestic Fresh Water - Control - Indication - Shore Filling -Water Quality (testing) -Tank Change Over - Safety	Think-Pair-Share Problem Solving Matching

Scenario	NA
Activity Steps	Instructor will explain the function of the domestic fresh water system on the submarine (2min). <u>Think-Pair-Share</u> Instructor will ask students, on their own and without talking, to list what they think the main components of the domestic fresh water system would be, on a

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	sheet of paper. Instructor will let the students know that it does not matter if they get any correct, this is just to get them thinking about the system (2min). Instructor will ask students to pair with another student to discuss and revise their list (3 min). Instructor will ask if any group thinks they have a correct list and have them list it on a white board. Instructor will correct the list on the board. -Tanks (x2) -Emergency Fresh Water Tanks (x2) -Compressed Air - ROD -Water Heater (x2) -Valves <u>Problem Solving (Build System)</u> Instructor will add the corresponding technical symbols next to the components, then ask students individually to combine the components into an operation system diagram to the best of their ability. Instructor can allow some class discussion, but each student should attempt their own diagram (10 min). Instructor will monitor students and correct as required. Once student have had 10 mins to attempt their own, the instructor will draw a simplified system diagram on the board, explain the function, water flow and answer questions. After presenting the simplified system, the instructor will ask students if they remember from the start of class what the fresh water is used for. As students give examples, the instructor will add them to the diagram on the board, until the complete submarine diagram is presented. Instructor can then present the official diagram for students to see. <u>Matching</u>

	Instructor displays numbered photos of fresh water components with a corresponding list of numbered component names. Students are to match the components with the photo on a sheet of paper (5 min). Instructor will monitor progress. Instructor will review answers. Instructor will show students location of components (using Virtual Submarine, 3D Sub model, or internal submarine diagram)

Resources	References
1)3D Submarine (if available) 2) Paper and pen/pencil 3)One computer with DWAN access 4) Whiteboard 5) EO 008.01 lesson	A1, A2, A3, B4, B90, B91, B107, B113, B137, B160.

Notes to Instructor
Material will be assessed in EC 008.03.

EO 008.01 – Operate Submarine Domestic Systems

008.01B – Basics of the Sanitary System

Conditions		**Sequence**
Classroom (S100)		After EO 008.01A
Time	**Classification**	**Instructor**
2 x 50 mins	Unclas	AILS, AIND, AINA, AILQ

Learning Objectives	**Method**
1)Function of the Sanitary System 2)Components of the Sanitary System - Tank - System Valves 3)Location of the Sanitary System Components - Tank - System Valves 4)Operation of the Sanitary System - Control - Indication - Blowing - Flushing Through - Safety 5)Function of Galley Slop Drain 6)Components of Galley Slop Drain - Tank - System Valves 7)Location of Galley Slop Drain Components - Tank - System Valves 8)Operation of Galley Slop Drain - Control - Indication - Blowing - Safety	Think-Pair-Share Problem Solving Matching

Scenario	NA
Activity Steps	Instructor will explain the function of the Sanitary system on the submarine (2min). <u>Think-Pair-Share</u> Instructor will ask students, on their own and without talking, to list what they think the main components of the sanitary system would be, on a sheet of paper. Instructor will let the students know that it does not matter if they get any correct this is just to get them thinking about the system (2min). Instructor will ask students to pair with another student to discuss and revise their list (3 min). Instructor will ask if any group thinks they have a correct list and have them list it on a white board. Instructor will correct the list on the board. -Tanks -Valves -Heads <u>Problem Solving (Build System)</u> Instructor will add the corresponding technical symbols next to the components, and ask students individually to combine the components into an operation system diagram, to the best of their ability. Instructor can allow some class discussion but each student should attempt their own diagram (10 min). Instructor will monitor students and correct as required. Once students have had 10 mins to attempt their own, the instructor will draw a simplified system diagram on the board, explain the function, flow, and answer questions. After presenting the simplified system, the instructor will ask students if they remember from the start of class what the sanitary system is used for. As students give examples, the instructor will add them to the diagram on the board, until the complete submarine diagram is presented.

	Instructor can then present the official diagram for students to see. <u>Matching</u> Instructor displays numbered photos of sanitary system components with a corresponding list of numbered component names. Students are to match the components with the photo on a sheet of paper (5 min). Instructor will monitor progress. Instructor will review answers. Istructor will discuss the Galley Slop Drain, function and components Instructor will show students location of components (using Virtual Submarine, 3D Sub model, or internal submarine diagram)

Resources	**References**
1)3D Submarine (if available) 2) Paper and pen/pencil 3)One computer with DWAN access 4) Whiteboard 5) EO 008.01 lesson	A1, A2, A3, B4, B90, B91, B107, B113, B137, B160.

Notes to Instructor
Material will be assessed in EC 008.03.

EO 008.02 – Operate Submarine Domestic Systems

008.02A – Basics of Garbage Ejector

Conditions		Sequence
Classroom		After EO 008.01B
Time	**Classification**	**Instructor**
1 x 50 mins	Unclas	AILS, AIND, AINA, AILQ

Learning Objectives	Method
1) Function of Garbage Ejector 2) Describe alongside state of the Garbage Ejector to include: -Flood Valves -Drain Valve -Breech Door -Muzzle Door -Flushing Valve -LP Bilge Suction -Test Cock and Reamer -Telegraph -Water Tight Integrity	Unstructured Brainstorm Interactive Lecture Think-Pair-Share

Scenario	N/A
Activity Steps	Instructor explains function of the garbage ejector and explains the issues around water tight integrity. <u>Interactive Lecture</u> Instructor will go through parts and operation of the garbage ejector and what to check on rounds.

Resources	References
1)Whiteboard 2) Paper and pen/pencil 3)One computer with DWAN access	A1, A2, A3, B4, B90, B91, B107, B113, B137, B160.

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4) EO 008.02 Lesson	

Notes to Instructor

Material will be assessed in EC 008.02.

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EC 008.02 – Operate Submarine Domestic Systems as a Harbour Watchkeeper

Scope	Conditions	Sequence
EO 008.01-02	Classroom	24 Hrs After EO 008.02A
Time	**Classification**	**Instructor**
1 x 50 mins	Unclass	AILS, AIND, AINA, AILQ

Scenario	N/A
Assessment Steps	Will be a written exam based on content from EO 008.01-02. Exam shall contain short answer questions and may contain multiple choice. Suggested breakdown of exam as follows: Sanitary System – 30% Galley Slop Drain – 30% Domestic Fresh Water – 30% Garbage Ejector – 10% INSTRUCTIONS GIVEN TO TRAINEE 1. EC 008.02 will be a written test based on content from EO 008.01-02. 2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam. 3. Do not open the booklet until you are told to do so. 4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet. 5. Work carefully. Do not spend to much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s). 6. Read each question carefully, and all responses fully, before answering the question. 7. If you have any questions during the exam, raise your hand. 8. References will/will not be provided. 9. Pass mark is 70% 10. You will have 1 x 50 mins period to complete this exam. 11. Do you have any questions before you start?

Resources	References
1) Classroom 2) One test per student 3) DS Solution for test	A1, A2, A3, B4, B90, B91, B107, B113, B137, B160.

Notes to Instructor
N/A

PO 009 Stand Watch as a Harbour Watchkeeper

1. Performance. (T0002, T0004, T0013, T0014, T0016, T0022) Stand Watch as a Harbour Watchkeeper

2. Conditions

   a) Given:

      (1) Submarine;

      (2) Publications and Drawings; and

      (3) Applicable References.

   b) Denied: Assistance.

   c) Environmental: Alongside Submarine.

3. Standard. The submariner shall, IAW references, (T0002, T0004, T0013, T0014, T0016) Stand Watch as a Harbour Watchkeeper to include:

   a) Conducting Rounds (Lower Deck Trot);

   b) Operating SCC;

   c) Conducting Routine Operations;

   d) Reacting to Onboard Emergencies; and

   e) (T0022) Responding to Changes in Force Protection Levels.

4. References.A1,A2,A3,A4,A5,A6,A7,A12,A13,B1,B2,B7,B8,B12,B13,B118,B120,B121, B122,B138, B139,B145,B146,B147,B148,B149,B150,B157.

5. Task Numbers. T0002, T0004, T0013, T0014, T0016, T0022

6. Training Limitations. Limited or no access to a submarine. Student exposure may be limited to shore based trainers and simulators.

EO 009.01 – Conduct Rounds as a Harbour Watchkeeper

009.01A – Perform Duties of Upper Deck Trot

Conditions		Sequence
Classroom		After EO 008.02A
Time	**Classification**	**Instructor**
3 x 50 mins	Unclas	AILS

Learning Objectives	Method
1) Ceremonial Procedures 2) Monitor Submarine Buoyancy a. Recording Draught Marks b. Reasons for Changes in Draught	Interactive Lecture

Scenario	N/A
Activity Steps	Instructor will introduce the students to the duties of an Upper Deck Trot. Upon completion of the lecture, instructors will provide confirmation questions in order to verify the students have understood the learning objectives.

Resources	References
1) Computer with DWAN 2) Projector 3) 009.01A Presentation	A1,A2,A3,A4,A5,A6,A7,A12,A13, B1,B2,B7,B8,B12,B13,B118,B120, B121,B122,B138,B139,B145,B146, B147,B148,B149,B150,B157

Notes to Instructor
Nil

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EO 009.01 – Conduct Rounds as a Harbour Watchkeeper

009.01B – Perform Duties of Lower Deck Trot

Conditions		Sequence
Classroom		After EO 009.01A
Time	**Classification**	**Instructor**
2 x 50 mins	Unclas	AILS

Learning Objectives	Method
1) Victoria Class Publications 2) Log Books 3) Round Procedures 4) Alarms and Locations	Interactive Lecture

Scenario	N/A
Activity Steps	Instructor will introduce the students to the duties of a Lower Deck Trot. Upon completion of the lecture, instructors will provide confirmation questions in order to verify the students have understood the learning objectives.

Resources	References
4) Computer with DWAN 5) Projector 6) 009.01B Presentation	A1,A2,A3,A4,A5,A6,A7,A12,A13, B1,B2,B7,B8,B12,B13,B118,B120, B121,B122,B138,B139,B145,B146, B147,B148,B149,B150,B157

Notes to Instructor
Nil

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EO 009.02 – Operate SCC as a Harbour Watchkeeper

009.02A – SCC Operation

Conditions		Sequence
Classroom and SCT		After EO 009.01B
Time	**Classification**	**Instructor**
4 x 50 mins	Unclas	AILS

Learning Objectives	Method
1) Monitor Alarms (Grade A, B, fire and flood) 2) Main, External, Steering & Hydroplanes Hydraulic Plant Alarms, Levels, and Pressures 3) Hydrolic Plant Control 4) LP Bilge system 5) HP Bilge and Ballast sytem 6) Operate Surveillance system	Interactive lecture EDI

Scenario	As a Harbour Watchkeeper you are required to operate the SCC
Activity Steps	Instructor will demonstrate SCC functions within the trainer's capability. Students are then given time to operate the SCC.

Resources	References
1) Computer with DWAN 2) Projector 3) 009.01C Presentation	A1,A2,A3,A4,A5,A6,A7,A12,A13, B1,B2,B7,B8,B12,B13,B118,B120, B121,B122,B138,B139,B145,B146, B147,B148,B149,B150,B157

Notes to Instructor
Nil

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EO 009.03 – Conduct Routine Operations as a Harbour Watchkeeper

009.03A – Routine Operations and MCC

Conditions		Sequence
Classroom / SCT		After EO 009.02A
Time	**Classification**	**Instructor**
4 x 50 mins	Unclas	AILS

Learning Objectives	Method
1) Operate MCC 2) Conduct Blowing Around 3) Raising/Lowering Masts and Periscopes	Interactive Lecture EDI

Scenario	N/A
Activity Steps	Instructor will introduce the students to Routine Operations as a Harbour Watchkeeper. Instructors will demonstrate procedures on the MCC within the trainers capability and allow the students to practice. Upon completion of the lecture, instructors will provide confirmation questions in order to verify the students have understood the learning objectives.

Resources	References
4) Computer with DWAN 5) Projector 6) 009.03A Presentation	A1,A2,A3,A4,A5,A6,A7,A12,A13, B1,B2,B7,B8,B12,B13,B118,B120, B121,B122,B138,B139,B145,B146, B147,B148,B149,B150,B157

Notes to Instructor
Nil

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EO 009.04 – React to Onboard Emergencies as a Harbour Watchkeeper

009.04A – Fire Fighting and DC Equipment

Conditions		Sequence
Classroom		After EO 009.03A
Time	**Classification**	**Instructor**
4 x 50 mins	Unclas	AILS

Learning Objectives	Method
1) Fire Fighting and DC Equipment 2) Immediate EOPs a. Discoverer Alongside b. Lower Deck Trot c. SCC	Interactive Lecture

Scenario	N/A
Activity Steps	Instructor will introduce the students to the submarine Fire Fighting and DC equipment and EOPs. Upon completion of the lecture, instructors will provide confirmation questions in order to verify the students have understood the learning objectives.

Resources	References
7) Computer with DWAN 8) Projector 9) 009.04A Presentation	A1,A2,A3,A4,A5,A6,A7,A12,A13, B1,B2,B7,B8,B12,B13,B118,B120, B121,B122,B138,B139,B145,B146, B147,B148,B149,B150,B157

Notes to Instructor
Nil

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EO 009. 04 – React to Onboard Emergencies as a Harbour Watchkeeper

009.04B – React to Onboard Emergencies at the SCC

Conditions		Sequence
SCT		After EO 009.04A/ Start EO 009.04B in Conjunction with EO 011.01A
Time	**Classification**	**Instructor**
60 x 50 mins=(3000 mins)	Unclas	AILS

Learning Objectives	Method
1) React to onboard emergencies as the SCC Watchkeeper to include: a. Collision b. Fire c. Flood d. HP Air Burst e. Hydraulic Burst f. Man Overboard g. Hazmat Spill h. Security Alert i. Casualty	EDI

Scenario	N/A
Activity Steps	Instructor will demonstrate the reactions to EOPs on the SCC then run students through EOPs. One period at the end will be used for a debrief and review.

Resources	References
1) Scenarios 2) Trainer	A1,A2,A3,A4,A5,A6,A7,A12,A13, B1,B2,B7,B8,B12,B13,B118,B120, B121,B122,B138,B139,B145,B146, B147,B148,B149,B150,B157

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Notes to Instructor
Nil

EO 009. 04 – React to Onboard Emergencies as a Harbour Watchkeeper

009.04C – DC Shool

Conditions		Sequence
DCTF		After EO 009.05A
Time	**Classification**	**Instructor**
16 x 50 mins=(800 mins)	Unclas	AILS

Learning Objectives	Method
1) Chemistry of Fire 2) Methods of Extinguishment 3) Fire Development 4) Special Considerations for Fire Suppression 5) Firefighing Terminology 6) Firefighing Equipment 7) Damage Control Equipment 8) Atmospheric Monitoring 9) Firefighting Techniques and Procedures 10) Capabilities and Limitations of Firefighting Systems 11) Location and Operations of Fitted and Portable Fire Fighing and DC Equipment 12) DC Organization	Interactive lecture EDI

Scenario	N/A
Activity Steps	This portion is conducted at the Damage Control Training Facility.

Resources	References
1) DCTF	A1,A2,A3,A4,A5,A6,A7,A12,A13, B1,B2,B7,B8,B12,B13,B118,B120,

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	B121,B122,B138,B139,B145,B146, B147,B148,B149,B150,B157

Notes to Instructor

Nil

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EC 009.04 – React to Onboard Emergencies as a Harbour Watchkeeper

Scope	Conditions	Sequence
EO 009.04	SCT	After EO 009.04B in Conjunction with EO 011.01A
Time	**Classification**	**Instructor**
20 x 50 mins	Unclas	AILS

Scenario	You are on watch as the SCC Watchkeeper during an Emergency.
Assessment Steps	This assessment will take place In the SCT. Students will be given time to practice all the EOPs and be critiqued during their practice runs so that they are aware of the standard that will be expected of them during the actual assessment. Students will be assessed individually.

Resources	References
1) Trainer 2) References	A1,A2,A3,A4,A5,A6,A7,A12,A13, B1,B2,B7,B8,B12,B13,B118,B120, B121,B122,B138,B139,B145,B146, B147,B148,B149,B150,B157

Notes to Instructor
Nil

Limitations	PC will be after Harbour Watchkeeper portion of OJPR
Impact of Failure	IAW NPTORD 4500-54

PROTECTED B (when complete)

Item	EC 009.04 React to Onboard Emergencies as a Harbour Watchkeeper	SAT	UNSAT	Remarks
1. Critical Factors - Safety				

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Personnel	☐	☐	**Critical**
Equipment	☐	☐	**Critical**
Water tight integrity	☐	☐	**Critical**
2. EOPS - Fire			
Makes correct pipe	☐	☐	
Stops ventilating	☐	☐	
Crash stops:			
-Ships ventilation	☐	☐	
-Battery ventilation	☐	☐	
-Fuel fire safe	☐	☐	
Line up trim system for fire fighting	☐	☐	
Contact shore firefighting (if alongside)	☐	☐	
3. EOPs - Flood			
Makes correct pipe	☐	☐	
Stops generating (if required)	☐	☐	
Stops ventilating	☐	☐	
Shut:			
- Induction hull valve	☐	☐	
- Vex hull valve	☐	☐	
-Induction/vex emergency flap vv Emergency hull shut down switch to all	☐	☐	
Crash Stops:			
-Ships ventilation	☐	☐	
- Battery ventilation	☐	☐	
- Fuel fire safe	☐	☐	
Shut and clip conning tower upper lid and main access hatch	☐	☐	
4. EOPs - Hydraulic Burst			
Makes correct pipe	☐	☐	
Stops generating (if required)	☐	☐	

Do not lower masts	☐	☐	
Stops ventilating	☐	☐	
Secure bilge and ballast system	☐	☐	
Crash Stops:			
-Ships ventilation	☐	☐	
- Battery ventilation	☐	☐	
- Fuel fire safe	☐	☐	
Line up the trim system for fire fighting	☐	☐	
5. EOPs - Collision in Harbour			
Makes correct pipe	☐	☐	
Stops generating (if required)	☐	☐	
Stops ventilating	☐	☐	
Shut:			
- Induction hull valve	☐	☐	
- Vex hull valve	☐	☐	
- Induction/vex emergency flap vv	☐	☐	
Crash Stops:			
- Ships ventilation	☐	☐	
- Battery ventilation	☐	☐	
- Fuel fire safe	☐	☐	
Shut and clip conning tower upper lid and main access hatch	☐	☐	
Check heel and trim	☐	☐	
6. EOPs - Hydraulic Burst			
Makes correct pipe	☐	☐	
Stops generating (if required)	☐	☐	
Do not lower masts	☐	☐	
Stops ventilating	☐	☐	
Secure bilge and ballast system	☐	☐	
Crash Stops:			
1 - Ships ventilation	☐	☐	

2 - Battery ventilation	☐	☐	
3 - Fuel fire safe	☐	☐	
Line up the trim system for fire fighting	☐	☐	
7. EOPs - HP Air Burst			
Shut all GIVs	☐	☐	
Stop generating (if required)	☐	☐	
Make correct pipe	☐	☐	

Score: /

Trainee must achieve 80% on non-critical factors **AND** all critical factors to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of this assessment with the candidate.		**Candidate** I have been debriefed on the result of this assessment	
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PROTECTED B (when complete)

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EO 009.05 – Respond to Changes in Force Protection Level

009.05A – Force Protection

Conditions		Sequence
Classroom		After EC 009.04
Time	**Classification**	**Instructor**
5 x 50 mins	Unclas	AILS

Learning Objectives	Method
1) General FP Knowledge 2) Upper Deck Sentry and Access Point 3) Bomb Threat Organization	Interactive Lecture

Scenario	N/A
Activity Steps	Instructor will introduce the students to the submarine Force Protection Organization. Upon completion of the lecture, instructors will provide confirmation questions in order to verify the students have understood the learning objectives.

Resources	References
1) Computer with DWAN 2) Projector 3) 009.05A Presentation	A1,A2,A3,A4,A5,A6,A7,A12,A13, B1,B2,B7,B8,B12,B13,B118,B120, B121,B122,B138,B139,B145,B146, B147,B148,B149,B150,B157

Notes to Instructor
Nil

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EC 009.05 - EC / PC Name

Scope	Conditions	Sequence
EO 009.01-03 and 05	Classroom S-100	After EO 009.04C
Time	**Classification**	**Instructor**
1 x 50 mins	Unclas	AILS

Scenario	N/A
Assessment Steps	Will be a written exam based on content from EO 009.01-03 and 009.05. Exam shall contain short answer questions, and may contain multiple choice. INSTRUCTIONS GIVEN TO TRAINEE 1. EC 009.05 will be a written test based on content from EO 009.01-03 and 009.05. 2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam. 3. Do not open the booklet until you are told to do so. 4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet. 5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s). 6. Read each question carefully, and all responses fully, before answering the question. 7. If you have any questions during the exam, raise your hand. 8. References will/will not be provided. 9. Pass mark is 70% 10. You will have 1 x 50 mins period to complete this exam. 11. Do you have any questions before you start?

Resources	References
1) Test 2) DS Solution	A1,A2,A3,A4,A5,A6,A7,A12,A13, B1,B2,B7,B8,B12,B13,B118,B120,

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	B121,B122,B138,B139,B145,B146, B147,B148,B149,B150,B157

Notes to Instructor
Nil

Limitations	Practical component will be assessed on OJPR
Impact of Failure	IAW NPTORD 4500-54

PO 010 Operate Submarine Escape and Rescue Systems

1. Performance. (T0001) Operate Submarine Escape and Rescue Systems

2. Conditions

   a) Given:

      (a) Submarine;

      (b) Publications and Drawings; and

      (c) Applicable References.

   b) Denied: Assistance.

   c) Environmental: Submarine at sea dived or surfaced, in all conditions.

3. Standard. The submariner shall, IAW references, (T0001) Operate Submarine Escape and Rescue Systems to include:

   a) Performing Submarine Escape and Rescue; and

   b) Conducting Submarine Surface Abandonment.

4. References. A1,A2,A3,A4,A7,A10,A11,A14,B11,B12,B13,B117,B118,B120,B121,B122.

5. Task Numbers. T0001

6. Training Limitations. Limited or no access to a submarine. Student exposure may be limited to shore based trainers and simulators.

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EO 010.01 – Perform Submarine Escape and Rescue

Lesson	Conditions	Sequence
010.01A	Classroom	Start PO 010/ After EO 009.04C
Time	**Classification**	**Instructor**
1 x 50 mins	UNCLASS	BSQ Submarine NAV COM

Learning Objectives	Method
Describe the various states of Submarine Accountability to include: ● SUBNOTE ● SUBCHECK ● SUBLOOK ● SUBMISS ● SUBSUNK	Interactive Lecture Guided Discussion

Scenario	Indications of a Distressed Submarine (DISSUB)
Activity Steps	<u>Interactive Lecture</u> Instructor will explain the importance of submarine accountability and effective communications and that the failure to receive an expected transmitted signal from a submarine, may well be the very first indication that a submarine is experiencing difficulties. Submarine Notice (SUBNOTE) - Submarines are routed by means of a SUBNOTE which is issued by CANSUBFOR. A SUBNOTE is a message that provides operational and movement instructions for submarines. The SUBNOTE accurately defines the route that the centre of the submarine moving haven will follow. Submarine Check Report (SUBCHECK) - Signal transmitted by a submarine at specified intervals to ensure CSF of the submarine's continued safety. CSF will instruct submarines to make SUBCHECK Reports at intervals specified in the SUBNOTE. No other signal received from a submarine may replace a SUBCHECK Report. Non-receipt of other anticipated signals should not

	normally give rise to undue concern. Although, in such circumstances, it may be appropriate to initiate a Submarine Safety Communication Check. SUBLOOK - code word of the procedures initiated by CSF when the safety of a submarine is in doubt, or when a Surfacing Signal, SUBCHECK, or Arrival Report is one hour overdue. SUBMISS - code word used when the initial search SUBLOOK has failed to establish the safety of the submarine, for an operation that will be executed in order to initiate a fully coordinated search, and for a submarine that is believed to be missing. SUBSUNK - code word of the signal originated by any unit or authority who has positive information that a submarine has sunk or when the DISSUB has been located. The signal will initiate a full-scale search and rescue operation if this has not already been done by declaration of SUBMISS. <u>Guided Discussion</u> Instructor will map out a DISSUB scenario while slowly guiding the class through the various steps of submarine accountability.

Resources	References
1) Classroom 2) OJT	A1,A2,A3,A4,A7,A10,A11,A14,B11,B12,B13,B117,B118,B120,B121,B122.

Notes to Instructor
The Escape and Rescue equipment location portion of the OJT will be assessed during the QSP OJPR process.

EO 010.01 – Perform Submarine Escape and Rescue

Lesson	Conditions	Sequence
010.01B	Classroom	After EO 010.01A
Time	**Classification**	**Instructor**
1 x 50 mins	UNCLASS	BSQ

Learning Objectives	Method
Explain the purpose of key Submarine Escape and Rescue equipment.	Interactive Lecture

Scenario	Distressed Submarine
Activity Steps	<u>Interactive Lecture</u> <u>Think pair share</u> Instructor will explain the purpose of the following equipment: a. All Red Cross Lockers to include: 1) Guard Cards 2) Tower Flood Hull Valve Ratchet 3) Spare Vent Cap 4) Last Man Straps b. SESSPE Suit and Storage Locations c. Emergency DISSUB Communications to include: 1) 639 Emergency Indicator Buoys 2) Operation of SEPIRBs 3) Personal Locator Beacons 4) Emergency UWTs d. Air Purification to include: 1) EO2 26H Candles 2) LIOH Curtains e. Atmosphere Monitoring to include: 1) ANALOX 2) Draeger Kits f. Emergency Drinking Water g. Portable Water Purification equipment h. Escape Rations i. Long Breathing Units (BU)

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	j. Emergency Pyrotechnic Stores

Resources	References
1) Classroom 2) Submarine (pending availability) 3) Photos (PowerPoint) 4) OJT	A1,A2,A3,A4,A7,A10,A11,A14,B11,B12,B13,B117,B118,B120,B121,B122.

Notes to Instructor
The Escape and Rescue equipment location portion of the OJT will be assessed during the QSP OJPR process.

EO 010.01 – Perform Submarine Escape and Rescue

Lesson	Conditions	Sequence
010.01C	Classroom	After EO 010.01B
Time	**Classification**	**Instructor**
1 x 50 mins	UNCLASS	BSQ

Learning Objectives	Method
Explain the line-up of the following equipment for submarine escape.	Interactive Lecture

Scenario	Distressed Submarine
Activity Steps	<u>Interactive Lecture</u> Instructor will explain the line-up of the following escape equipment IAW VCS Guard Cards: a. Built In Breathing System (BIBS) b. Hood Inflation System (HIS) c. Escape Tower preps Scenarios 4 books.

Resources	References
1) Classroom 2) Submarine (pending availability) 3) Virtual Sub/ 3D Sub	A1,A2,A3,A4,A7,A10,A11,A14,B11,B12,B13,B117,B118,B120,B121,B122.

Notes to Instructor
The Escape and Rescue equipment location portion of the OJT will be assessed during the QSP OJPR process.

EO 010.01 – Perform Submarine Escape and Rescue

Lesson	Conditions	Sequence
010.01D	Classroom	After EO 010.01C
Time	**Classification**	**Instructor**
1 x 50 mins	UNCLASS	BSQ

Learning Objectives	Method
Describe Key Submarine Escape and Rescue Personnel Positions and Duties	Interactive Lecture

Scenario	Distressed Submarine
Activity Steps	<u>Interactive Lecture</u> IAW Victoria Class Submarine Guard Cards, the instructor will instruct students to find, describe, and key in Guard Cards for the Escape and Rescue Personnel and Team Duties. Students will be given a section of time to read and paraphrase. Student will, in there own words, explain to the class. Topics to be included: a. Senior Survivor b. Tower Operating Team c. Water / Rations / Stores d. Communications

Resources	References
1) Classroom 2) Submarine (pending availability) 3) Virtual Sub/ 3D Sub	A1,A2,A3,A4,A7,A10,A11,A14,B11,B12,B13,B117,B118,B120,B121,B122.

Notes to Instructor
The Escape and Rescue equipment location portion of the OJT will be assessed during the QSP OJPR process.

EO 010.01 – Perform Submarine Escape and Rescue

Lesson	Conditions	Sequence
010.01E	Classroom	After EO 010.01D
Time	**Classification**	**Instructor**
1 x 50 mins	Unclass	BSQ

Learning Objectives	Method
Describe ELSS POD Delivery	Interactive Lecture

Scenario	Distressed Submarine
Activity Steps	<u>Interactive Lecture (combine with lesson E, same structure)</u> IAW Victoria Class Submarine Guard Cards, describe ELSS POD Delivery to include: a. Escape Tower Preparation b. Escape Tower Operation c. POD Contents

Resources	References
1) Classroom 2) Submarine (pending availability) 3) Virtual Sub/ 3D Sub	1) 103 2) Guard Cards

Notes to Instructor
The Escape and Rescue equipment location portion of the OJT will be assessed during the QSP OJPR process.

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EO 010.01 – Perform Submarine Escape and Rescue

Lesson	Conditions	Sequence
010.01F	Classroom	After EO 010.01E
Time	**Classification**	**Instructor**
1 x 50 mins	Unclass	BSQ

Learning Objectives	Method
Describe Tower Escape	Interactive Lecture

Scenario	Distressed Submarine
Activity Steps	<u>Interactive Lecture (PowerPoint)</u> Describe Tower Escape IAW VCS Guard Cards to include: a. Contaminated Environment b. Last Man Procedures c. Surfacing Drills

Resources	References
1) Classroom 2) Submarine (pending availability) 3) Virtual Sub/ 3D Sub	A1,A2,A3,A4,A7,A10,A11,A14,B11,B12,B13,B117,B118,B120,B121,B122.

Notes to Instructor
The Escape and Rescue equipment location portion of the OJT will be assessed during the QSP OJPR process.

EO 010.01 – Perform Submarine Escape and Rescue

Lesson	Conditions	Sequence
010.01G	Classroom	After EO 010.01F
Time	**Classification**	**Instructor**
1 x 50 mins	Unclass	BSQ

Learning Objectives	Method
Describe Submarine Rescue	Interactive Lecture

Scenario	Distressed Submarine
Activity Steps	<u>Interactive Lecture (PowerPoint)</u> Describe Submarine Rescue to include: a. Atmospheric and Water Tight Integrity Considerations b. Preparing the ATP Hatch and Aft Escape Tower to accept a Submarine Rescue Vehicle or Submarine Rescue Chamber c. Draining the ATP Hatch and Aft Escape Tower Coffer Dam

Resources	References
1) Classroom 2) Submarine (pending availability) 3) Virtual Sub/ 3D Sub	1) 103 2) Guard Cards

Notes to Instructor
The Escape and Rescue equipment location portion of the OJT will be assessed during the QSP OJPR process.

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EO 010.01 – Perform Submarine Escape and Rescue

Lesson	Conditions	Sequence
010.01H	WPET Facility	After 011.04D
Time	**Classification**	**Instructor**
32 x 50 mins	Unclass	WPET Staff

Learning Objectives	Method
Escape Traning	Interactive Lecture EDI

Scenario	Distressed Submarine
Activity Steps	Interactive Lecture as per WPET Staff Curriculum.

Resources	References
N/A	A1,A2,A3,A4,A7,A10,A11,A14,B11,B12,B13,B117,B118,B120,B121,B122.

Notes to Instructor
The Escape and Rescue equipment location portion of the OJT will be assessed during the QSP OJPR process.

NTDC(A) Approval – 21/10/2020
NPTG HQ Approval 24/02/2020
Version No: 2.1
MITE Crse ID :109201

EO 010.02 – Conduct Submarine Surface Abandonment

Lesson	Conditions	Sequence
010.02A	Classroom	After EO 010.01G
Time	**Classification**	**Instructor**
1 x 50 mins	Unclass	BSQ

Learning Objectives	Method
Submarine Surface Abandonment Equipment and Procedures	Interactive Lecture

Scenario	Submarine encounters a situation where the crew is required to abandon (Distressed Submarine).
Activity Steps	<u>Interactive Lecture</u> Identify Submarine Surface Abandonment equipment to include: a. Submarine Surface Abandonment Suit b. 10 Man Portable Life Rafts Explain Submarine Surface Abandonment IAW VCEOP Abandon Ship

Resources	References
1) Classroom 2) Submarine (pending availability) 3) Virtual Sub/ 3D Sub	A1,A2,A3,A4,A7,A10,A11,A14,B11,B12,B13,B117,B118,B120,B121,B122.

Notes to Instructor
The Escape and Rescue equipment location portion of the OJT will be assessed during the QSP OJPR process.

EC – Conduct Submarine Surface Abandonment

Scope	**Conditions**	**Sequence**
EO 010.02	Classroom S-100	24 Hrs After EO 010.02
Time	**Classification**	**Instructor**
1 x 50 mins	Unclass	AILS

Scenario	N/A
Assessment Steps	Will be a written exam based on content from EO 010.02. Exam shall contain short answer questions and may contain multiple choice. INSTRUCTIONS GIVEN TO TRAINEE 1. EC 010.02 will be a written test based on content from EO 010.02. 2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam. 3. Do not open the booklet until you are told to do so. 4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet. 5. Work carefully. Do not spend to much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s). 6. Read each question carefully, and all responses fully, before answering the question. 7. If you have any questions during the exam, raise your hand. 8. References will/will not be provided. 9. Pass mark is 70% 10. You will have 1 x 50 mins period to complete this exam. 11. Do you have any questions before you start?

Resources	**References**
1) Test 2) DS Solution	A1,A2,A3,A4,A7,A10,A11,A14,B11,B12,B13,B117,B118,B120,B121,B122.

Notes to Instructor
Nil

Limitations	Nil
Impact of Failure	IAW NPTORD 4500-54

PO 011 Operate as a Safe Submariner at Sea

1. Performance. Operate as a Safe Submariner at Sea

2. Conditions

   a) Given:

      (1) Submarine;

      (2) Publications and Drawings; and

      (3) Applicable References.

   b) Denied: Assistance.

3. Environmental: At sea, dived or surfaced, in all conditions.

4. Standard. The submariner shall, IAW references, (T0002, T0003, T0004, T0009, T0010, T0011, T0012, T0016, T0017, T0018) Operate as a Safe Submariner at Sea, to include:

   a) Operating the Submerged Signal Ejector at Sea;

   b) Operating OMC at Sea;

   c) Reacting to Onboard Emergencies at Sea; and

   d) Conducting a Sea Watch.

5. References.
A1,A2,A3,A6,A7,A8,B1,B3,B119,B120,B127,B128,B139,B142,B143,B144,B150.

6. Task Numbers. T0002, T0003, T0004, T0009, T0010, T0011, T0012, T0016, T0017, T0018

7. Training Limitations. Limited or no access to a submarine. Student exposure may be limited to shore based trainers and simulators.

EO 011.01 - Operate as a Safe Submariner at Sea

011.01A – Operate the SSE in all Modes

Conditions		**Sequence**
SCT Trainer and Classroom S-100		After EC 009.04B
Time	**Classification**	**Instructor**
18 x 50 mins	Unclass	

Learning Objectives	**Method**
1)Safety Precautions 2)Modes of Operation 3)SSE Misfire Procedures.	EDI

Scenario	The Submarine is off the coast of Nova Scotia, conducting sea trials after a work period. You are in the WSC when you receive an order from the control room, to test the SSE in all modes before the Submarine conducts a deep dive, as part of the trials.
Activity Steps	1. The student will be given access to the SSE trainer in S-100, and videos on the operational modes of the SSE. 2. The Instructor will follow the Explain/Demonstrate/ Imitate method to instruct the students in all modes of operation of the SSE. 3. The student will be expected to fire the SSE in all modes IAW with SUBWOPS to include: a. Manual Sequencing b. Local Operate Panel (Auto) c. Emergency Hand (Spanner) to include system line up and d. Conduct SSE Misfire procedures. 4. The student will be given a demo of each procedure followed by a number of practice runs. 5. The student will have a practical test by performing all modes of operation on the SSE in the S-100 SSE trainer.

Resources	**References**

1)SSE Trainer in S-100 2) Videos	A1,A2,A3,A6,A7,A8,B1,B3,B119,B120, B127,B128,B139,B142,B143,B144,B150.

Notes to Instructor
Nil

EC 011.01– Operate the Submerged Signal Ejector at Sea

Scope	**Conditions**	**Sequence**
EO 011.01	SSE Trainer (S-100)	After EO 011.01A
Time	**Classification**	**Instructor**
20 x 50 mins	Unclas	AILS

Scenario	While at sea the sub has run aground and is now sitting on the bottom. The senior survivor directs you to fire two SLF mark 2 red flares from the SSE.
Assessment Steps	Instructor will run one student through the EC at a time. The EC will be conducted in two parts. Part one will consist of item one and two (to “Operate SSE using manual spanners”); students will be given 40 min.The second part will consist of item 3, “Conduct rounds of the WSC as a Harbour Watchkeeper”, students will be given 10 min. Part 1 Instructor will read out the scenario and direct the student to begin the EC. Part 2 Instructor will ask the student to show what they would check during rounds as a harbour watchkeeper and identifiy stated items.

Resources	**References**
1) SSE Trainer in S-100	A1,A2,A3,A6,A7,A8,B1,B3,B119,B120, B127,B128,B139,B142,B143,B144,B150.

Notes to Instructor
First flare will be fired using manual sequencing with misfire drill. Second flare will be fired upon loss of hydraulics.

Limitations	N/A
Impact of Failure	IAW NPTORD 4500-54

Item	EC 011.01 Performance Statement	SAT	UNSAT	Remarks
1. Safety				
Personal		☐	☐	**critical**
Equipment		☐	☐	
Water Tight Integrity		☐	☐	**critical**
2. Line the system up for full operational checks				
Line all valves associated with SSE		☐	☐	
Operate SSE using Manual Sequencing		☐	☐	
Conduct SSE Misfire Routine		☐	☐	
Conduct CP for Loss of Hydraulics		☐	☐	**critical**
Operate SSE using Manual Spanners		☐	☐	**critical**
3. Conduct rounds of the WSC as a Harbour Watchkeeper				
Torpedo Tube State for Safety		☐	☐	
State of SSE		☐	☐	
WSC General Safety		☐	☐	
Identify all Pyrotechnics and Ammunition Storage.		☐	☐	
Identify the Function of all Pyrotechnics.		☐	☐	
Identify Flood Arrangements and Locations of Ammunition Lockers.		☐	☐	

Score: /14

Trainee must achieve 8/10 SATs on non-critical factors **AND** all critical factors to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of EC 011.01 with the candidate.		**Candidate** I have been debriefed on the result of EC 011.01.	
EC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

NTDC(A) Approval – 21/10/2020
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EO 011.02 – Operate OMC
011.02A – Components of the OMC

Conditions		Sequence
Classroom S-100		After EO 010.02
Time	**Classification**	**Instructor**
2 x 50 mins	Unclas	AILS

Learning Objectives	Method
1) Components of the OMC	Interactive Lecture

Scenario	N/A
Activity Steps	Instructor will introduce the students to the OMC. Upon completion of the lecture, instructors will provide confirmation questions in order to verify the students have understood the learning objectives.

Resources	References
1)Computer with DWAN 2) Projector 3) 011.02 lesson	A1,A2,A3,A6,A7,A8,B1,B3,B119,B120, B127,B128,B139,B142,B143,B144,B150

Notes to Instructor

NTDC(A) Approval – 21/10/2020
NPTG HQ Approval 24/02/2020
Version No: 2.1
MITE Crse ID :109201

EO 011.03 - React to Onboard Emergencies at Sea

011.03A – Damage Control Organization at Sea

Conditions		Sequence
Classroom S-100		After 011.02A
Time	**Classification**	**Instructor**
2 x 50 mins	Unclas	AILS

Learning Objectives	Method
1) Damage Control Organization at Sea 2) Duties and Responsibilities of: a. Discoverer b. Attack BA c. HQ1 d. On Scene Coordinator e. Attack team f. BA monitor g. BA Controller 3) HQ1 Publications	Interactive Lecture

Scenario	N/A
Activity Steps	Instructor will introduce the students to the OMC. Upon completion of the lecture, instructors will provide confirmation questions in order to verify the students have understood the learning objectives.

Resources	References
1) Computer with DWAN 2) Projector 3) 011.03A Lesson	A1,A2,A3,A6,A7,A8,B1,B3,B119,B120 ,B127,B128,B139,B142,B143,B144,B150

Notes to Instructor
Nil

NTDC(A) Approval – 21/10/2020
NPTG HQ Approval 24/02/2020
Version No: 2.1
MITE Crse ID :109201

EO 011.03 - React to Onboard Emergencies at Sea

011.03B – Atmospheric Monitoring

Conditions		Sequence
Classroom S-100		After 011.03A
Time	**Classification**	**Instructor**
2 x 50 mins	Unclas	AILS

Learning Objectives	Method
1) Atmospheric Monitoring 2) Operation of Drager 3) Smoke Clearance Methods	Interactive Lecture EDI

Scenario	N/A
Activity Steps	Instructor will discuss atmospheric monitoring and demonstrate/explain the use of the Drager. Students will each use the Drager as demonstrated by the instructor. Instructor will discuss the methods of smoke clearing and differences from surface ships.

Resources	References
1) Computer with DWAN 2) Projector 3) 011.03B lesson	A1,A2,A3,A6,A7,A8,B1,B3,B119,B120 ,B127,B128,B139,B142,B143,B144,B150

Notes to Instructor
Nil

NTDC(A) Approval – 21/10/2020
NPTG HQ Approval 24/02/2020
Version No: 2.1
MITE Crse ID :109201

EO 011.04 – Conduct a Sea Watch

011.04A – React to Changes in Submarine States and Routines

Conditions		Sequence
Classroom S-100		After EO 011.03B
Time	**Classification**	**Instructor**
8 x 50 mins	Unclas	AILS

Learning Objectives	Method
1) Submarine States and Routines 2) Actions for Changes in States and Routines 3) Watch and Station Bill	Interactive Lecture

Scenario	N/A
Activity Steps	Instructor will introduce the students to Submarine States and Routines. Upon completion of the lecture, instructors will provide confirmation questions in order to verify the students have understood the learning objectives.

Resources	References
1) Computer with DWAN 2) Projector 3) 011.04A Lesson	A1,A2,A3,A6,A7,A8,B1,B3,B119,B120, B127,B128,B139,B142,B143,B144,B150

Notes to Instructor
Nil

NTDC(A) Approval – 21/10/2020
NPTG HQ Approval 24/02/2020
Version No: 2.1
MITE Crse ID :109201

EO 011.04 – Conduct a Sea Watch

011.04B – R2MR

Conditions		**Sequence**
Classroom S-100		After 011.04F
Time	**Classification**	**Instructor**
20 x 50 mins = 1000 mins	Unclas	Guest Lecture

Learning Objectives	**Method**
1) As Guest Lecturer Curriculum	Lecture

Scenario	N/A
Activity Steps	This lecture will be conducted by a civilian Mental Health Clinician or a Submarine Physician's Assistant as per their curriculum.

Resources	**References**
1) Computer with DWAN 2) Projector	N/A

Notes to Instructor
Ensure guest lecture is booked

NTDC(A) Approval – 21/10/2020
NPTG HQ Approval 24/02/2020
Version No: 2.1
MITE Crse ID :109201

EO 011.04 – Conduct a Sea Watch

011.04C – Physican's Assistant Brief

Conditions		Sequence
Classroom S-100		After 011.04A
Time	**Classification**	**Instructor**
4 x 50 mins	Unclas	PA

Learning Objectives	Method
1) As PA's curriculum	Lecture

Scenario	
Activity Steps	This lecture will be conducted by a Submarine Physician's Assistant as per their curriculum.

Resources	References
1) Computer with DWAN 2) Projector	N/A

Notes to Instructor
Ensure guest lecture is booked

NTDC(A) Approval – 21/10/2020
NPTG HQ Approval 24/02/2020
Version No: 2.1
MITE Crse ID :109201

EO 011.04 – Conduct a Sea Watch

011.04D – Sea Training (Submarine) Coxn Brief

Conditions		Sequence
Classroom S-100		After 011.04C
Time	**Classification**	**Instructor**
2 x 50 mins	Unclas	AILS

Learning Objectives	Method
1) As Coxn's Curriculum	Lecture

Scenario	N/A
Activity Steps	This lecture will be conducted by the submarine Sea Training Coxn as per their curriculum.

Resources	References
1) Computer with DWAN 2) Projector	N/A

Notes to Instructor
Ensure guest lecture is booked

NTDC(A) Approval – 21/10/2020
NPTG HQ Approval 24/02/2020
Version No: 2.1
MITE Crse ID :109201

EO 011.04 – Conduct a Sea Watch

011.04E – Newly Qualified Submariner Brief

Conditions		Sequence
Classroom S-100		After EO 010.01H
Time	**Classification**	**Instructor**
2 x 50 mins	Unclas	AILS

Learning Objectives	Method
N/A	Lecture

Scenario	N/A
Activity Steps	This lecture will be conducted by a newly qualified Submariner as per their curriculum.

Resources	References
1) Computer with DWAN 2) Projector	N/A

Notes to Instructor
Ensure Guest Lecturer is Booked

NTDC(A) Approval – 21/10/2020
NPTG HQ Approval 24/02/2020
Version No: 2.1
MITE Crse ID :109201

EO 011.04 – Conduct a Sea Watch

011.04F – CANSUBFOR Chief Petty Officer

Conditions		Sequence
Classroom S-100		After 011.04E
Time	**Classification**	**Instructor**
2 x 50 mins	Unclas	AILS CPO

Learning Objectives	Method
1) As per the CPO's curriculum.	

Scenario	
Activity Steps	This lecture will be conducted by the CANSUBFOR Chief Petty Officer or designate as per their curriculum.

Resources	References
1) Computer with DWAN 2) Projector	N/A

Notes to Instructor
Ensure Guest Lecturer is Booked

NTDC(A) Approval – 21/10/2020
NPTG HQ Approval 24/02/2020
Version No: 2.1
MITE Crse ID :109201

EC 011.04 – Conduct a Sea Watch

Scope	Conditions	Sequence
EO 011.04	Classroom S-100	24 Hrs After EO 011.04A
Time	**Classification**	**Instructor**
1 x 50 mins	Unclas	AILS

Scenario	N/A
Assessment Steps	Will be a written exam based on content from EO 011.04. Exam shall contain short answer questions and may contain multiple choice. INSTRUCTIONS GIVEN TO TRAINEE 1. EC 011.04 will be a written test based on content from EO 011.04. 2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam. 3. Do not open the booklet until you are told to do so. 4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet. 5. Work carefully. Do not spend to much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s). 6. Read each question carefully, and all responses fully, before answering the question. 7. If you have any questions during the exam, raise your hand. 8. References will/will not be provided. 9. Pass mark is 70% 10. You will have 1 x 50 mins` period to complete this exam. 11. Do you have any questions before you start

Resources	**References**
1)Test 2)DS Solution	A1,A2,A3,A6,A7,A8,B1,B3,B119,B120, B127,B128,B139,B142,B143,B144,B150

Notes to Instructor
Nil

Limitations	Nil
Impact of Failure	IAW NPTORD 4500-54

NTDC(A) Approval – 21/10/2020
NPTG HQ Approval 24/02/2020
Version No: 2.1
MITE Crse ID :109201

ANNEX A – TASK LIST

TASK NO	TASK STATEMENT	PO
T0001	Escape/Rescue from Disabled Submarine	010
T0002	Execute Casing Party Duties	011
T0003	Operate Fire Fighting Equipment	009 / 011
T0004	Operate Air Systems	003
T0005	Operate Hydraulic Systems	006
T0006	Operate Trim, Ballast and Bilge Systems as a Harbour Watchkeeper	005
T0007	Operate Planes and Steering	006
T0008	Operate One Man Control (OMC)	011
T0009	Operate Submerged Signal Ejector (SSE)	007 / 011
T0010	React to Changes in Submarine States and Routines	011
T0011	Carry out the Duties of Lower Deck Trot	009
T0012	Carry out the Duties of Upper Deck Trot	009
T0013	Handle Ammunition / Pyrotechnics	007
T0014	React to Onboard Emergencies	009 / 010
T0015	Operate Submarine Domestic Systems	008
T0016	Operate Garbage (Gash) Ejector	008

NTDC(A) Approval – 21/10/2020
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Version No: 2.1
MITE Crse ID :109201

T0017	Operate Electrical Systems	002
T0018	Operate Submarine Cooling Systems as a Harbour Watchkeeper	004
T0019	Operate fitted Combat Systems Equipment	007
T0020	Operate Mast and Periscopes	007
T0021	Conduct Atmosphere Monitoring	011
T0022	Respond to Changes in Force Protection Levels	009

ANNEX B – RECORD OF PROCEEDINGS

BOARD DATES AND LOCATION

25 February – 15 March 2019 (All Board Members)
11 June – 13 June 2019 (Chair, TDO, NTDC A Staff, and NFSA Instructional Staff)
Building S-17
Naval Training Development Center Atlantic
Stadacona, CFB Halifax
Halifax, NS

BOARD COMPOSITION

Chair – Lt(N) Bryson, CSF Sustainment

Secretary – MS Maciuk, NAV COMM HMCS Windsor

TDO – SLt Guthrie, NTDC(A)

SME – LCdr Munro, CSF Sustainment

SME – CPO1 McGuire, Coxswain Sea Training (Submarines)

SME – CPO2 Jensen, Coxswain HMCS Corner Brook

SME – CPO2 Niezen, MAR TECH Coord HMCS Chicoutimi

SME – CPO2 Byrnes, C MAR TECH MECH Sea Training (Submarines)

SME – CPO2 Munro, C Sonar Op Sea Training (Submarines)

SME – PO1 Anthony, W Eng Manager SUB SAR Coord SST (East)

SME – MS Jacobson, MAR TECH MECH HMCS Windsor

SME – S1 Hebert, Sonar Op Watch Leader HMCS Chicoutimi

SME – S1 Luck Lam, MAR TECH MECH HMCS Corner Brook

Instructor – PO1 Coady MAR TECH ELECT, BSC Instructor

Instructor – MS Cummings MAR TECH MECH, BSC Instructor

Alternate Service Delivery (MDA) Rep – to be captured in the Record of Proceedings

Guests – On a case by case basis to be captured in the Record of Proceedings

BOARD OBJECTIVE

1. Conduct an in-depth review of the existing Basic Submarine Qualification (BSQ), Qualification Standard Plan (QSP), and associated On Job Performance Record (OJPR). The aim is to remove the focus on the Lower Deck Trot (LDT) and return it to a "Qualified in Submarines", “Safe Submariner” package.

2. Explore the feasibility of Walkthroughs with a Board for NCMs to determine if this can be conducted in conjunction/ or aligned with ALDE (2nd OOW) for Submarine Officers.

BOARD CONTEXT

1. The current focus of the BSQ program is to prepare trainees to perform the duties and responsibilities of an LDT, a process achieved by:

   - successfully completing the Basic Submarine Course;
   - successfully completing all Performance Objectives (PO’s), Enabling Objectives (EO’s), and departmental walkthroughs of the OJPR package;
   - consolidation of PO’s and EO’s through credible time (15 days) at sea; and
   - successfully challenging an AILS “Dolphin Board,” which in essence is the Performance Check (PC).

2. The Board will be reviewing the BSQ in it’s entirety to examine and determine the following:

   - can the training process be streamlined and consolidated without compromising the safe submariner standard;

- is there to much instruction on a system purpose, system component(s),and system operation;
- are PO and EO signing authorities clearly delineated;
- is the training sequence/ flow right;
- if something is removed from the QSP – how is the delta made up;
- the current OJPR is 10 weeks – is there enough time allotted;
- how many days constitutes the right amount of sea days – can it be condensed through the use of trainers or models offering the right fidelity;
- is there one standard that's 100% consistent across the Force – coast to coast – boat to boat. How do we capture and keep the standard – use of checklists, question banks, or a list of specific items;
- Insert the new R2MR curriculum spanning 2.5 days specific to the submarine environment. This also includes the creation of a new Submarine Environment MITE Code;
- requirement to remove submarine escape training pool phase prior to receiving identical training during WET; and
- requirement to remove MP 5 training from Force Protection.

3. In addition to the above:

- this will be an excellent opportunity to houseclean the document to reflect such items as the latest establishment and MOSID changes;
- how Alternate Service Delivery (ASD) will play a role in the future; and
- how can we best incorporate Computer Based Training (CBT), virtual reality training, or videos into the course

DISCUSSIONS/DECISIONS

Day One – Monday 25 February

INTRODUTION

The Board commenced with the Chair welcoming all members, giving an introduction, asking each members to provide a quick brief on their background, and any other related QSP experience. The Chair then briefed the Board on the rationale for convening the Basic Submarine Qualification Writing Board (BSQWB) (captured above). The roles of each of the participants were vehemently stressed by the Chair with a reminder of how their participation would be key to the success of the BSQWB. The Chair stated how very fortunate

the Board was to have such a broad and excellent representation of ranks, coasts, MOSID's,and experience. As such, participants were reminded to keep an open mind, be candid, and share relevant experiences. It was further discussed that during the course of the Board, additional personnel have already been invited to attend as guest SMEs, from which the Board will surely benefit from the added experiences and expertise.

The TDO, SLt Guthrie provided an excellent overview of the QSP process. He described the various quality control phases of CFITES, while at the same time, providing an example of Performance Orientated Training and its various components. He stated that as the Board progressed, there would be a reiteration of the various phases of CFITES. Prior to beginning this important task, the Chair asked if there were any questions. With none, the Board commenced its work.

OCCUPATIONAL SPECIALTY SPECIFICATION (OSS) – TASK LIST REVIEW

The Chair informed the Board that he had previously reviewed the OSS together with the TDO. He went on to state that both the skills and knowledges were fine, but the Military Occupational Structure and Prerequisites would need to be amended to reflect current policy, and the latest MOSID structure. He stated that the Task List would have to be reviewed for a train/ non-train decision, or slightly modified to reflect the realigned POs and EOs. Moreover, the Chair acknowledged that as things unfold, the OSS would more than likely be revisited and there may be further proposed amendments.

Accordingly, the Board proceeded with a comprehensive line by line review of the task list.

TASK NUMBER	TASK ELEMENT	PO
T 0011	**Operate Submarine Signal Ejector.** Change to **"Operate Submerged Signal Ejector"**	007 / 011
T 0018	**Operate the Garbage Ejector.** Change to **"Conduct Rounds on the Garbage Ejector"**	008
T 0022	**New Respond to Changes in Force Protection Levels.** Add as a new task.	009

Upon review of the task list, the Board recommended the following amendments:

- in T 0011 change submarine to submerged in order to reflect the actual name of the equipment;
- in T 0018 change Operate to Conduct, as the only one who actually operates the Garbage Ejector is the Submarines Coxswain; and
- add new task T 0022 Respond to Changes in Force Protection Levels in order to support the duties and responsibilities of the Harbour Watchkeeper;

One of the main objectives of the Board was to re-sequence POs and EOs, and front end load system knowledge for flow, ease, and substance. As a result, the Board determined it would be best, that this be done first, in order to give the members a solid foundation to work from. Given that, the Board determined the following:

PO 001 **Identify Submarine Construction (New)**
EO 001.01 Describe Submarine Layout (From Old EO 008.01)
EO 001.02 Identify Submarine Components (From Old EO 008.01)
EO 001.03 Describe Buoyancy and Stability (From Old EO 005.04)

Note: Although not a performance based PO, the Board added to provide students with a solid foundation prior to commencing submarine systems. These EOs were embedded in other POs and subsequently moved to PO 001.

PO 002 **Operate Electrical Systems as a Harbour Watchkeeper (Old PO 001)**
EO 002.01 Conduct Rounds on the Main Battery System as a Harbour Watchkeeper
EO 002.02 Conduct Rounds on the Submarine DC Electrical System as a Harbour Watchkeeper

EO 002.03 Conduct Rounds on the Submarine AC Electrical System as a Harbour Watchkeeper

PO 003 Operate Air Systems as a Harbour Watchkeeper
EO 003.01 Operate HP Air System as a Harbour Watchkeeper
EO 003.02 Operate LP Air System as a Harbour Watchkeeper

PO 004 Operate Submarine Cooling System as a Harbour Watchkeeper
EO 004.01 Operate Sea Water Cooling System as a Harbour Watchkeeper
EO 004.02 Operate Fresh Water Cooling System as a Harbour Watchkeeper
EO 004.03 Operate Chilled Water Cooling System as a Harbour Watchkeeper

PO 005 Operate Trim, Bilge, and Ballast Systems as a Harbour Watchkeeper
EO 005.01 Operate Trim System as a Harbour Watchkeeper
EO 005.02 Operate Bilge and Ballast Systems as a Harbour Watchkeeper

PO 006 Operate Hydraulic Systems as a Harbour Watchkeeper (OLD PO 002)
EO 006.01 Operate Hydraulic Systems
EO 006.02 Operate Steering and Hydroplanes

Note: Moved from PO 002 to PO 006 as Hydraulic Systems were viewed by students as one of the most complex engineering systems.

PO 007 Operate Combat Systems as a Harbour Watchkeeper
EO 007.01 Operate the Communications Equipment as a Harbour Watchkeeper
EO 007.02 Conduct Rounds on Weapon Systems and Small Arms
EO 007.03 Conduct Rounds on the Tactical Weapons Systems

PO 008 Operate Submarine Domestic Systems as a Harbour Watchkeeper (OLD PO 006)
EO 008.01 Operate Submarine Domestic Systems
EO 008.02 Conduct Rounds on the Garbage Ejector

PO 009 Stand Watch as a Harbour Watchkeeper (NEW)
EO 009.01 Conduct Rounds as a Harbour Watchkeeper
EO 009.02 Operate SCC as a Harbour Watchkeeper
EO 009.03 Conduct Routine Operations as a Harbour Watchkeeper
EO 009.04 React to Onboard Emergencies as a Harbour Watchkeeper

EO 009.05 Respond to Change to Force Protection Levels
EO 009.06 Stand Duty Watch as a Harbour Watchkeeper

Note: Once all of the front end loaded POs and EOs are completed, this PO is designed to enable the trainee to fulfil the duties and responsibilities as a Harbour Watchkeeper.

PO 010 Operate Submarine Escape and Rescue Systems (NEW)
EO 010.01 Perform Submarine Escape and Rescue (Old EO 010.01)
EO 010.02 Conduct Submarine Surface Abandonment (Old EO 010.01)

Note: Given the importance of Submarine Escape and Rescue the Board felt that this be a separate PO rather than combining it with EOPs.

PO 011 Operate as a Safe Submariner at Sea (NEW)
EO 011.01 Operate the Submerged Signal Ejector
EO 011.02 Describe the OMC
EO 011.03 React to Onboard Emergencies at Sea
EO 011.04 Conduct a Sea Watch

Note: This PO was created to give the student a 100% hands on experience at sea and captures such key items as reacting to onboard emergencies, experiencing snorting, returning to periscope depth, emergency go deep, emergency surfacing evolutions, noise husbandry, and domestics. In addition, this PO will have the trainee responding to various submarine states, rotating through 35 Bulkhead, SCC 2, and MCC 2 Watchkeeping positions.

Day Two – Tuesday 26 February

The Chair began by recapping the previous day – fully ensuring that the Board was content with the re-sequencing of the POs and EOs. With all members satisfied, the Board then went on to review the OJPR – Annex E and work back to the POs – a re-engineering application of sorts.

Day Three – Wednesday 27 February

Today, the Board resumed the PO review starting with PO 007 – Operate Combat Systems. The Board, but most notably, CSE SMEs strongly believed that there was far to much expected from a trainee in this PO, that was completely inconsequential to developing a safe submariner at sea. Consequently, teaching points were either simplified, moved elsewhere – to another PO where it made more sense, or for those deemed unnecessary, removed. As an example, external

communications was removed in its entirety as the only crew members that operate the equipment are NAV Comms.

Day Four – Thursday 28 February

Today the Board moved on to PO 003 Operate Air Systems and with the exception of the Emergency Blows, it was deemed that all of the other items could in fact be conducted while alongside. Keeping the same PO number (003) the performance statement was changed to Operate Air Systems as a Harbour Watchkeeper. Given its importance, Emergency Blows were moved to the "at sea portion".

Today's Board concluded early because of the CANSUBFOR Town Hall.

Day Five – Friday 01 March

Today, the Board welcomed both Capt (N) Robinson Commander CANSUBFOR (CCSF) and CPO1 Mouton CANSUBFOR CPO (CSF CPO) who were both able to take time out of their busy schedule for a visit. The Chair briefed both on the Board's progress thus far discussing how the qualification process be separated into two segments, with two distinct MITE Codes. Both CCSF and the CSF CPO agreed that front loading as much theory as possible – with practical training left for the at sea time was definitely the right way ahead, and aligned with what they had envisioned. When the topic of submarine virtual reality training came up, CCSF discussed and strongly agreed that there should be a requirement for all trainees to have either a tablet or laptop assigned to them. The CSF CPO is going to investigate whether CSF can support this.

Day Six – Monday 04 March

As three representatives from Sea Training (Submarines) (ST(S)) the Chair cited that in the interest of capitalizing on the Mechanical SME, today's aim would be to tackle PO 006 – Operate Hydraulic Systems as a Harbour Watchkeeper.

The Board concluded at 1030 because of a winter storm that resulted in the Base closing.

Day Seven – Tuesday 05 March

Today, the Board picked up where we left off prior to the Base closing hoping to make up some lost ground. After a prolonged discussion on the advantages of trainees knowing each and every

single hydel block, and root valve"cold", a decision was made to annotate the OJPR, whereby, the trainee is NOT required to memorize every single hydel valve on the block. Rather, they should locate the hydel block, understand the general purpose of that block, and its respective root valve. For ease and streamlining, the Board separated hydraulics by each compartment/ space and merged all signature blocks for that space into one chunk. In this manner, it allows the trainee to be examined by the designated signing authority on a space by space basis and not individual blocks. The Board also changed the action "locates" to "identify" because when you locate something, you are just sighting it, but to identify something, you are both sighting it and describing the item you are sighting. It was decided that all Cross Connections need to be put into the teaching points, so that it is understood that you can cross-connect systems in an emergency for redundancy purposes.

Day Eight – Wednesday 06 March

Today the Board moved onto PO 005 -- Operate Trim, Bilge, and Ballast Systems as a Harbour Watchkeeper, PO 006 -- Operate Hydraulic Systems as a Harbour Watchkeeper, and PO 008 – Operate Submarine Domestic Systems as a Harbour Watchkeeper. With the exception of some minor changes to Task (actions) – (operate to describe), grammar corrections, and consolidation of signature blocks, there were no major modifications.

Day Nine – Thursday 07 March

Today the Board carried on with reviewing the OJPR.

Day Ten – Friday 08 March

The Board commenced the day reviewing PO 009, Execute Duties of a Member of the Watch at Sea, and decided on reclassifying it to – PO 011, Operate as a Safe Submariner at Sea. The Board went to build a series of EO's within the PO, ensuring to include HISB and R2MR, among other important items such as: Operate the SSE, Operate the OMC, React to Onboard Emergencies, and Conduct a Watch at Sea. The Board included a form of "round robin" for trainees in order to experience different evolutions at sea, by compartment, and watchkeeping spaces. The Board was also fully cognizant that this PO would be revisited often, and populated with items to be conducted at sea.

The Board strongly supported the benefit of having guest speakers such as the CSF Chief, Submarine Escape and Rescue SME, a Submarine PA, Sea Training (Submarine) Coxswain, as well as a newly qualified submariner.

Day Eleven – Monday 11 March

The Chair brought the Board to order and commenced the day by recapping the previous week's proceedings. HMCS Windsor's PA was able to attend the forenoon in order to review and offer SME advise on medical aspects of BSC. This included matters ranging from importance of personal hygiene in a submarine environment; to knowing where the medical kit is located. Because of its nature, the Board reiterated that a Submarine PA is the only one that should be speaking on this important topic. The Board spent the remainder of the day discussing what is pertinent, and what should be taught in class versus captured in the OJPR.

Day Twelve – Tuesday 12 March

The Chair commenced the day reviewing PO 011, Operate as a Safe Submariner at Sea. Upon discussing Casing procedures, the Board came to the conclusion that although very important, there is no requirement for a trainee to conduct a casing crawl because of the associated hazards.

Day Thirteen – Wednesday 13 March

The Chair brought the Board to order and started things off with PO 010, Operate Submarine Escape and Rescue Systems. As this was just a case of moving the material from existing EOs it was more of a verification exercise. However, the Underwater Telephone (UWT) was a major point of discussion and was brought up during escape and rescue review. In the end, the Board concluded that all submariners should know how to operate UWT's in emergency. Since the UWT is operated frequently during diving stations it was decided to capture this in PO 011, Operate as a Safe Submariner at Sea.

Day Fourteen – Thursday 14 March

Today's deliberations focused solely on trainee assessment. Board members recollected the countless discussions surrounding both the POs and EOs, with regard to whether or not a trainee is required to be AILS qualified, to fulfill the duties and responsibilities of a Harbour Watchkeeper. The Board considered all of the factors at length, and the consensus was that a trainee does NOT need to be AILS "Dolphin" qualified in order to stand Harbour Watchkeeper.

As all system knowledge had been front end loaded, and the previous standard was not compromised (in fact some cases more rigorous); the Board was very satisfied with the decision going forward. Moreover, the Harbour Watchkeeper would be considered the trainees "first aiming point" in the OJPR process, and from there, will continue to gain valuable platform experience – knowing the boat cold. In effect, the Harbour Watchkeeper qualification would be a separate MITE Code required prior to embarking on PO 011. The Safe Submariner – at sea portion would strictly be all hands – allowing the trainee to experience the sensation of being at sea. This would include:

- exercising acquired submarine system knowledge and practices from the Harbour Watchkeeper competency;
- participating in at sea emergencies by cycling through various positions and operating key DC equipment;
- cycling through key watchkeeping positions such as SSC 2 and MCC 2; and
- reacting to submarine states and routines.

In order to achieve this, the OJPR would be separated into the following two objectives:

Harbour Watchkeeper – Successful completion of OJPR POs 001 through 009 followed by two walkthroughs:

Lower Deck Trot Walkthrough – conducted by either a HOD or CHOD; and

SCC Walkthrough – conducted by either the MSEO, MAR Tech Coord, or Senior MAR Tech (Elect / Mech).

Successful completion of these two walkthroughs assign MITE code ID (TBD) enabling the individual to perform the duties and responsibilities of a Submarine Harbour Watchkeeper.

Safe Submariner - Successful completion of OJPR POs 010 and 011 followed by a series of walkthroughs:

Four Consolidation Walkthroughs – conducted by the Coxn, MAR Tech Coord, CSEC, and Senior MAR Tech (Elect / Mech); and

Executive Officers Walkthrough – to satisfy the Commanding Officer that the individual can perform effectively and safely as an individual of a submarine crew.

Successful completion of the Executive Officers Walkthrough assigns MITE code ID 109285 (existing AILS) – enabling the individual to perform the duties and responsibilities of a Safe Submariner at Sea.

With the new process for qualifying, the Board believed that six months was ample time to complete the package, but that can be revisited at a later time once new statistics are presented.

The Board was also adamant that going to sea is an **absolute** requirement for the completion of OJPR POs 010 and 011 to ensure the trainee can function as a Safe Submariner. The standard of a minimum of 15 days at sea in order to consolidate OJPR POs 010 and 011 is still required.

Follow On Meeting 12 – 13 June

The Chair met over the course of two days with both NFS(A), Instructional Staff, and NTDC(A) Staff in order to review QSP Firefly conversion progress. This was also an excellent opportunity to brief both the Division Cdr and staff on outstanding items from when the formal Board last met.

RECOMMENDATIONS

1. Throughout the Board process, members kept a running record of the following recommendations:

    - complete the conversion of the QSP from its current format to Firefly;
    - develop a new MITE Code for the Harbour Watchkeeper competency;
    - develop a new MITE Code for Submarine Environmental R2MR competency;
    - maximum use of the following SMEs:
        - R2MR Team;
        - Submarine PA;
        - Sea Training (Submarines) Coxn;
        - CSF CPO; and
        - new submariner;
    - amend the following documentation:
        - 48 Hour Package; and
        - VCSSOs Ch 12;
    - revamp MCS Trainee Tracker to reflect New QSP POs and EOs;
    - to ensure that Sonar Ops receive the necessary OMC training, including formal helm training on AILO (Sonar Op) Course;

- explore the possibility of issuing students a laptop or tablet IOT access Virtually Submarine Training software; and
- develop a series of question banks for consistency.

2. Once approved by the NTDC(A) Cdr and NPTG, recommendations:

- conduct a Pilot Course 26 Sep – 13 Dec 19;
- extend the qualification process from the current 10 weeks to six months;
- explore the viability of bi-coastal instruction (fully cognizant that the trainers reside on the east coast);
- develop a communication plan; and
- devise a CANSUBFOR implementation/ transition plan.

ANNEX C – SCALARS

Not Applicable

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ANNEX D – COSTING INFORMATION

Include information about cost in whichever format is applicable.
This annex is optional (**Not** to be deleted).
Costing is not applicable for QSP-SUB-BSQ (AILS)

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ANNEX E – ASSESSMENT PLAN

EC/PC	Scope Covered	Purpose	How	Pass/Fail Criteria
EC 001.03	EO 001.01 – EO 001.03	To assess student's knowledge of submarine construction.	Written	Student must obtain a pass mark of 70%
EC 002.03	EO 002.01 – EO 002.03	To assess student's knowledge of submarine electrical systems.	Written	Student must obtain a pass mark of 70%
EC 003.02	EO 003.01- EO 003.02	To assess student's knowledge of submarine air systems.	Written	Student must obtain a pass mark of 70%
EC 004.03	EO 004.01- EO 004.03	To assess student's knowledge of submarine cooling systems.	Written	Student must obtain a pass mark of 70%
EC 005.02	EO 005.01- EO 005.02	To assess student's knowledge of submarine trim, bilge, and ballast systems.	Written	Student must obtain a pass mark of 70%
EC 006.01	EO 006.01- EO 006.02	To assess student's knowledge of submarine hydraulic systems.	Written	Student must obtain a pass mark of 70%
EC 007.02	EO 007.02	To assess student's knowledge of submarine weapon handling system.	Written	Student must obtain a pass mark of 70%
EC 007.03	EO 007.01 and EO 007.03	To assess student's knowledge of submarine combat systems.	Written	Student must obtain a pass mark of 70%
EC 008.02	EO 008.01- EO 008.02	To assess student's knowledge of submarine domestic systems.	Written	Student must obtain a pass mark of 70%
EC 009.04	EO 009.04	To assess student's reaction to onboard emergencies as a harbour watchkeeper.	Practical (Trainer)	Student must obtain a pass mark of 80% of all factors and all critical factors

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EC 009.05	EO 009.01- EO 009.03 and EO 009.05	To assess student's harbour watchkeeper knowledge.	Written	Student must obtain a pass mark of 70%
EC 010.02	EO 010.02	To assess student's surface abandonment knowledge.	Written	Student must obtain a pass mark of 70%
EC 011.01	EO 011.01 and EO 007.02	Ta assess student's operation of (and rounds on) the SSE.	Practical (Trainer)	8/10 SATs on non-critical factors and all critical factors IAW EC 011.01 Checklist
EC 011.04	EO 011.02- EO 011.04	To assess student's sea watch knowledge.	Written	Student must obtain a pass mark of 70%
PC 001	PO 001- PO 009	To assess student's harbour watchkeeper performance.	Practical (Walkthough)	
PC 002	PO 010 – PO 011	To assess student's safe submariner performance.	Practical (Walkthough)	

ANNEX F – TEL DEVELOPMENT OPPORTUNITIES

Title	Type	Description of Requirement	Lesson	SOR #
3D Printed Models of Submarine	Models	These models will be used for orientation and possibly scenario based instruction (table top) in order to provide students with visual representation of 3D battlespace.	All lessons	
Improvments to Virtual Submarine	DLN	Increased interactivity with the Virtual Submarine will allow more performance based training and assessment.	All lessons	

ANNEX G – SCHEDULING INFORMATION

Scheduling Information- BSQ													
Lesson		Location	IT Assets					Footprint			Instructor		
#	Time	Location	DWAN	CSNI	SSA	Proj	TEL	Sec Class	WB	Desks	Type	Qual	Sec Cl
RESIDENTIAL PHASE													
001.01A	100	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
001.02A	100	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
001.03A	100	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
EC 001.03	50	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
002.01A	150	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
002.01B	150	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
002.02A	100	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
002.02B	100	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
002.03A	150	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
EC 002.03	50	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
003.01A	200	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
003.01B	100	Classroom (S-100)	1	0	0	1`		UNC	1	20		AILS	UNC
003.02A	100	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
003.02B	100	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
003.02C	100	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC

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003.02D	50	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
EC 003.02	50	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
004.01A	100	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
004.01B	50	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
004.02A	100	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
004.02B	100	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
004.03A	100	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
004.03B	50	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
EC 004.03	50	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
005.01A	200	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
005.02A	300	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
005.02B	250	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
EC 005.02	50	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
006.01A	100	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
006.01B	100	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
006.01C	50	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
006.01D	100	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
006.01E	50	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
006.01F	100	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC

006.02A	150	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
006.02B	100	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
EC 006.02	50	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
007.01A	100	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
007.01B	100	TWS	0	0	0	0		UNC	0	0		AILS	UNC
007.01C	50	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
007.01D	50	TWS	0	0	0	0		UNC	0	0		AILS	UNC
007.02A	150	Classroom (S-100) and WH&DS trg	1	0	0	1		UNC	1	20		AILS	UNC
007.02B	150	Classroom (S-100) and WH&DS trg	1	0	0	1		UNC	1	20		AILS	UNC
007.02C	500	Classroom (S-100) and WH&DS trg	1	0	0	1		UNC	1	20		AILS	UNC
007.02D	100	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
EC 007.02	50	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
007.03A	100	Classroom (S-100) and TWS or Fred Sherwood Trg	1	0	0	1		UNC	1	20		AILS	UNC
EC 007.03	50	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
008.01A	100	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
008.01B	100	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
008.02A	50	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
EC 008.02	50	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC

009.01A	150	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
009.01B	100	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
009.02A	200	SCT/Classroom	1	0	0	1		UNC	1	20		AILS	UNC
009.03A	200	SCT/Classroom	1	0	0	1		UNC	1	20			
009.04A	200	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
009.04B	3000	SCT	0	0	0	1		UNC	0	0		AILS	UNC
009.04C	800	DCFT	-	-	-	-		UNC	-	-			UNC
EC 009.04	1000	SCT	0	0	0	0		UNC	0	0		AILS	UNC
009.05A	250	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
EC 009.05	50	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
010.01A	50	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
010.01B	50	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
010.01C	50	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
010.01D	50	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
010.01E	50	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
010.01F	50	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
010.01G	50	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
010.01H	800	WPET	0	0	0	0		UNC	0	0		AILS	UNC
010.02A	50	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
EC 010.02	50	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC

011.01A	900	Classroom (S-100) and WH&DS trg	1	0	0	1		UNC	1	20		AILS	UNC
EC 011.01	1000	WH&DS trg	1	0	0	1		UNC	1	20		AILS	UNC
011.02A	100	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
011.03A	100	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
001.03B	100	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
011.04A	400	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
011.04B	1000 R2MR	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS PA or Civilan	UNC
011.04C	200	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
011.04D	100	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
011.04E	100	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
011.04F	100	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC
EC 011.04	50	Classroom (S-100)	1	0	0	1		UNC	1	20		AILS	UNC

ANNEX I – ON JOB PERFORMANCE RECOR

Basic Submarine Qualification OJPR

MITE: 122073– Submarine Harbour Watchkeeper

MITE: 109285 – Safe Submariner at Sea

_____

(Service Number) (Rank) (Last Name) (Initials)

_____

Date Training Started Expected Date of Completion

THIS OJPR MUST BE COMPLETED IN ACCORDANCE WITH THE DETAILED INSTRUCTIONS BELOW.

INTRODUCTION

1. This OJPR summarizes the pertinent information, data, and documents required by training officers and supervisors when organizing, conducting, and tracking the OJPR of trainees. The aim of this OJPR is to specify the standards of performance that the trainee must achieve during the OJT portion of the Basic Submarine Qualification (BSQ).

INSTRUCTIONS

2. This OJPR shall be used by HMC Submarines as the primary document governing the content, progress tracking, organization, and conduct of the BSQ OJPR. Requirements of the BSQ OJPR consist of the Performance Factors listed herein, on board training time, two Harbour

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Watchkeeper Walkthroughs, four Consolidation Walkthroughs, and finally, a formal AILS Executive Officers Walkthrough.

3. This OJPR shall be completed within six months after the residential Basic Submarine Qualification portion of the course.

4. The Submarine's Training OPI for the OJPR must maintain Training Tracker fields as outlined in Monitor Mass.

5. As the trainee completes each Performance Factor to the required standard, the assessor possessing the proper signing authority will date, print, and sign their names on the applicable item of the OJPR.

TRAINING OBJECTIVES

6. The OJPR is separated into the following two objectives:

a. **Harbour Watchkeeper** – Successful completion of OJPR POs 001 through 009 followed by two walkthroughs:

1) Lower Deck Trot Walkthrough – conducted by either a HOD or CHOD; and

2) SCC Walkthrough – conducted be either the MSEO, MAR Tech Coord or Senior MAR Tech (Elect / Mech).

Successful completion of these two walkthroughs assign MITE code ID 122073 enabling the individual to perform the duties and responsibilities of a Submarine Harbour Watchkeeper.

b. **Safe Submariner** - Successful completion of OJPR PO's 010 and 011 followed by a series of walkthroughs:

1) Four Consolidation Walkthroughs – conducted by the Coxn, MAR Tech Coord, CSEC, and Senior MAR Tech (Elect / Mech); and

2) Executive Officers Walkthrough – to satisfy the Commanding Officer that the individual can perform effectively and safely as a member of a submarine crew.

Successful completion of the Executive Officers Walkthrough assigns MITE code ID 109285 – enabling the individual to perform the duties and responsibilities of a Safe Submariner at Sea.

Note: PO's 001 thru 011 must be completed prior to the Executive Officers Walkthrough.

SEA TIME

7. Going to sea is an absolute requirement for the completion of OJPR PO's 010 and 011 to ensure the trainee can function as a Safe Submariner. Ordinarily, the standard of a minimum of 15 days at sea in order to consolidate OJPR PO's 010 and 011 is required. However, should the

Commanding Officer be satisfied with the trainee's performance at sea, the qualification will be awarded effective the date of the Executive Officers Walkthrough.

PROGRESS REVIEW

8. Progress reviews shall be carried out every two weeks by the Divisional Officer and Section Head (NCMs) or Training Officer (Officers).

Print Rank, Name and Position	Date	Signature

WALKTHROUGHS

9. Harbour Watchkeeper – Individuals must complete two separate walkthroughs. For the Lower Deck Trot, all HOD's and CHOD's are responsible for this walkthrough. For the Ship Control Console, the MSEO, MAR Tech Coord, Senior MAR TECH (Elect), and/ or Senior MAR TECH (Mech) are responsible for this walkthrough. The table below shall be used to record the completion of both Harbour Watchkeeper Walkthroughs.

10. Consolidation – Individuals must complete four consolidation walkthroughs. The Executive, Combat Systems, Mechanical, and Electrical Systems. In addition to the Coxswain for the Executive Walkthrough, HOD's will normally delegate the conduct of walkthroughs to the CSEC, MAR Tech Coord, and Senior MAR Tech (Elect) / (Mech) accordingly. They may however delegate signing authority to other suitably qualified individuals, as required. The table below shall be used to record the completion of the four Consolidation Walkthroughs.

Note: For the purposes of the MAR Tech Walkthroughs, the Senior MAR Tech Walkthrough shall be conducted by the opposite technical stream of the MAR Tech Coord. For example, in the event that the MAR Tech Coord is from the Mech stream, than the Senior MAR Tech Elect shall conduct the Electrical Walkthrough.

WALKTHROUGH	DATE/PRINT/SIGN	SIGNING AUTH
HARBOUR WATCHKEEPER		
Lower Deck Trot		HOD / CHOD
Ship Control Console		MSEO / MAR Tech Coord Senior MAR Tech

		(Elect) Senior MAR Tech (Mech)
CONSOLIDATION		
Executive		Coxswain CHOP's
Combat Systems		CSEC
Mechanical Systems		MAR Tech Coord Senior MAR Tech (Mech)
Electrical Systems		MAR Tech Coord Senior MAR Tech (Elect)

BASIC SUBMARINE QUALIFICATION TRAINING COMPLETION CERTIFICATION

_____

(Service Number) (Rank) (Last Name) (Initials)

Individual above has successfully passed the Basic Submarine Qualification Walkthrough and has qualified for the nomination of VICTORIA CLASS SUBMARINE QUALIFIED (NQUAL AILS) effective _____.

(Print Name, Rank and Title) (Sign) (Date)

Section Head (NCMs) or HOD (Officers)

(Print Name and Rank) (Sign) (Date)

Executive Officer

(Print Name and Rank) (Sign) (Date)

Commanding Officer

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PO 001 Identify Submarine Construction

PERFORMANCE FACTORS	DATE/PRINT/SIGN	SIGNING AUTH
EO 001.01 – Describe Submarine Layout		
1. Describe Submarine Construction to include: a. External Tank locations b. Compartment layout c. Water Tight Integrity and the operation of all doors and hatches d. Fuel Group layout		QUALIFIED SUBMARINER
2. Describe Routine Casing Procedures to include: a. Safety precautions, hazards , and considerations b. Proper authorization both at sea and alongside c. Purpose of a Casing Crawl		CASO Scratcher
EO 001.02 – Identify Submarine Components		
1. Locate the following Submarine External Hull and Casing fittings to include: a. Torpedo Embarkation Hatch b. ATP Hatch c. Fwd Escape Tower Hatch d. Fwd and Aft SSE e. Surface Mufflers f. Fin Door g. Upper Conning Tower Hatch h. Main Access Hatch i. SBS Well j. Aft Escape Tower Hatch k. Number 1,2,3,4 Main Vents l. Active, Intercept & Ranging Array m. Both 2073 10KHz Pingers n. BQQ-10 Sail o. Fairleads p. Bollards q. Capstans r. High and Low Salvage Connections		CHOPs UC1

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2. Locate the following Shore Connections to include: a. Sewage Connection b. Fwd DC Shore Connection c. Aft DC Shore Connection d. Fwd First Filling e. Subsequent Filling f. Fire Fighting Connection g. Fresh Water Filling h. Battery Ventilation Overboard Discharge i. Chilled Water j. LP Bilge k. Lubrication Oil Filling l. AC Shore Connection m. Nitrogen Filling n. HP Air Connections o. Hydraulic Oil		QUALIFIED SUBMARINER
3. Locate the following Masts and Periscopes to include: a. Mast Head Steaming Light b. Search Periscope c. Radar Mast d. Comms Mast e. Attack Periscope f. Induction Mast g. Exhaust Mast h. VEX Mast		QUALIFIED SUBMARINER

PO 002 Operate Electrical Systems as a Harbour Watchkeeper

PERFORMANCE FACTORS	DATE/PRINT/SIGN	SIGNING AUTH
EO 002.01 – Conduct Rounds on the Main Battery Systems as a Harbour Watchkeeper		
1. Describe the Main Battery to include: a. Battery Compartments b. Number of Battery Cells c. Voltage range d. Identify Fwd / Aft Battery Hatches		MAR TECH ELEC
2. Conduct a Battery Dip by entering the Battery Compartment to include: a. Proper permission b. Proper pipe c. Safety precautions d. Dipping procedures (including manual) e. Log entries		MAR TECH ELEC
3. Conduct Battery Shutdown procedures for a Battery Fire to include: a. Isolate Fwd / Aft Battery Ventilation System b. Isolate Battery Agitation Bulkhead Isolations		MAR TECH ELEC
4. Locate all Hydrogen Detection Units		MAR TECH ELEC
5. Locate the Hydrogen Clearance Fans and describe the following: a. Purpose b. How and where to operate c. When operated		MAR TECH ELEC

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6. Locate the Battery Fans and describe the following: a. Purpose b. Power Supply c. Control and Indication d. When operated		MAR TECH ELEC
7. Change over from Battery Ventilation to Hydrogen Clearance, or vice versa		MAR TECH ELEC
8. Operate Battery Agitation system and explain when it would be required		MAR TECH ELEC
9. Operate Battery Cooling System and explain when it would be required		MAR TECH ELEC
EO 002.02 – Conduct Rounds on the Submarine DC Electrical System as a Harbour Watchkeeper		
1. Describe the function of the Main VPDC System to include: a. Voltage range of the system b. Location of Fwd / Aft DC Shore Supply Connections c. Location of both Battery Switchboards d. Location of Fwd / Aft DC Load Centres and all Equipment and Starters fed from each with their respective Re-set buttons		MAR TECH ELEC

PERFORMANCE FACTORS	DATE/PRINT/SIGN	SIGNING AUTH
2. Identify the Fwd/Aft Battery Switchboards to include: a. Isolating Links b. Shore Supply Links c. Mid-Point Links d. Auxiliary Power Fuse Board to include both Normal and Alternate Supplies to 140 KW MGs and Load Centers e. Main Power Breakers f. Auxiliary Power Breakers g. Risers from the Battery to the Switchboard		MAR TECH ELEC
3. Describe how to switch on Secondary Lighting		MAR TECH ELEC
4. Describe the Control and Indication functions for both the Main and Auxiliary Power Breakers to include: a. Positions of Operation: 1) Local 2) Remote b. Positions of Indication: 1) SCC 2) MCC 3) Switchboards		MAR TECH ELEC
24 VDC System		
5. Describe the function of the 24 VDC System to include: a. Grade One (Battery Back-up) b. Grade Two		MAR TECH ELEC
6. Locate the 24 VDC Distribution / Charging Panel and 24 VDC LP Battery		MAR TECH ELEC
7. Conduct procedure for an Earth Check on the 24 VDC System		MAR TECH ELEC

PERFORMANCE FACTORS	DATE/PRINT/SIGN	SIGNING AUTH
EO 002.03 – Conduct Rounds on the Submarine AC Electrical System as a Harbour Watchkeeper		
1. Locate and describe the 440 VAC System to including the following components: a. 140KW MG Sets and Starters b. AC Shore Supply Hull Connections c. AC Switchboards d. AC Shore Supply Link Box e. AC Load Centers (4)		MAR TECH ELEC
2. Describe Submarine Emergency Power and locate the associated Cables		MAR TECH ELEC
3. Locate the following major equipment and starters fed by the 440 VAC System to include: a. Fridge Compressors b. Trim Pumps c. Transformer Rectifier Units d. Agitation Blower e. Hydraulic Pumps f. Static Frequency Converters g. Air Treatment Units h. LP Bilge Pump i. Chiller Units j. Fresh Water Cooling Pumps k. Sea Water Cooling Pumps l. Battery Cooling Pump		MAR TECH ELEC
4. Locate the Ventilation Control Panel and describe what items are operated from the Panel.		MAR TECH ELEC
5. Locate the Galley Isolation Switch		MAR TECH ELEC

PO 003 Operate Air Systems as a Harbour Watchkeeper

PERFORMANCE FACTORS	DATE/PRINT/SIGN	SIGNING AUTH
EO 003.01 – Operate HP Air System as a Harbour Watchkeeper		
1. State the following HP Air System Functions: a. HP Air including Operation b. Normal Blow c. Emergency Blow d. Integrity Valves		MAR TECH MECH
2. Locate and operate GIVs, Bottle Group Hull Valves, and #1 and #2 Bottle Group Pressure Gauges		MAR TECH MECH
3. Locate and operate Emergency Blow and Normal Blow in hand [**for exercise**] along with the associated Pressure Gauges		MAR TECH MECH
4. Locate Weapons Firing Cross-Connection to Isolations		MAR TECH MECH W ENG Fire Control
5. Locate Ships Services Fwd Cross-Connection to include the following supplies: a. SCBA Charging b. SSE Reducer c. BIBS Charging Connection d. Fwd Main AV & B Reducer and Isolation Valve e. Captain's Valve HPA 819 f. Port / Stbd Cross-Connection By-Pass Valve HPA 840		MAR TECH MECH
6. Locate 35 Bulkhead HP Air Isolation Valves		MAR TECH MECH
ATP Space		
7. Locate Emergency Bottle Group Hull Valve		MAR TECH MECH

PERFORMANCE FACTORS	DATE/PRINT/SIGN	SIGNING AUTH
Mastwell Space		
8. Locate D-Tank Reducer		MAR TECH MECH
Control Room		
9. Locate the Control Room Cross-Connection including supply to the following: a. D-Tank Reducer b. Emergency Blow Control c. After Hydroplanes in Air Emergency d. Desiccators Panel		MAR TECH MECH
10. Locate Control Room HP Air Local Pressure Gauges and 35 Bulkhead and 56 Bulkhead Isolation Valves		MAR TECH MECH
Engine Room		
11. Locate 56 Bulkhead HP Air Isolation Valves		MAR TECH MECH
12. Locate Engine Cross-Connections to include the following: a. Diesel Air Start Reducing Station b. SCBA Charging c. Cross-Connection Isolations d. SFU 90 Bottle Charging Connection		MAR TECH MECH
Motor Room		
13. Identify HP Air Compressor Cross-Connections to include the following: a. HP Air Compressor Discharge Valve b. Shore Charging Hull Connection		MAR TECH MECH

PERFORMANCE FACTORS	DATE/PRINT/SIGN	SIGNING AUTH
14. Conduct an air charge on the HP Air System IAW VCSOPs to include the following: a. HP Air Compressors b. Shore Charging c. Identify Drier components to include: 1) Towers 2) Tower Drains 3) Tower Isolation 4) Valves 5) Control Panel		MAR TECH MECH
15. Locate the Motor Room System Cross-Connection to Include the following: a. Supply to Aft SSE b. Captain's Valve HPA 299 c. Aft Main AV & B Reducer d. Isolation Valve e. BIBS Charging Connection f. Port / Stbd Cross-Connection HPA 296 g. Reactivating Air Supply		MAR TECH MECH
16. Locate and operate GIVs, Bottle Group Hull Valves, and #3 and #4 Bottle Group Pressure Gauges		MAR TECH MECH
17. Demonstrate Charging SCBA in accordance with VCSOP's		MAR TECH MECH W ENG Fire Control
EO 003.02 – Operate LP Air System as a Harbour Watchkeeper		
WSC		
1. Locate WSC AV & B Reducer Supplies to include the following: a. Fwd Trim Tanks b. WRTs c. Sewage Tank d. EBS (including pressure)		MAR TECH MECH

PERFORMANCE FACTORS	DATE/PRINT/SIGN	SIGNING AUTH
2. Locate the following Pressure Gauges: a. EBS b. Main AV & B		MAR TECH MECH
3. Locate the Compartment Salvage Air (Compartment Blow) to Control Room		MAR TECH MECH
4. Locate 35 Bulkhead AV & B Isolation Valves		MAR TECH MECH
ATP Space		
5. Locate Trim Tank Vent & Blow Valves		MAR TECH MECH
2 Deck FWD 35 Bulkhead		
6. Locate Sewage Tank Vent & Blow Valves		MAR TECH MECH
7. Locate 35 Bulkhead Isolation EBS Valves		MAR TECH MECH
Mastwell Space		
8. Identify Galley Slop Tank Vent & Blow Valve		MAR TECH MECH
Control Room		
9. Identify 35 Bulkhead AV & B Isolation Valve		MAR TECH MECH

PERFORMANCE FACTORS	DATE/PRINT/SIGN	SIGNING AUTH
10. Identify Control Room AV & B Reducers including: a. Slop Tank Drain b. EBS Reducer and Pressure Gauge c. M Port Tank Vent and Blow Valves including IPB d. FW/DW Reducer		MAR TECH MECH
11. Identify 56 Bulkhead AV & B and Salvage Air (Compartment Blow) Isolations		MAR TECH MECH
Engine Room		
12. EBS Reducer and Pressure Gauge		MAR TECH MECH
Motor Room		
13. Identify Aft Main AV & B Reducer and Gauge		MAR TECH MECH
14. Identify Motor Room AV & B Reducers to include Pressure for Trim Tanks		MAR TECH MECH
15. Locate and describe the function of the Inflatable Shaft Seal		MAR TECH MECH

PO 004 Operate Submarine Cooling Systems as a Harbour Watchkeeper

PERFORMANCE FACTORS	DATE/PRINT/SIGN	SIGNING AUTH
EO 004.01 – Operate Sea Water Cooling System as a Harbour Watchkeeper		
1. Locate the following components supplied by the Sea Water Cooling System to include: a. Battery Cooling Heat Exchanger SW 229 b. FW / SW Heat Exchanger SW 218 / 219 c. SAN System SW 402 to include Relief Valve		MAR TECH MECH
2. Operate Sea Water Pumps to include: a. Local and Remote Control Modes of Operation b. Harbour Line-up		MAR TECH MECH
EO 004.02 – Operate Fresh Water Cooling System as a Harbour Watchkeeper		
1. Locate the following equipment supplied by the Fresh Water Cooling System to include: e. Fwd 140 KW MG Air Cooler AW 603 f. Chilled Water Plant Condensers AW 409 / 410 g. Aft 140 KW MG Air Cooler AW 215 h. Port / Stbd HP Air Compressors AW 207 / 208		MAR TECH MECH
2. Operate Fresh Water Pumps to include: a. Local and Remote control Modes of Operation b. Harbour Line-up		MAR TECH MECH
3. Top up Fresh Water Expansion Tanks IAW VCSOPs and demonstrate proper reading of the Sight Glass		MAR TECH MECH
EO 004.03 – Operate Chilled Water Cooling System as a Harbour Watchkeeper		
1. Conduct Harbour Rounds on AC Plants		MAR TECH MECH

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2. Top up Chilled Water Expansion tank IAW VCSOPs and demonstrate reading of Sight Glass		MAR TECH MECH
3. Locate the following equipment supplied by the Chilled Water Cooling System to include: a. Ships Refrigeration Plant b. ATU's c. Main Hydraulic Plant Rep Tank Cooler d. External Hydraulic Plant Rep Tank Cooler e. Steering & Hydroplanes Hydraulic Plant Rep Tank Cooler		MAR TECH MECH
4. Locate the following equipment supplied by the Chilled Water Cooling System Components to include: a. Shore Supply Connections AC 421 / 425 b. AC Plants c. Control and Indication (Local / MCC)		MAR TECH MECH

PO 005 Operate Trim, Bilge and Ballast Systems as a Harbour Watchkeeper

PERFORMANCE FACTORS	DATE/PRINT/SIGN	SIGNING AUTH
EO 005.01 – Operate Trim System as a Harbour Watchkeeper		
1. State the functions of the Trim System including Firefighting		MAR TECH MECH
2. Locate the following Trim System Dip Rods, Tanks, and Tank Isolation Valves to include: a. WRT Port/Stbd Dip Rod b. FWD Trim Tanks WS 801/802 c. TOTs WS 803/804 d. M Port WS 626 e. M Stbd WS 627 (Trim) and WS 611 (Ballast) f. R Port WS 406 g. R STBD WS 405 h. After Trim Tanks WS 201/202 i. D-Tank Hull and Back Up Value WS 616/617 and Location (D-Tank inside of #4 OFT)		MAR TECH MECH
3. Locate the following Fire Fighting arrangements associated with the Trim System to include: a. Supply to Fwd Center Fed Hose Reel WS 701 and location b. Trim System Supply to AFF System AFF 605 c. Supply to Engine Room d. Spray AFF 602/601 e. Supply to AFT Center Fed Hose Reel AFF 603 f. Shore Fire Fighting Hull Valve AFF 606 g. WSC Spray SWS 802		MAR TECH MECH

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PERFORMANCE FACTORS	DATE/PRINT/SIGN	SIGNING AUTH
4. Locate Trim System Bulkhead Isolation Valves WS 809 / 629		MAR TECH MECH
EO 005.02 – Operate Bilge and Ballast Systems as a Harbour Watchkeeper		
1. State the function of the HP Bilge and HP Ballast System (including associated Priming Systems) and locate the following valves: a. HP Bilge Hull and Back Up Valves BS 401 / 402 b. HP Bilge-Ballast System Cross- Connect BS 411 c. LP-HP Bilge System Cross Connect BS 406 d. HP Ballast Hull and Back Up Valves WS 601 / 602		MAR TECH MECH
2. State the function of the LP Bilge System and locate the following valves: a. 35 Bulkhead Isolation BS 801 / BS 606 b. WSC Bilge Suction BS 802 c. LP Bilge Hose Connection BS 805 d. Garbage Ejector LP Bilge Suction BS 613 e. Mastwell Bilge Suction BS 604 f. Hose Connection BS 605 g. AMS Bilge Suction BS 602 h. Hose Connection BS 603 i. Engine Room Bilge Suction BS 414 j. LP Bilge Hose Connection Fwd BS 415 / Aft BS 436 k. LP Bilge Overboard Discharge Hull Valve (Engine Room) l. Motor Room Hose Connection BS 202 m. Clean / Dirty Bilge Tanks including Dip Rods		MAR TECH MECH
3. Identify #1 and #2 Battery Sump Suctions including: a. Reasons for Shut and Locked b. Location of the Key c. Pumping Routine		MAR TECH MECH

PERFORMANCE FACTORS	DATE/PRINT/SIGN	SIGNING AUTH
4. Operate LP Bilge System, to include: a. Pump Bilge to Clean/Dirty Bilge Tank b. Precautions and Limitations		MAR TECH MECH

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PO 006 Operate Hydraulic Systems as a Harbour Watchkeeper

PERFORMANCE FACTORS	DATE/PRINT/SIGN	SIGNING AUTH
EO 006.01 – Operate Hydraulic Systems as a Harbour Watchkeeper		
NOTE: FOR THIS PO TRAINEES ARE **NOT** EXPECTED TO IDENTIFY EACH SPECIFIC HYDEL VALVE BUT RATHER EACH HYDEL BLOCK		
WSC		
1. Locate the following Main and External Cross Connections: a. Supply (Main/External) HS 801 b. Return(Main/External) HS 802		MAR TECH MECH
2. Locate the following Return Relief Valves including pressure: a. Main HS 833 b. External EHS 845		MAR TECH MECH
3. Locate Supply to the following: a. Anchor / Windlass EHS 803 b. Capstan EHS 801		MAR TECH MECH W ENG Fire Control
4. Locate #1 and #2 Main Vents and confirm state		MAR TECH MECH
5. Locate Main and External Pressure Gauges		MAR TECH MECH
6. Locate HS 839 (Root Valve) including the two Hydel Blocks: a. Supply to GIVs b. FWD Trim Tanks c. MBT Vent Valve		MAR TECH MECH

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7. Locate HS 821 (Root Valve) including the two Hydel Blocks: a. WRT Isolation Valve (Stbd) b. #1 and #2 MBT Blow Valves c. Fwd Trim Tank Isolation Valve d. WRT Isolation Valve (Port)		MAR TECH MECH
8. Locate supply to Weapons Discharge EHS 830		W ENG Fire Control
2 DECK AFT 35 BLKHD		
9. Operate 35 Bulkhead Door in hand and by Local Operating Switch.		QUALIFIED SUBMARINER
10. Locate Induction Hull Valve Ballast Pump Hull Valve and two Accumulators (Fan Space)		MAR TECH MECH
AMS		
11. Conduct rounds on Main and External Power Plants, describe and locate the following: a. Rep Tank Level b. Unloader Valves c. SIRCO Control Units d. Plant Accumulators and Nitrogen Bottle e. Plant By-pass f. Pressure Gauges g. Heat Exchanger h. Plant Trips		MAR TECH MECH
12. Explain Unloader Operation including the following: a. Two Manual Unloading Hydels b. Manual control of plant in 24V Grade 2 failure		MAR TECH MECH

PERFORMANCE FACTORS	DATE/PRINT/SIGN	SIGNING AUTH
13. Explain indication of oil emulsification		MAR TECH MECH
CONTROL ROOM		
14. Locate 35 Bulkhead Isolation Valves (Main and External Hydraulics)		MAR TECH MECH
15. Locate HS 619 Root Valve including associated Hydel Block consisting of the following Hydels: a. M Port Vent b. D Tank Ballast Suction c. Conning Tower Flood / Drain d. ATP Bilge Suction e. WSC Bilge Suction f. WTB 35 Door HS 611		MAR TECH MECH
16. Locate Root Valve to Conning Tower Upper Lid EHS 652		MAR TECH MECH
17. Locate Root Valve to Comms / Radar Mast EHS 605		MAR TECH MECH
18. Locate Root Valve to Search / Attack Periscope HS 677		MAR TECH MECH
19. Locate Root Valve HS 621 including associated Hydel Block consisting of the following Hydels: a. Snort Drain 1 b. Fwd Trim Pump By-pass c. M Stbd Trim Isolation d. M Port Trim Isolation e. M Stbd Ballast Isolation		MAR TECH MECH

PERFORMANCE FACTORS	DATE/PRINT/SIGN	SIGNING AUTH
20. Locate Root Valve HS 641 including associated Hydel Block consisting of the following Hydels: a. Fwd trim Pump suction b. Fwd Trim Pump Tank Suction c. Fwd Trim Pump Tank Discharge d. D-Tank Blow e. Fwd Trim Pump Discharge		MAR TECH MECH
21. Locate Root Valve HS 681 Supply to HS 659 and HS 660		MAR TECH MECH
22. Locate Root Valve HS 660 including associated Hydel Block consisting of the following Hydels: a. Snort Induction Hull Valve b. VEX Hull Valve c. D-Tank Flood Hull Valve d. HP Ballast Back Up Valve		MAR TECH MECH
23. Locate Root Valve HS 601 Supply to BWA		MAR TECH MECH
24. Locate Root Valve HS 659 including associated Hydel Block consisting of the following Hydels: a. Ballast Pump Hull Valve b. D-Tank Vent c. D-Tank Flood Back-up Valve		MAR TECH MECH
25. Locate Root Valve EHS 603 including associated Hydel Block consisting of the following Hydels: a. Diesel Exhaust Mast b. Snort Induction Mast		MAR TECH MECH
26. Locate Root Valve HS 701 including associated Hydel Block consisting of the following Hydels: a. Motor Room Rapid Suction b. HP / LP Bilge Cross Connect c. Ballast Pump By-pass d. Ballast Pump Suction		MAR TECH MECH

PERFORMANCE FACTORS	DATE/PRINT/SIGN	SIGNING AUTH
27. Locate Fire Isolation Valve Hydels consisting of the following Hydels: a. EHP 624 b. MHP 624		MAR TECH MECH
28. Locate Main and External Return Line relief valves to include: a. HS 676 b. EHS 644		MAR TECH MECH
29. Locate 56 Bulkhead Main / External Isolation Valves		MAR TECH MECH
30. Locate Snort Induction Inboard Drain Valve By-pass HS 631 and Supply to Conning Tower Hatch Isolating Valve EHS 652 (EOP – Valves used for Collapsing Hydraulic Plants)		MAR TECH MECH
31. Locate Main / External System Pressure Gauges		MAR TECH MECH
ENGINE ROOM		
32. Locate Root Valve HS 586 including associated Hydel Block consisting of the following Hydels: a. HP Bilge Flushing Inlet Valve b. HP Bilge Flushing Stop c. Mast Well Bilge Suction d. AMS Rapid Suction		MAR TECH MECH
33. Locate Root Valve HS 560 Supply to WTD 56 Bulkhead		MAR TECH MECH
34. Operate 56 Bulkhead Door in Hand and by Local Operating Switch		MAR TECH MECH

PERFORMANCE FACTORS	DATE/PRINT/SIGN	SIGNING AUTH
35. Locate Root Valve HS 410 Supply to HP Bilge Back up including Accumulator		MAR TECH MECH
36. Locate Root Valve HS 401 including associated Hydel Block consisting of the following Hydels and Accumulator: a. Port Ships SW Inlet Hull Valve b. Port Ships SW Outlet Back Up Valve c. Stbd Ships SW Inlet Back Up Valve d. Stbd Ships SW Outlet Hull Valve		MAR TECH MECH
37. Locate Root Valve HS 421 including associated Hydel Block consisting of the following Hydels and Accumulators: a. Port Ships SW Inlet Back Up Valve b. Port Ships SW Outlet Hull Valve c. Stbd Ships SW Inlet Hull Valve d. Stbd Ships Outlet Back Up Valve		MAR TECH MECH
38. Locate Root Valve HS 481 including Accumulator and HP Bilge Pump Hull Valve		MAR TECH MECH
39. Locate Root Valve HS 491 including associated Hydel Block consisting of the following Hydels: a. Trim / Bilge Cross Connection b. HP Bilge Pump By-pass c. LP Bilge Overboard Discharge d. Bilge / Ballast Cross Connection		MAR TECH MECH
40. Locate Root Valve HS 510 including associated Hydel Block consisting of the following Hydels: a. R Stbd Trim Isolation b. R Port Trim Isolation c. R Port Ballast d. HP Bilge Pump By-pass		MAR TECH MECH

PERFORMANCE FACTORS	DATE/PRINT/SIGN	SIGNING AUTH
41. Locate Root Valve HS 540 including associated Hydel Block consisting of the following Hydels: a. Stbd Interspace Drain b. Stbd Silencer Drain c. Port Interspace Drain d. Port Silencer Drain		MAR TECH MECH
42. Locate Root Valve HS 520 including associated Hydel Block consisting of the following Hydels: a. LP Bilge Tank Discharge b. LP Bilge Tank Suction c. Dirty Bilge Tank Suction d. Clean Bilge Tank Suction		MAR TECH MECH
43. Locate Root Valve HS 491 including associated Hydel Block consisting of the following Hydels: a. Trim / Bilge Cross-Connect b. HP Bilge Pump By-pass c. LP Bilge Overboard Discharge d. Bilge Pump to Ballast Pump Cross Connect		MAR TECH MECH
44. Locate Root Valve HS 461 including associated Hydel Block consisting of the following Hydels including accumulator: a. Stbd DSW Inlet Hull Valve b. Stbd DSW Outlet Back Up Valve c. Port DSW Inlet Hull Valve d. Port DSW Outlet Back Up Valve		MAR TECH MECH
45. Locate Root Valve HS 441 including associated Hydel Block consisting of the following Hydels including accumulator: a. Stbd DSW Inlet Back Up Valve b. Stbd DSW Outlet Hull Valve c. Port DSW Inlet Back Up Valve d. Port DSW Outlet Hull Valve		MAR TECH MECH

PERFORMANCE FACTORS	DATE/PRINT/SIGN	SIGNING AUTH
MOTOR ROOM		
46. Locate Root Valve HS 291 including associated Hydel Block consisting of the following Hydels including accumulator: a. Stbd DG Hull Valve b. Port DG Back Up Valve		MAR TECH MECH
47. Locate Root Valve HS 281 including associated Hydel Block consisting of the following Hydels including accumulator: a. Stbd Aft Trim Tank Isolation Valve b. Port Aft Trim Tank Isolation Valve c. Aft Trim Pump Tank suction Valve d. Aft Trim Pump By-pass Valve e. Trim Pump Discharge Valve f. Aft Trim Pump Isolation Valve g. Aft Trim Pump Isolation Tank Valve		MAR TECH MECH
48. Locate Root Valve HS 241 including associated Hydel Block consisting of the following Hydels: a. #4 MBT Port and Stbd Vent Valve b. #3 MBT Stbd Vent Valve c. #3 MBT Port Vent Valve d. #3 MBT Blow Valve e. #4 MBT Blow Valve f. #4 GIV g. #3 GIV		MAR TECH MECH
49. Locate Root Valve HS 274 Supply to Aft SSE		MAR TECH MECH
50. Locate Root Valve HS 237 Supply to Turning Gear		MAR TECH MECH
51. Locate Root Valve EHS 211 Supply to Aft Capstan		MAR TECH MECH

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52. Locate Root Valve EHS 201 Supply to Towed Array Cable Release, including Hydel		MAR TECH MECH
53. Locate Return Line Relief Valves: a. Main HS 208 b. External EHS 217		MAR TECH MECH
54. Conduct Rounds on Steering & Hydroplane Power Plant, to include: a. Rep Tank Level b. Locate AC pump c. Locate DC pump d. Locate Unloader Valves e. Locate SIRCO Control Units f. Locate Plant Accumulators/ Nitrogen Bottles g. Locate Plant By-pass h. Locate Pressure Gauges i. Locate Heat Exchanger		MAR TECH MECH
55. Locate Hydraulic Storage Tank		MAR TECH MECH
56. Locate and confirm status of: a. #3 Main Vents b. #4 Main Vent		MAR TECH MECH
EO 006.02 – Operate Steering & Hydroplanes as a Harbour Watchkeeper		
WSC		
1. Locate Supply to Fwd Hydroplane (Main) HS 831 Fwd Hydroplane (External) EHS 840		MAR TECH MECH
ATP		
2. Locate and explain Safety Considerations while working around the Fwd Hydroplanes		MAR TECH MECH

PERFORMANCE FACTORS	DATE/PRINT/SIGN	SIGNING AUTH
ENGINE ROOM		
3. Locate Steering & Hydroplanes Plant Fire Isolation Hydel HHP 212		MAR TECH MECH
MOTOR ROOM		
4. Locate Root Valve HS 265 Supply to Steering System		MAR TECH MECH
5. Locate Dive Stop Actuator By-pass Valve HSD 240 (EOP – Valves used for Collapsing Hydraulic Plants)		MAR TECH MECH
6. Locate and explain Safety Considerations working around the Aft Steering & Hydroplanes		MAR TECH MECH

PO 007 Operate Combat Systems as a Harbour Watchkeeper

PERFORMANCE FACTORS	DATE/PRINT/SIGN	SIGNING AUTH
EO 007.01 - Operate the Communications Equipment as a Harbour Watchkeeper		
1. Describe the functions of the Internal Communication Systems		W ENG Comms
2. Locate and describe the following components of the Internal Communications System: a. VICES b. VCS 446 c. DC / Maintenance Net d. Command Net e. PRCs		W ENG Comms
3. Operate the following Internal Communications Systems: a. VICES b. VCS 446 c. DC/maintenance Net d. Command net e. PRCs		W ENG Comms
4. Locate, describe and operate the External VHF System		DWS
EO 007.02 – Conduct Rounds On the Weapon Systems and Small Arms as a Harbour Watchkeeper		
1. Describe the function of the Submerged Signal Ejector		W ENG Fire Control
2. Locate and describe the following components of the Submerged Signal Ejector: a. Interlocks b. Air System c. Hydraulic System d. Local Operators Panel e. Sequencing manifold		W ENG Fire Control

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3. Locate and describe the various Pyrotechnics and Ammunition to include the following: a. Operation b. Safety c. Types d. Function e. Construction f. Flooding arrangements and stowage g. Accidental Activation (VCEOP)		W ENG Fire Control
4. Conduct Weapons rounds of the WSC to include: a. Torpedo Tube Status: 1) Outboard Operating Pressure 2) Rear Door Tally 3) Rear Door Chained and Locked b. Secondary Display Decoder Indications c. Hydraulic Control Packs Line-up d. WSC Bilge e. Tube Operator Console f. Submerged Signal Ejector Breach Shut g. Mk 48 Torpedo Stowage h. Otto Fuel / HAZMAT Spill (VCEOP) i. Ammunition/Pyrotechnics Lockers Secured		W ENG Fire Control

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5. Locate and describe the major components of the Weapons Handling Discharge System to include the following: a. Torpedo Tube layout b. Tanks to include: 1) WRTT 2) TOT 3) WTT c. ATP to include: 1) Drains 2) Inflatable Seals 3) Inflatable Seal Reducing Station 4) ATP 821 Valve Tally 5) Firing Air Reducing Stations 6) TOT Vents 7) ATP Bilge		W ENG Fire Control
6. Locate and Describe Small Arms types to include the following: a. C8 b. Sig Sauer c. Signal Flare pistols d. Line Throwing Devices e. Stowage / Security Location Small Arms Log Book		W ENG Fire Control
EO 007.03 – Conduct Rounds On The Tactical Weapons Systems as a Harbour Watchkeeper		
1. Locate and describe the following components of the fitted Communications Suite to include: a. Navigation RADAR b. Sea Search 2 c. Fitted Masts d. Emergency Portable RADAR (FURUNO)		W ENG Comms

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2. Locate and describe the components of the fitted Acoustic Suite to include: a. 2073 Emergency Underwater Telephone b. 2008 Underwater Telephone c. BQQ-10 d. Echo Sounder e. SDM (SHINNADS Dual Monitor) f. INS (Inertial Navigation System) g. NIDS (Navigational Information Display Console) h. TTVC (Tactical Television Console) i. CCS 876 (Fire Control System) j. Weapons System Data Bus k. Periscopes		W ENG Sonar

PO 008 Operate Submarine Domestic Systems as a Harbour Watchkeeper

PERFORMANCE FACTORS	DATE/PRINT/SIGN	SIGNING AUTH
EO 008.01 – Operate Submarine Domestic Systems as a Harbour Watchkeeper		
1. Describe the function of the following Domestic Systems: a. Sewage b. Galley Slop c. Fresh Water to include requirement with ammunition embarked		MAR TECH MECH W ENG Fire Control
2. Conduct Flush Through Soil Pipe Procedure: a. Shut SAN 809 WSC Drain Isolation b. Open SAN 810 Sea Water Flushing		W ENG Fire Control
3. Identify Sewage / Slop Tank Level Indications to include: a. Local Dip Rod b. Local Full / Empty Indicator		Qualified Submariner
4. State the positions for the Ammo Locker Flood Valves to include: a. WSC Ammo Locker FWC 801 CO's Cabin / WSC b. Ammo Locker Flood 2 Deck Passage Way either side of 35 Bulkhead c. Ammo Locker Flood Motor Room either side of 56 Bulkhead		W ENG Fire Control

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5. Identify Domestic Fresh Water Tanks to include location, dip rods, and level indicators for the following: a. Port Fresh Water b. Stbd Fresh Water		MAR TECH MECH
6. Describe changing over Fresh Water tanks		MAR TECH MECH
7. Locate and state the purpose of the Submarine Reverse Osmosis Desalination Unit		MAR TECH MECH
EO 008.02 – Conduct Rounds on the Garbage (Gash) Ejector as a Harbour Watchkeeper		
1. State the function of the Garbage Ejector and Describe the importance of Water Tight Integrity (WTI)		DWS
2. Describe the alongside state of the Garbage Ejector to include: a. Flood Valves b. Drain Valves c. Breech Door d. Muzzle Door e. Flushing Valve f. LP Bilge Suction g. Test Cock and Reamer h. Telegraph		DWS

PO 009 Stand Watch as a Harbour Watchkeeper

PERFORMANCE FACTORS	DATE/PRINT/SIGN	SIGNING AUTH
EO 009.01 – Conduct Rounds as a Harbour Watchkeeper		
1. Conduct Setting the Trot and correct Rounds Procedures IAW the Lower Deck Trot Sheet to include the following: a. Perform a Set of Rounds utilizing the Lower Deck Trot Sheet b. Locate and describe the following Duty Watch Books to include: 1) MSEO's Night Orders 2) Turn Over Sheets 3) Visitor / Contractor Log 4) Small Arms Rounds 5) Ships Log 6) DWS Pack 7) SCANDEX 8) Lock Out Tag Out 9) Hot Work / Gas Free Chits 10) Fridge Log 11) Man Aloft Chits 12) Chiller Rounds Log 13) Divers Down		DWS
2. Submarine Buoyancy to include: a. Recording Draught Marks b. Describing reasons for changes in the Submarine's Draught		DWS
EO 009.02 – Operate SCC as a Harbour Watchkeeper		
1. Monitor all Alarms to include Grade A, B, Fire and Flood		QUALIFIED SUBMARINER

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2. Monitor Main, External, and Steering & Hydroplanes Hydraulic Plant Alarms, Levels, and Pressures to include: a. Rep Tanks Low Level Grade B b. Rep Tanks High Level Grade B c. Low Operating Pressures Grade A / B		MAR TECH MECH
3. Operate Hydraulic Plant Control to include: a. Harbour VCSOP Line-up b. Demonstrate Plant Operation c. Main Plant Pressures 1) Line High 2) Line Low 3) Auto d. External Plant Pressures: 1) Auto 2) Line e. Steering & Hydroplanes Plant Pressures: 1) Auto 2) Line f. Fire Isolations Valves IPB's g. Pressure Gauges		DWS DWS
4. Perform the following HP Air System functions to include: a. Operate GIV IPBs b. Monitor HP Air Bottle Group Pressure Gauges c. Monitor MBT Blows d. Identify Emergency Blow Quick Acting Valves e. Identify Lock / Unlock Switches f. Monitor Emergency Bottle Group Pressure Gauges		QUALIFIED SUBMARINER

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5. Perform the following Trim System functions to include: a. Operate Trim Pump Start / Stop Remote (where to find indication - Dowty) b. Line up Trim System for Firefighting from M Port. c. Line up Trim System for Firefighting from Aft Stbd Trim Tank d. Line up Trim System for Firefighting from Shore Supply e. Revert Trim System to Normal Harbour VCSOP Line Up f. Monitor Flow Meters		DWS
6. Operate LP Bilge System to include: a. Pump Bilges to Clean and Dirty Bilge Tanks b. Operate Wandering Leads c. Monitor Tank Content Gauges		QUALIFIED SUBMARINER
7. Operate HP Bilge and Ballast System to include: a. Priming Pumps b. Pumping to Sea (for VCEOP) c. Hull and Back Up Valves d. Emergency Hull Valve Shutdown Switch e. Monitor Flow Meter		OPCERT
8. Operate Surveillance System to include: a. Navigation of Pages b. Acknowledge Alarms c. Inhibit / Enable Alarms d. Surveillance Modes 1,2,3 & 5		DWS
9. Identify components of the Mimic Panel		DWS
EO 009.03 – Conduct Routine Operations as a Harbour Watchkeeper		

PERFORMANCE FACTORS	DATE/PRINT/SIGN	SIGNING AUTH
1. Perform the following functions at the MCC IAW VCSOPs to include: a. Monitor Fire Detection System Alarm b. Operate Battery Ventilation c. Operate Battery Agitation d. Operate Battery Cooling e. Operate Sea Water Cooling f. Operate Fresh Water Cooling g. Operate HP Air Charge		DWS
2. Conduct Blowing Around IAW VCSOP's to include: a. Proper Voice Procedures b. LP Blower Checks c. Induction System d. VEX System		DWS
3. Raising / Lowering Masts and Periscopes IAW VCSOP's to Include: a. Man Aloft Considerations b. Check Bridge and Fin Clear c. Lock Out / Tag Out d. RadHaz e. Pre-wet		DWS
4. Describe Submarine CSE Evolutions Alongside to include: a. Ammunitioning b. Tube Water Shots c. SSE Water Shots d. Man Aloft		CSEO CSEC
5. Describe Submarine Seamanship Evolutions Alongside to Include: a. Cold Move b. Warping the Submarine c. Receiving a Vessel Alongside		CHOPS UC1

PERFORMANCE FACTORS	DATE/PRINT/SIGN	SIGNING AUTH
EO 009.04 – React to Onboard Emergencies as a Harbour Watchkeeper		
1. Identify and locate onboard Damage Control equipment to include: a. Thermal Imaging Camera b. SCBA and 50ft Extension c. Force Entry Tools d. Bunker Gear e. EBS Masks f. EEBD (Emergency Escape Breathing Device) g. Smoke Curtains h. EBS / Hydraulic Burst Bags i. Attack BA Bags j. DC Tool Roll k. ERT Bags (Mech / Elect / CSE) l. Fire Extinguishers m. Submarine Incident Board n. WSC Re-entry Poles		MAR TECH (Fire King) MAR TECH (Fire King)
2. Demonstrate use of EBS and Fleeting procedures		QUALIFIED SUBMARINER
3. Locate and describe onboard Medical Equipment to include the following: a. AED b. O2 Kit c. Spare O2 d. Backboard e. Trauma Kits (2) f. Dental Kit g. Cadaver Bags h. Extrication Kit i. Suction Unit j. PA Bag k. Incubator		Submarine PA CCT Member

PERFORMANCE FACTORS	DATE/PRINT/SIGN	SIGNING AUTH
4. Locate and describe the operation of the following Fitted Firefighting Equipment to include: a. SFU 90 b. Centre Fed Hose Reels c. Halon d. WSC Spray e. Engine Room Drench System f. Ammunition Flood Valve g. Firefighting Valve Line-up alongside		MAR TECH Fire King
5. Describe the Bomb Threat Organization IAW VCSSO's to Include: a. Bomb Box Location b. Initial Actions c. Zones d. Search Cards e. Key Responsibilities		DWS
6. Respond to HISB Incident		COXN
React to onboard emergencies IAW VCEOPs as the SCC Watchkeeper		
1. Collision		DWS
2. Fire		
3. Flooding		
4. HP Air Burst		
5. Hydraulic Burst		
6. Man Overboard		
7. Hazmat Spill		
8. Security Alert		

PERFORMANCE FACTORS	DATE/PRINT/SIGN	SIGNING AUTH
9. Casualty		
React to onboard Emergencies IAW VCEOPs as the Discoverer		
1. Collision		DWS
2. Fire		
3. Flooding		
4. HP Air Burst		
5. Hydraulic Burst		
6. Man Overboard		
7. Hazmat Spill		
8. Security Alert		
9. Casualty		
EO 009.05 – Respond to Changes in Force Protection Levels as a Harbour Watchkeeper		
1. Describe Response to a change in Force Protection Levels to include: a. Initial Actions b. VCS FP manning IAW Watch c. Use of Force (ROE cards) d. Location of FP Equipment e. Halifax/Esquimalt Communications f. Key Responsibilities		OPS O/CHOPS
EO 009.06 – Stand Duty Watch as a Harbour Watchkeeper **** MINIMUM OF FIVE 24 HOUR WATCHES ****		
1. Stand Double Banked Duty Watch in Home Port #1		DWS
2. Stand Double Banked Duty Watch in Home Port #2		DWS

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3. Stand Double Banked Duty Watch in Home Port #3		DWS
4. Stand Double Banked Duty Watch in Home Port #4		DWS
5. Stand Double Banked Duty Watch in Home Port #5		DWS

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EO 010.01 – Perform Submarine Escape and Rescue		
2. Describe Submarine Accountability to include: a. Sub Note b. Sub Check c. SUBLOOK d. SUBMISS e. SUBSUNK		NAV COMM
3. Locate and explain the following escape and rescue equipment: k. All Red Cross Lockers to include: 1) Guard Cards 2) Tower Flood Hull Valve Ratchet 3) Spare Vent Cap 4) Last Man Straps l. SESSPE Suit and Storage Locations m. Emergency DISSUB Communications to include: 1) 639 Emergency Indicator Buoys 2) Operation of SEPIRBs 3) Personal Locator Beacons 4) Emergency UWTs n. Air Purification to include: 1) EO2 26H Candles 2) LIOH Curtains o. Atmosphere Monitoring to include: 1) ANALOX 2) Dräger Kits p. Emergency Drinking Water q. Portable Water Purification Equipment r. Escape Rations s. Long Breathing Units (BU's) t. Emergency Pyrotechnic Stores		COXN Escape 2 I/C

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4. Explain the line-up of the following equipment for submarine escape: a. Built In Breathing System (BIBS) b. Hood Inflation System (HIS) c. Escape Tower Preps		COXN Escape 2 I/C
5. Describe Key Escape and Rescue Personnel Positions and Duties		COXN Escape 2 I/C
6. Describe ELSS POD Delivery		COXN Escape 2 I/C
7. Describe Tower Escape from a Contaminated Environment		COXN Escape 2 I/C
8. Last Man Escape Procedures		COXN Escape 2 I/C
9. Explain Escape Surfacing Drills		COXN Escape 2 I/C
10. Describe Submarine Rescue to include: a. Atmospheric and Water Tight Integrity considerations b. Preparing the ATP Hatch and Aft Escape Tower to accept a Submarine Rescue Vehicle or Submarine Rescue Chamber c. Draining the ATP Hatch and Aft Escape Tower coffer dam		COXN Escape 2 I/C
EO 010.02 – Conduct Submarine Surface Abandonment		
1. Identify Submarine Surface Abandonment equipment to include: a. Submarine Surface Abandonment Suit b. 10 Man Portable Life Rafts		COXN Escape 2 I/C

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2. Explain Submarine Surface Abandonment IAW VCEOP Abandon Ship		COXN Escape 2 I/C

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EO 011.01 – Operate the Submerged Signal Ejector at Sea ****** SHALL BE DONE WHILE AT SEA ******		
1. Fire a pyrotechnic or Water shot from the Submerged Signal Ejector IAW SUBWOPS to include: a. Safety Precautions b. Control and Indication c. Modes of Operation: a) Manual Sequencing b) Local Operator Panel (Auto) c) Emergency Hand (Spanner) to include system line-up d. Conduct Submerged Signal Ejector Misfire procedures		W Eng Fire Control
EO 011.02 – Describe the OMC		
1. Describe the function of the OMC		Watchkeeper Dived
2. Identify the following components of the OMC to include: a. Stick Wheel (Yoke) b. Afterplanes in Air Emergency c. Propulsion Panel and Orders to include modes and groups d. The 4 panels (helmsman, command, propulsion and autopilot) e. Heel and trim		Watchkeeper Dived
3. Describe the following Conning Orders and response to include: a. Steer a course/ course changes b. Depth changes c. Change ordered speeds and maintain speed		Watchkeeper Dived

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4. Locate and describe the components of the Main Propulsion System to include: a. Propulsion Switchboard b. Main Motor c. Shaft Line		MAR TECH ELEC MAR TECH MECH
5. Describe the modes of operation of the Propulsion System: a. Primary b. Secondary c. Manual Mode 1 d. Manual Mode 2		MAR TECH ELEC
EO 011.03 – React to Onboard Emergencies at Sea		
1. Identify and describe the function of atmospheric monitoring equipment		QUALIFIED SUBMARINER
2. Operate Dräger Hand Pump with associated Gas Tubes		QUALIFIED SUBMARINER
3. Conduct and report the following: a. Electrical Audits b. Phase One Damage Checks c. Phase Two Damage Checks d. Recommissioning and Re-pressurization of Hydraulics		CSEC PO1 MAR TECH ELEC Wrecker

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4. Describe the Submarine Damage Control Organization and responsibility to include: a. HQ1 b. ERT c. SCOOW d. On Scene Coordinator e. Support Party Leader f. Attack BA g. Attack Teams h. 35 and 56 Bulkhead Watchkeeper i. Casualty Clearing Team		Member of HQ1
5. Describe DC Communications to include: a. VICES b. VCS 446 c. DC Net d. Command Net e. PRCs		Member of HQ1
6. Describe the two primary methods of Submarine Smoke Clearance to include: a. Location of all Smoke Clearance Cards b. Procedure as directed by either HQ1 or SCOOW		QUALIFIED SUBMARINER
REACT TO ONBOARD EMERGENCIES AT SEA IAW VCEOPS AS THE FOLLOWING ****** SHALL BE DONE WHILE AT SEA ******		
1. Discoverer		MAR TECH Coord
2. Attack BA		MAR TECH Coord
3. Attack Team Member		MAR TECH Coord
4. Sentry		MAR TECH Coord

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5. 35 Bulkhead		W Eng
6. 56 Bulkhead		Control Room Watchkeeper
7. ERT Member		Member of ERT
8. Atmospheric Monitor		QUALIFIED SUBMARINER
9. Plotter / HQ1		NAV Comm
EO 011.04 – Conduct a Sea Watch		
1. Conduct Submarine Personal Accountability to include: a. Casing Board and Bridge Board Procedure b. Cycling through the Conning Tower c. Transiting through 56 Bulkhead while Generating / Snorting while requesting permission from MCC 1 or MCC 2		HODS/CHODS

PERFORMANCE FACTORS	DATE/PRINT/SIGN	SIGNING AUTH
2. React to Changes in Submarine States and Routines IAW VCSSOs to include describing the following: a. Special Sea Duty b. Harbour Stations c. River Routine d. Fog Routine e. Blind Pilotage Team f. Transit Routine g. Passage Routine h. Patrol Routine i. Action Stations j. Diving Stations k. Open Up For Dive l. Attack Teams m. Emergency Stations n. Watch Dived o. Watch Snorting p. Quiet States q. Verification Muster		CHOPS

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3. Identify the importance of Lighting Discipline during Periscope Operations to include: a. Red Lighting b. Black Lighting c. White Lighting d. Compartment Lighting		QUALIFIED SUBMARINER
4. Describe the duties and responsibilities of the Upper Deck Trot IAW VCSSO's.		DWS
EVOLUTIONS AT SEA ****** SHALL BE DONE WHILE AT SEA ******		
1. Observe Putting on a Snort in the Control Room		SCC 1
2. Observe Putting on a Snort in the Motor Room		MCC 1
3. Observe Return to Periscope Depth in the Control Room		2OOW
4. Observe Weapons Quarters in the WSC		W ENG Fire Control
5. Observe Emergency-Go-Deep in the Control Room		2OOW
6. Observe Open Up For Dive in 1 of 3 locations		XO / CSEO / MSEO Wrecker / Horse / TI
7. Observe / Experience Surfacing in Normal or Emergency		HOD / CHOD
8. Participate in Noise Rounds		SONAR Watch Leader
9. Observe Blowing of Slop Drain and Sewage		W ENG MAR TECH

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10. Assist with Garbage Management		COXN
11. Operate Emergency Underwater Telephone		W ENG MAR TECH SONAR OP
12. Stand Watch as a Double Bank 35 Bulkhead Watchkeeper		CHOD
13. Stand Watch as a Double Bank SCC 2		CHOD
14. Stand Watch as a Double Bank MCC 2		CHOD

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ANNEX J – MAIN REFERENCES

A1 = CF or other official Canadian government publication

A2 = Second and subsequent references in any category

B = Official Allied Force or Allied government publication

C = Commercial publication

A1	Victoria Class Submarine Standing Orders (VCSSOs)
A2	Victoria Class Emergency Operating Procedures (VCEOPS)
A3	Victoria Class Standard Operating Procedures (VCSOPs)
A4	CFCD 108 Naval Boarding and Force Protection
A5	Manual of Ceremony for HMC Ships, Submarines and Naval Reserve Divisions 2004-11-11
A6	MARLANTORD 24-3
A7	C-23-045-000/AG-004 Damage Control Sea, Volume 4, Submarines
A8	Submarine Workup Guide (SWG)
A9	CFCD 105 Seamanship and Rigging Manual
A10	CFCD 103 Submarine and Escape and Rescue Manual
A11	SUBSMASH COMMPLANS
A12	Harbour Control Instructions (HCI)

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A13	Naval Force Protection SOPs (NFPSOPs)
A14	CANCOM CONPLAN 10251/10 (CF Response to SUBSAR)
A15	MARCORDS
A16	C-03-005-033/AA-000 NEM Naval Engineering Maintenance Vol.1 Mar Sys Eng
A17	C-02-040-007/TS-001 General Safety Publications
A18	C-02-040-009/AG-001 DND General Safety Standards
A19	C-82-007-001/AF-000 Ref Book on Petroleum & Associated Products
B1	BRF 1966 (1) 01 General Data and References
B2	BRF 1966 (1) 04 Bulkheads and Partitions
B3	BRF 1966 (1) 05 Mooring & Towing Fittings, Anchor handling & Storage
B4	BRF 1966 (1) 07 Domestic Equipment
B5	BRF 1966 (1) 08 Weapons, Ammunition & Pyrotechnics Storage
B6	BRF 1966 (1) 09 Depth Gauges & Inclinometers
B7	BRF 1966 (2) 01 Fire prevention and detection System
B8	BRF 1966 (2) 04 Firefighting Systems
B9	BRF 1966 (3) 01 Escape arrangements
B10	BRF 1966 (3) 02 Built-in breathing & Hood Inflation Systems
B11	BRF 1966 (3) 03 Escape equipment
B12	BRF 1966 (3) 04 External Salvage self-salvage & compartment testing arrgts
B13	BRF 1966 (3) 05 Salvage Hull Valve

B14	BRF 1966 (4) 01 VPDC Distribution and shore Supplies
B15	BRF 1966 (4) 02 Earth Monitoring Equipment
B16	BRF 1966 (4) 03 Battery Switchboards
B17	BRF 1966 (4) 05 DC Circuit Breakers
B18	BRF 1966 (4) 06 Generators
B19	BRF 1966 (4) 07 Generator air coolers
B20	BRF 1966 (4) 09 Generator switchboards
B21	BRF 1966 (4) 10 DC Load Centers
B22	BRF 1966 (5) 01 Hydrogen Monitoring
B23	BRF 1966 (5) 02 Battery Ventilation and Hydrogen Clearance
B24	BRF 1966 (5) 03 Battery Ventilation Isolators & Selectors
B25	BRF 1966 (5) 04 Battery Cooling and Top Up System
B26	BRF 1966 (5) 05 Battery Agitation System
B27	BRF 1966 (5) 06 Battery Coolant Pump
B28	BRF 1966 (5) 07 Battery Exhaust Fan Motor
B29	BRF 1966 (5) 08 Battery Exhaust Fan Motor Starter
B30	BRF 1966 (6) 01 24V DC System
B31	BRF 1966 (6) 02 Transformer Rectifier Units
B32	BRF 1966 (6) 03 24V DC Distribution Panels
B33	BRF 1966 (7) 01 440V 60 HZ System

B34	BRF 1966 (7) 02 200/115V 400HZ System
B35	BRF 1966 (7) 04 140KW Generators
B36	BRF 1966 (7) 05 MG air cooler
B37	BRF 1966 (7) 06 140KW Control Panel
B38	BRF 1966 (7) 07 140KW Starter
B39	BRF 1966 (7) 08 AC Switchboards
B40	BRF 1966 (7) 09 AC Load centers
B41	BRF 1966 (7) 10 AC Load fuse panels
B42	BRF 1966 (8) 01 Diesel Engine Paxman Valenta
B43	BRF 1966 (9) 02 Introduction to Power Generation, Propulsion and Control
B44	BRF 1966 (9) 03 Propulsion Control
B45	BRF 1966 (9) 05 Propulsion Motor
B46	BRF 1966 (9) 06 Propulsion Motor Blower and Starter
B47	BRF 1966 (9) 07 Propulsion Motor Air Cooler
B48	BRF 1966 (9) 08 SSMG
B49	BRF 1966 (9) 09 SSMG Starter
B50	BRF 1966 (9) 10 SSMG Control Panel
B51	BRF 1966 (9) 11 Field Supply Converter
B52	BRF 1966 (9) 12 Propulsion Switchboard
B53	BRF 1966 (10) 01 Introduction to Power Generation, Propulsion and Control

B54	BRF 1966 (10) 02 Thrust Block and Shaft
B55	BRF 1966 (10) 06 Propeller Assembly
B56	BRF 1966 (11) 01 HP Air and Nitrogen Systems
B57	BRF 1966 (11) 02 SBS Air System
B58	BRF 1966 (11) 03 Main Vent and HP Blow System
B59	BRF 1966 (11) 04 LP Blow and Ventilation System
B60	BRF 1966 (11) 05 Auxiliary Vent and Blow System
B61	BRF 1966 (11) 06 EBS System
B62	BRF 1966 (11) 07 HP Air Compressor
B63	BRF 1966 (11) 08 HP Air Compressor Motor
B64	BRF 1966 (11) 09 HP Air Compressor Motor Starter
B65	BRF 1966 (11) 10 HP Air Compressor Control Panel
B66	BRF 1966 (11) 11 LP Blower
B67	BRF 1966 (11) 12 LP Blower Motor
B68	BRF 1966 (11) 13 LP Blower Motor Starter
B69	BRF 1966 (11) 14 HP Air Drier
B70	BRF 1966 (11) 17 Air System Valves
B71	BRF 1966 (11) 18 Air System Pressure Controls
B72	BRF 1966 (11) 19 MBT vent Valves
B73	BRF 1966 (12) 01 Ships Ventilation System

B74	BRF 1966 (12) 06 Air Treatment Units
B75	BRF 1966 (13) 01 Chilled Water System
B76	BRF 1966 (13) 02 Chilled Water Plant
B77	BRF 1966 (13) 03 Control and Indication
B78	BRF 1966 (13) 04 Chilled Water Pump
B79	BRF 1966 (13) 05 Chilled Water System Valves
B80	BRF 1966 (14) 01 Main Hydraulic System
B81	BRF 1966 (14) 02 External Hydraulic System
B82	BRF 1966 (14) 04 Main and External Hydraulic Power Plant Systems
B83	BRF 1966 (14) 05 Main and External Hydraulic Pumps
B84	BRF 1966 (14) 07 Hydraulic Systems Portable Hand Pump
B85	BRF 1966 (14) 08 Hydraulic Pump Discharge Manifold
B86	BRF 1966 (14) 09 Hydraulic Accumulators
B87	BRF 1966 (14) 10 Hydraulic Systems Selectors
B88	BRF 1966 (14) 11 Hydraulic Systems Valves
B89	BRF 1966 (15) 01 Trim System
B90	BRF 1966 (15) 02 Bilge System
B91	BRF 1966 (15) 03 Ballast System
B92	BRF 1966 (15) 04 Priming Systems HP Bilge and Ballast Pumps
B93	BRF 1966 (15) 05 Trim and LP Bilge Pumps

B94	BRF 1966 (15) 06 HP Bilge and Ballast Pumps
B95	BRF 1966 (15) 07 HP Bilge and Ballast Pumps: Motors
B96	BRF 1966 (15) 08 HP Bilge and Ballast Pumps: Motor Starters
B97	BRF 1966 (15) 09 Priming Systems Pumps
B98	BRF 1966 (15) 10 Priming Systems Pumps: Motors
B99	BRF 1966 (15) 11 Priming Systems Pumps: Motor Starter
B100	BRF 1966 (15) 12 HP Bilge and Ballast Systems: Flexible Pipe Couplings
B101	BRF 1966 (15) 13 HP Bilge and Ballast Systems: Pressure Compensating Units
B102	BRF 1966 (15) 14 Trim, Bilge and Ballast Systems: Flow meters
B103	BRF 1966 (15) 15 Ballast System Combined Hull & Back-up Valve Actuators
B104	BRF 1966 (15) 16 Trim, Bilge & Ball Sys. Remotely Controlled Valve
B105	BRF 1966 (15) 17 LP Bilge System: Screw-down Swing Check Valve
B106	BRF 1966 (16) 01 Ships Sea-Water Cooling System
B107	BRF 1966 (16) 02 Sanitary System
B108	BRF 1966 (16) 03 Sea-Water Cooling System pumps
B109	BRF 1966 (16) 04 Sea-Water Systems: Valves
B110	BRF 1966 (16) 05 Sea-Water Cooling System: Flexible Pipe Couplings
B111	BRF 1966 (16) 06 Sea-Water Cooling System: Pressure Compensating Units
B112	BRF 1966 (17) 01 Ships FW System
B113	BRF 1966 (17) 02 Domestic FW System

B114	BRF 1966 (17) 03 Ships FW Cooling System: Pumps
B115	BRF 1966 (17) 04 Ships FW/SW Heat Exchangers
B116	BRF 1966 (17) 05 Flow-Regulating Valves
B117	BRF 1966 (20) 01 Surveillance System Configuration
B118	BRF 1966 (21) 01 Submarine Control Console
B119	BRF 1966 (21) 02 One Man Control Console
B120	BRF 1966 (21) 04 Machinery Control Console
B121	BRF 1966 (21) 05 Compartment Flood Alarms
B122	BRF 1966 (21) 06 Grade A and B Alarms
B123	BRF 1966 (22) 03 Rudder and Hydroplanes: Control System
B124	BRF 1966 (22) 04 Rudder and Hydroplanes: Power Plant System
B125	BRF 1966 (22) 06 Rudder and Hydroplanes: Power Plant Equipments
B126	BRF 1966 (22) 07 Rudder and Hydroplanes: Power Plant DC Motor
B127	BRF 1966 (22) 08 Rudder and Hydroplanes: Power Plant DC Motor Starter
B128	BRF 1966 (22) 12 Auto Pilot, Rudder and Planes Electronic Enclosures
B129	Intentionally Left Blank
B130	BRF 1966 (23) 01 Snort Induction Mast and Head Valve
B131	BRF 1966 (23) 02 Diesel Exhaust Mast
B132	BRF 1966 (23) 03 Ventilation Exhaust Mast and Check Valve
B133	BRF 1966 (23) 04 Masts: Ram-Type Elevating Gear

B134	BRF 1966 (23) 05 Masts: Screw-Type Elevating Gear
B135	BRF 1966 (23) 06 Mast Control and Indication Circuits
B136	BRF 1966 (23) 07 Mast Control Valves
B136	BRF 1966 (23) 07 Mast Control Valves
B137	BRF 1966 (25) 09 Garbage Ejector
B138	BRF 1966 (26) 01 Water Tight Doors and Operating Equipment
B139	BRF 1966 (26) 02 Hatches and Doors
B140	BRF 1966 (27) 01 BWA Aerial Transfer System
B141	BRF 1966 (27) 02 BWA Aerial Transfer Equipment
B142	BRF 1966 (28) 01 Capstan
B143	BRF 1966 (28) 02 Windlass
B144	BRF 1966 (28) 03 Anchor and Towing Slips
B145	BRF 1966 (29) 01 Water Tight Bulkhead 35
B146	BRF 1966 (29) 02 Water Tight Bulkhead 56
B147	BRF 1966 (29) 03 Pressure Hull: Fwd Dome to WTB 35
B148	BRF 1966 (29) 04 Pressure Hull: WTB 35 to WTB 56
B149	BRF 1966 (29) 05 Pressure Hull: WTB 56 to After Dome
B150	BRF 1966 (31) 01 Internal Communications Systems
B151	BRF 2075 Main Battery
B152	BRF 2075 (K) Main Battery

B153	BRF 2475 Submarine VLF/LF Aerial Outfit Fin and Buoy Mounted
B154	Intentionally Left Blank
B155	BRF 4500 (529) MK 36 SFC
B156	BRF 4500 (701) Batteries
B157	BR 2203 Ships Husbandry Manual
B158	BR 2217 (1) Pyro Misc. Stores
B159	BR 2217 (9) SSE MK9
B160	BR 3623 (01) Ships Systems
B161	BR 3623 (02) Propulsion Systems
B162	BR 3623 (03) Tactical Weapons Systems
B163	BR 3623 (04) Electrical Generation and Distribution Systems
B164	BR 3623 (7C) Air System Composite
B165	BR 8070 (2A) (04) SUBWOPs
B166	BR 8070 (2C) (1) SUBWOPs
B167	BR 8212 (2) Vol. 1 Operating Handling System
B168	BR 8212 (2) Vol. 2 Operating Launch System
B169	BR 3623 7A Diesel System Composite
B170	BR 3623 7B Water System Composite
B171	BR 3623 7D Hydraulic System Composite
B172	BR 3623 7E Hull System Composite

B173	BRF 1966 (18)
B174	BRF 1966 (24)
B175	BRF 1966 (30)
C1	Propulsion Power Statement
C2	Victoria Class EMS Environmental Management System